Research Project

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A Study on the

“STOCK PRICE PREDICTION USING LSTM AND GRU”

Research Project submitted to Jain Online (Deemed-to-be University)in partial fulfillment of the requirements for the award of

Master of Computer Application

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DECLARATION

I ***BHAGYASHREE PUJARI,*** hereby declare that the Research Project Report titled ***"STOCK PRICE PREDICTION USING LSTM AND GRU"*** has been prepared by me under the guidance of ***DR. SHALINI SURESH.*** I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of degree of Master of Computer Application by Jain University, Bangalore. I have undergone a project for a period of Eight Weeks. I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

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CERTIFICATE

This is to certify that the Research Project report submitted by Ms. ***BHAGYASHREE PUJARI*** bearing USN: 231VMTHR0202 on the title *"**STOCK PRICE PREDICTION USING LSTM AND GRU**"* is a record of project work done by her during the academic year 2023-24 under my guidance and supervision in partial fulfilment of Master of Computer Application. s

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Date: 15/05/2025

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DR. SHALINI SURESH

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EXECUTIVE SUMMARY

This project explores the use of deep learning methods, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) neural networks, in predicting stock market prices. The stock market is a highly dynamic and complex system that depends on a wide range of factors such as economic data, investors' attitudes, and global events, making prediction difficult but highly rewarding for investors and financial experts. The study will create a trusted and precise model of forecasting using past stock market information and contemporary machine learning techniques to guide investors in making well-informed choices, reducing risks, and maximizing returns.

The research employs a large data set collected from Yahoo Finance, with almost 900,000 records that capture important characteristics like open, close, high, low prices, and volumes. The raw data are subjected to rigorous preprocessing, such as cleaning, normalization, and feature engineering, to improve the quality and significance of the inputs for the models. Feature selection algorithms assist in extracting the most useful factors that affect stock prices, facilitating the building of better predictive models. Deep learning architectures—LSTM and GRU—are selected based on their capacity to learn long temporal relationships and nonlinear interactions in sequence data. Data is split into training and validation sets, 80% for training and 20% for validation. The models are trained through Python and Jupyter Notebook environments, minimizing their error of prediction through parameter optimization. Performance measurement utilizes metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

The outcomes show that LSTM and GRU models perform better than conventional regression-based methods, with LSTM showing marginally higher accuracy by efficiently modeling the complex temporal structures of stock price changes. GRU also performs competitively with lower computational complexity. These models effectively model market trends and volatility, providing a richer representation than basic models. This method provides a number of benefits, including increased predictive accuracy, flexibility towards learning from new data, and possibility for real-time predictions to inform automated trading models. Additionally, the incorporation of risk management methods such as stop-loss limits and portfolio diversification

reduces possible financial loss, enhancing the pragmatic applicability of the models for investors.

Yet, the project also has some limitations. The accuracy of predictions relies significantly on the quality of the data, and incomplete or noisy data negatively impacts the outcomes. The deep learning models are computationally demanding, and they need substantial processing power and training time. Moreover, such models are "black boxes," and they provide limited interpretability, which can decrease the level of confidence among users who want to see transparent decision-making tools. In summary, this project shows how LSTM and GRU networks offer strong instruments for stock market forecasting using their ability to handle sequential financial data. Though issues like data quality, computational complexity, and interpretability still pose challenges, the outcomes highlight the potential of these methods in enabling better and more informed investment choices. Potential future studies could further improve model resilience by including additional data sources such as news sentiment and social media trends, creating hybrid models, and increasing transparency to encourage wider usage in financial markets.

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ABSTRACT

Stock market prediction using machine learning has gained significant attention in recent years due to its potential to provide valuable insights for investors and traders. This study explores the application of machine learning algorithms, specifically Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), for predicting stock market trends. The research aims to develop an accurate and reliable prediction model by leveraging historical stock market data and employing advanced machine learning techniques. The proposed methodology involves data preprocessing, feature selection, model training, and evaluation. The dataset, obtained from Yahoo Finance, consists of approximately 9 lakh records of stock market data, including date, symbol, open, close, low, high, and volume. The data is transformed into a data frame using the Pandas library in Python and normalized using the sk learn library. The dataset is divided into training and testing sets, with the experiment set comprising 20% of the data. Two architecture models, Regression-based Model and LSTM, are employed for prediction. The Regression-based Model is used for predicting continuous values through given autonomous values using a linear function, while LSTM architecture is capable of identifying trends and changes that may not follow a regular pattern. The results indicate that LSTM outperforms the Regression-based Model, suggesting its ability to capture interrelationships within the data. The study highlights the significance of machine learning in stock market prediction, as it can assist investors in making informed decisions and maximizing profits. The proposed approach demonstrates the potential of LSTM and CNN in analyzing complex stock market data and providing accurate predictions. Future research directions include incorporating additional data sources, exploring hybrid models, and enhancing the interpretability of the predictions to further improve the reliability and practicality of the stock market prediction system.

CHAPTER 1

INTRODUCTION

INTRODUCTION

1.1 General

The stock market is affected by financial fundamentals, news cycles, and consumer mood. This strategy has so many parts that beginner investors and financial specialists may struggle to forecast stock market behaviour. People say that the stock market is unstable and hard to predict. But technologies like machine learning and artificial intelligence have made it possible to study the stock market and predict what will happen in it in new ways. This study looks at how machine learning might be able to predict changes in stock prices. The main goal of the research project is to build a strong and advanced prediction model that can accurately and regularly guess how assets will do based on a lot of data analysis. With this study, the most important things for the forecast model are being looked for. It will also look at the best machine-learning methods for numerical datasets that show how stock prices change over time. These types of advances in accuracy of predictions could have ramifications beyond just individuals' profit margins: they could impact market strategies and significant investment decisions that might change the way the market works. We have set two main goals for this project: finding the most important features quickly so they can be used as inputs in a prediction model, and correctly predicting how stock prices will move in the future using examples from any mix of supervised and unsupervised learning. In the end, we will judge how accurate and reliable the model we built is. It's possible that accurate statements about the stock market could have a big effect on the sellers and brokers who trade on the market. The impacts of improved prediction accuracy could travel beyond personal profits to affect strategies in the markets, or even collective investment decisions driving market dynamics. Ultimately, with developments in machine learning technology the stock market forecasting is headed in a different direction and businesses and investors will be incorporating new machine learning methods into their practices. In this way, the investment landscape as a whole may cultivate a more data-backed way of making decisions. The information gathered from the research could help people better understand the stock market. The main goal of this project is to choose the best factors to put into the prediction model and compare the most reliable and accurate machine learning methods to see how well the prediction model works. If well-calibrated at guessing how the stock market will shift, it could obviously make money for both the product

seller and the reseller operating in the market. Furthermore, these types of changes in processes for gauging how accuracies in estimates may exist maybe part of a larger societal change with regards to what roles people play in the marketplace and how significant investment decisions may exist. Ultimately, utilizing specific advanced machine learning techniques to predict the stock market may be representative of changes in how buyers learn about the market and make subjective decisions, and thus could make investing more data driven. to the existing knowledge surrounding financial data and machine learning methods. It is, however, often argued that the prediction is chaotic in nature rather than just random. This also means there can always be a structure or theme to the chaos, and can often be understood by studying a tremendous amount of previous data, and through a more systematic investigation in the financial market in question. One simple method to present or explain these complex processes is through machine learning. In other words, in order to predict a market value close to an item's actual value, and ultimately provide accurate prediction estimates. There are increasing numbers of professionals interested in machine learning methods for prediction of the stock market. This is due to how it can help accomplish quick inspections of market behaviours and trends, and have accurate results. There is a clear connection between technologies and finance and the impact on us, they highlight both the continued improvement of our prediction analytics, and what implications they could have on consumer spending behaviour in the future. The importance of this study comes from the fact that accurate stock market predictions could help buyers, traders, and financial experts. If buyers can correctly guess how a company will do, they may be able to make better decisions about whether to buy or sell, increasing benefits while reducing risks. Furthermore, the information gathered from this project could help banks and experts understand the stock market with more knowledge and therefore provide quality advice to their clients. The plan for this project is to collect and pre-process former stock market data, extract the most relevant features, develop a machine learning model and train it, and finally evaluate accuracy. The study will be based on a single stock or a small stock universe to be able to analyse more thoroughly and ultimately build a predictive model with higher precision. Altogether, the main goal of this endeavour is to contribute to the body of knowledge surrounding the application of machine learning in economics and banking. This project stands to provide something new regarding the relationship between market data and machine learning

information systems possibly leading to new thinking about how to understand the stock market and benefit buyers and professionals wishing to make sound investment decisions.

Traders and students have been interested in the stock market for a long time. Things change all the time on the stock market, so it's hard to say what will happen next. There were old-fashioned ways to use statistics to look at how markets moved, but they didn't work very well because markets are always changing. Machine learning has grown into a powerful way to look at huge amounts of difficult data. There is a good way to guess how markets will move with machine learning. Complex stock market trends and connections may be found with machine learning. Patterns seen in the past could help machine learning systems predict how the stock market will move in the future. With this information, buyers may be able to make better decisions about when to buy, keep, or sell a business, which could lead to higher profits. The study's goal is to find the most important parts of a prediction model. Picking the right features is very important when making a machine learning model because. The project will also think about which machine learning method is best for putting the forecast model into action, since every program has its own pros and cons. Checking and evaluating how accurate the model is would also form an important part of the two assessments. Checking the accuracy of the model will utilize different techniques, including mean absolute error, mean squared error, and association factors. The accuracy of the model will help determine if it can ultimately be trusted and used in real life. Why is this research valuable? Because both buyers and practitioners stand to benefit from more accurate predicting of stocks. With credible predictions of how a stock price will trade, buyers will be in a better position to make sounder decisions about buying or selling, which may allow for a greater profit margin. In addition, with such large quantities of data produced from the project, it is possible that many financially institutions, and experienced analyst will utilize this fresh information to de-construct and work with the complexities of the stock market. Subsequently, this will potentially allow them to provide smarter investment decisions on behalf of their clients who rely heavily on their insights. In summary, this comprehensive study seeks to examine the intricate application of different cutting-edge algorithms.

1.2 Brief Information About Significance of Predicting the Stock Market

If we have a good idea of where the stock market is going, then we can save unproductive expense by not making non-productive investments and we can save that money for better investment opportunities. Stock Market Prediction (SMP) however is defined by the statistical approach that takes rich historical data and statistical patterns and applies them to predicting a direction of the stock market that allows investors to make decisions about when to buy and/or sell, based on an empirical basis. Prediction allows for a reduction of risk for investors, and more confidence in their investments. If they could predict goals before they hit them, they might avoid loss of capital. So, all these aspects, are part of SMP. We used previous data, and predicted future trends, this is SMP. Predicting the stock market is important because it can help not only the investor, but also the financial institution, and ultimately the economy. If stock market predictions are accurate, they can help the investor make smart decisions to maximize their potential rewards, while minimizing risk. Financial institutions and analysts can utilize predictions and make more accurate recommendations to their clients resulting in greater reputational and credibility value. Predicting the stock market can also have larger economic possibilities. An efficient stock market which appropriately identifies the value of various companies can create positive economy-wide levels of investment and economic growth. Erroneous forecasts or failure to efficiently price stocks in the stock market could result in major losses to the economy and loss of investors' confidence. Additionally, stock market predictions can allow for trading strategies and risk management strategies to be developed. By making accurate predictions of market trends, one can develop more efficient trading techniques and financial institutions can develop better risk management applications. Generally speaking, the ability to develop accurate and reliable predictions about the fluctuations and trends of the stock market can offer tremendous benefits not only for individual investors who are looking to gain maximum returns on investment, but also to financial institutions who rely on stock market prediction to develop their plans of action and find the best way to manage risk.

Finally, although, the trade and investment returns that can be achieved via accurate stock market predictions is where the benefits of accurate stock market forecasts surface, the advantages for the overall economy can be far greater, as accurate stock market predictions can

be instrumental in promoting stability in the market, reducing risk and providing more efficient resource allocation in the economy, thereby emphasizing the significance of stock market predictions as an important area of research and analysis within the finance discipline.

Predicting the stock market helps consumers and financial institutions make smarter investment decisions. Predictions can help investors purchase or sell stocks at the right moment, maximising returns. It can also inform financial organizations of the best time to make investments in specific stocks or trade shares of stocks in specific industries to fulfill their investment goals. Anticipating activity in the stock market can also have beneficial results in terms of the overall stability of the financial system. Good forecasting can help financial regulators and policymakers discern risk factors in the market and closely consider actions to ameliorate them with appropriate or effective delays. Good future prospects can also help both individuals and financial organizations avert risky investments that can lead to huge losses and results in instability in the financial system. Additionally, predicting what will occur with the stock market can guide organizations regarding their working. By reviewing historical trends and discerning future trends, firms can best make economically sound decisions. For example, whether to allocate more funds to research and development, whether to expand operations or develop opportunities in new markets. To recap, buyers in the market, financial institutions, legislators and politicians, all need to know about forecasting and why stock market forecasting is important. Buyers would make better buying decisions; financial institutions would be able to better manage their risk; regulators would be able to identify possible systemic risks; and, it would help firms decide their long-term desirables which would be good for the stock market overall.

1.3 Approach of Predict the Stock Market Using Machine Learning

To make our estimates more accurate and reliable, we will carefully and methodically plan and carry out a strategy with several separate yet connected steps. This will be done using advanced machine learning techniques. After that, we'll keep our clients up to date on the stock market as it changes and moves. The first thing to do is gather information. The stock market will give us information about the companies we are making forecasts about in the past. This information will have news stories, stock prices, trade volumes, financial measures, and other important details. Once we have systematically gathered and gotten the data using a range of

different approaches, the next step will be a thorough preprocessing phase to make sure that the dataset is complete and in a structured format that can be quickly and effectively processed by a number of advanced machine learning and analytics tools designed for machine learning tasks. Preprocessing consists of data cleaning, feature selection, feature engineering, and normalisation. The following phase is feature selection, which means choosing the most important features to use in your prediction model using statistical methods like correlation analysis, PCA, or other valid means. Following a systematic procedure will really help us determine the most significant variables affecting stock prices and stock market performance indicators. After we have finished racing through the feature selection phase, we will turn to the task of finding the best machine learning approach to use in our predictive model, a decision which will certainly be informed by the characteristics of the problem we are dealing with and the data that we have. Many types of machine learning techniques can be used, but the ones most often used to guess what will happen in the stock market are regression analysis, decision tree algorithms, support vector machines (SVMs), to name a few. Each has good and bad points. There is one more thing we need to do before we add new data to the model and train it. In other words, we will split the information into training and testing sets that have already been handled. This will provide us with sufficient data to train our model and allow us to adjust some of the model parameters. Additionally, we will also adjust the program's hyperparameters to help make the model more accurate when it predicts what will occur when run on a separate data set. Once the model is trained, we will check its predictability using measures including mean absolute error, mean square error, and correlation coefficients. This will allow us to more concretely understand the model's predictability measures for predicting stock prices and impending movements in the overall market. In the end, we should have a predictive model that not only provides accurate forecasts, but is reliable so that it can be used to provide an ongoing forecast of trending stock market movement in real-time. We will continue to improve the accuracy and efficiency of the model, so we will continue to adjust and improve the model which will draw from newly obtained data and user-provided input, as part of a recursive improvement process. In summary, our methodology for predicting stock market movement through machine learning consists of an ordered process of collecting and preprocessing raw data, determining features of interest, choosing a machine learning algorithm, constructing, and testing the model.

To further elaborate on the approach, we will be taking to predict the stock market using machine learning, let's delve deeper into each of the steps:

- Data Collection: In the subsequent phase of our research methodology, we shall embark upon the comprehensive endeavor of gathering and aggregating data from a multitude of diverse sources that encompass a wide array of financial websites, specialized data providers, as well as reputable news outlets that disseminate pertinent information relevant to the financial domain. The information that we are going to be collecting will be quite elaborate time-series data, where the changes and movements of stock prices, along with other important and relevant features of the financial instruments are accurately and meticulously recorded at specified time intervals for analysis and interpretation. The process collection of data will be the basis for the subsequent analysis, furthering our understanding of the market dynamics to come to an informed conclusion based on the observed trends and patterns in the time-series data.
- Data Preprocessing: A dataset must be cleaned and pre-processed before being used for machine learning methods, which means cleaning and pre-processing to make sure it is complete and suitable for machine learning methods. This involves, but is not limited to, checking for missing values, removing the outliers, and transforming the dataset into a format usable in the machine learning proces.
- Feature Selection: After data cleaning, it is important to determine the most important features in predicting stock prices. There are various methods to choose the most relevant variables. We can use several feature selection methods, including correlative analysis and PCA (Principal Component Analysis).
- Model Selection: We will find the right machine learning method for the predictive model after choosing the traits that are important. There are several methods that will be compared, such as regression, decision trees, SVM, and neural networks. The best technique will be chosen based on how the problem is structured and how the information is structured.
- Model Training: After model selection, we will conduct training using preprocessed data by dividing the dataset into test and training data, where model parameters are fine-tuned

through the training data, while also adjusting the hyper parameters of the algorithm to improve the models accuracy.

- Model Evaluation: Absolute mean error (MAE), mean squared error (MSE), and relationships will be used to find the best-fitting model. These measures show how well the models worked after they were trained.
- Model Deployment: When we have built and tested the model, we will embed it within a web-based application or a mobile application that will allow traders and investors to create real-time forecasts.
- Model Refinement: Ultimately, we will continuously improve the model's precision and efficiency by continually bringing in new data and user data. This means providing the model with new information, and adding more functionality over time to improve its robustness and accuracy.

Before we can use machine learning to guess what will happen in the stock market, we need to gather and prepare data, pick features that are useful, train and test the model, use new data to test its forecasts, and then make changes to the model based on new data and user comments.

1.4 Challenges Facing the Use of Machine Learning in Stock Market Prediction

Pick the right features, train and test the model, let the model make predictions on new data, and then update the model with new data and user feedback. This is how machine learning can be used to guess what will happen in the stock market.

- **Data Quality and Availability:**

The accuracy of machine learning models depends on the quality and quantity of data. However, financial data is usually messy and incomplete, which makes it very hard to make accurate predictions. Furthermore, not all data providers have reliable data and the quality of the data differs from one provider to another. Additionally, missing important data with regards to insider trading will also affect the performance of fundamentals machine learning models.

- **Overfitting:**

  Overfitting is when a machine learning algorithm is carefully matched to a single dataset to the point where it can't be used on new data. This is a big problem when it comes to the stock market, which is naturally not fixed, so trends from the past may not happen again . As a result, the challenge of developing a model that can provide reliable predictions about future price movements is greatly magnified.

- **Black Box Models:**

  Machine learning algorithms are often labeled "black boxes" due to their intrinsic inability to be interpreted. The term "black box" implies that understanding the reasoning behind a specific prediction made by the model is non-trivial and may raise concerns related to the model's explain ability and interpretability. The inability for interpretation can be unsettling in the finance business, where it is often essential to understand the reasons for predictions.

- **Market Efficiency:**

  The equity market is efficient, meaning that share prices are already accounting for all available information. Therefore, learning models would struggle to consistently beat the market. As other traders and investors implement machine learning models, the equity market will become more efficient, making it even harder to out-maneuver others.

- **Human Bias:**

  The quality of the data used to teach machine learning algorithms determines how well they work. Biases in humans can have a big effect on the quality of the training dataset. Let's say that the data that was used to build a model was biased towards a certain business or market area. That means the model might not be able to tell you what prices will be in other places. In the same way, a model might not be capable to correctly predict how prices will move in the future if the information it was trained on isn't spread out over time and has a time limit. To sum up, machine learning could change how stock market predictions are made, but it needs to be fixed before it can make all of its benefits known. It can be hard to use machine learning to guess what will happen in the stock market because of things like bad data or not having enough of it, black box models that don't explain anything, markets that aren't working well, and mistakes made by people.

1.4.1 Role of machine learning in stock market prediction

One place where machine learning has come in handy is the stock market. Trends can now be found in large amounts of data thanks to machine learning. This has changed how experts guess that stocks will move. There are many things that machine learning systems can use to make good guesses about what stock prices will be in the future. These include market data from the past, reader feedback on news stories, and many more. One great thing about using machine learning to predict the stock market is that it can handle very large and difficult datasets. A lot of information is made every day by financial markets, such as past price data, trade amounts, economic indicators, news stories, and more. It's possible that these files will be hard for traditional statistics tools to work with. These pieces of information can be used by machine learning systems like neural networks to come up with useful results. The connections and trends that machine learning models can find are so complicated that people can't see them. In this way, machine learning models can look at many variables at once, find very small links between them, and make more accurate predictions. Systems that can learn on their own might be capable to do things like look for things. The study of how people feel about something is a further method that machine learning is used to guess what will happen in the stock market. Figure out how people feel about the market with machine learning. To do this, they check news stories, social media, and other sources to see how people feel about different stocks and the market as a whole. This helps buyers and sellers because it shows how the stock market feels in general. This can cause the price of a stock to change. Mood research can also make models for machine learning systems more complex, which could help them make more accurate predictions. Finally, reinforcement learning is another type of machine learning that has been used to guess what will happen in the stock market. They learn by interacting with their environment and figuring out what works and what doesn't based on what they do. As stock traders, for example, these models change how they spend based on the prizes or punishments they have gotten for their actions.

- Choosing which features to use and reducing the number of dimensions: a number of machine learning methods can automatically find key factors or features that impact stock prices. The number of dimensions in the information will simply go down, which will make the forecasting models faster and more accurate.

- Risk assessment and portfolio optimization: Advanced machine learning techniques can be utilized to facilitate risk assessment through measuring volatility and possible decrease on individual stocks or investment portfolios. They can also assist investors in portfolio optimization through identifying optimal risk-return properties of asset allocation.
- Programs that use machine learning can quickly and in real time look at huge amounts of data and decide what to do with it. This is known as high-frequency trade. Very short-term mistakes in bear markets are utilised by high-frequency trading models to make money, as well as very small price changes.
- Detection of market anomalies: Machine learning algorithms have the ability to detect market irregularities or unusual patterns in stock prices that differ from expected behavior. Detecting irregularities could represent significant events (e.g., announcements, unexpected earnings, or fraud). By detecting anomalies, it provides valuable insight for traders and investors.
- For example, data in the stock market changes over time and follows a set structure. Models for machine learning, like recurrent neural networks (RNN) and long short-term memory (LSTM), are better at guessing what will happen with data that is organised over time. By looking at market data from the past, these time-based processes try to guess what stock prices will be in the future.
- Algorithmic trading: By examining and recognizing market trends, automated trading strategies using machine learning algorithms can execute trades based on criteria. This method helps to eliminate the influences of potential biases and emotional responses that human traders can introduce into the process, thereby providing a more objective and disciplined alternative approach to decision-making.
- Adaptive frameworks: Machine learning frameworks are able to learn and change over time with new data. They have the ability to learn from users and market reactions, allowing them to develop and change their predictions and actions. The frameworks "addictiveness" allows for continual advantage of being relevant and responsive amid changing market conditions.
- Market sentiment assessment: advanced machine learning algorithms can process social media, news stories, and other text data to assess market sentiment at the present moment.

Machine learning algorithms capture the collective sentiment of market actors and also capture price movements arising from psychology and sentiment.

Retrospective studies and performance evaluation: Machine learning algorithms may be evaluated retrospectively using historical datasets to analyze the performance and evaluate the predictive power of the algorithms. This step helps to improve the models while providing detail about the strengths and weaknesses of the models prior to use in real-time trading.

In summary, machine learning constitutes a range of applications to predict stock market direction, from feature selection, risk assessment, anomaly detection, algorithmic trading, and sentiment analysis. By applying data analytics and algorithms, machine learning could help improve decision-making related to the stock market and improve investment strategies.

1.5 Machine learning powerful tool for predicting stock market trends

Machine learning can now help you figure out how the stock market will move. Most of the info that comes from the financial markets is huge. Programs that use machine learning can look at this data and find patterns, trends, and market moves that people probably won't be able to see. Artificial neural networks are a type of machine learning that is often used to guess what will happen on the stock market. The idea behind ANNs is that they should work like the brain does, with layers of linked nodes that look at and process information. The neural networks can be taught with old market data to find trends or links between things like stock prices, trade numbers, economic factors, and more. The network can be used to guess where the markets will go in the near future after it has been properly trained. The support vector machine (SVM) is another type of machine learning that is often used to guess what will happen in the stock market. SVMs are a type of guided learning algorithm that can really help you separate pieces of data into groups. An SVM would be taught to look at market data from the past and tell us if our stock will go up or down. We could use this to guess what will happen on the stock market. Scent analysis is another, more general use of machine learning algorithms. It looks at news stories and social media to see how people feel about a business or the business itself. In evaluating this data, machine learning algorithms could identify a potential market moving event before it happens, which would help investors become more informed traders. Overall, machine learning appears to be useful in predicting stock market

movements, however it is important to note that predictive algorithms cannot guarantee future events. Market conditions can change rapidly and unexpectedly, and even the best machine learning algorithms may struggle to adapt. However, by combining machine learning with human judgment and responsible risk management practices, investors can extract important information from the markets also engage in smarter investment layouts. ANNs and SVMs aren't the only machine learning methods that buyers can think about when they want to predict the stock market. Decision trees, random forests, and deep learning networks are some others. In addition to technical indicators, investors could also use other machine learning algorithms with fundamental data or alternative forms of data such as satellite images or social media sentiment.

Every day, machine learning methods are much better at handling big amounts of data than humans are. This is one of their best features. This means that buyers can look at a lot more info more quickly and make decisions as soon as they notice changes in the market. In addition, machine learning models can figure out complicated trends and connections in the data that might be hard for people to see or describe. For instance, a machine learning model might figure out that certain stocks do well in certain weather situations or that there is a link between a stock's success and all the talk on social media. Even so, the data that is used to teach machine learning models is what makes them good. When you use old data that doesn't reflect how the market is now, you might end up with models that can't tell you what the market will do in the future. Also, machine learning algorithms can overfit, which happens when the old data used to train the model is applied too tightly. This means that the models can't find new structures or behaviours in new data sets. Therefore the choice and pre-processing of models developed in the machine learning process is important, as well as periodically testing the model with future datasets. To sum up, machine learning has plenty of potential utility for predicting trends in stock markets, but its credibility as a strategy should be supported by human experience and risk management considerations. Machine learning algorithms have the capacity to analyze unprecedented volumes of data, which could yield some useful inferences about the market which in turn may help with the investors investment decisions.

1.5.1 Benefits of using machine learning for stock market prediction

- The stock market and the rest of the world have changed since machine learning came along. There is a lot of information out there in the world of stock trading, and buyers and traders can use machine learning tools to sort it out and use it to learn more and make better choices. There are several good things about using machine learning to guess how the stock market will move in the future:
- Making choices based on facts: A lot of real-time and past data is used by machine-learning models. This includes stock prices, financials, news stories, social media opinion, and macroeconomic factors. When these data points are looked at by machine-learning models, they can find trends and other connections that a human expert might miss. This method based on data helps buyers make smarter choices because they are better.
- Increased accuracy and efficiency: Traditional stock market analysis requires analysts to review large datasets manually and create reports. It is a slow process and can lead to human errors. However, machine-learning algorithms are able to evaluate large datasets in real-time, detect complicated patterns, and make further accurate predictions, all much more efficiently. This allows traders to better respond to changes in the market and make quicker investment decisions.
- Pattern Recognition and Prediction Modelling: One of the best things about machine learning algorithms is that they can find patterns and trends in large amounts of data. It can find and look at trends in the past that involve trade numbers, stock prices, and other factors. When a machine learning model sees these trends, it can tell when stock prices will change, which is very useful for investors who need to know where the market is going and how to position their investments properly.
- Risk assessment and portfolio optimization: Investment professionals can leverage algorithms based on machine learning to develop and evaluate their investment strategies based on exposure to risk. Machine learning algorithms are capable of analyzing historical data and current market environments to assess various risks: market risks related to volatility in equities as well as individual risks at a company level, or both. Equally importantly, machine learning models can also assist in improving the portfolios managers create to achieve the best possible asset allocations and diversification consistent with the investor's objectives while remaining within risk tolerance levels.

- Sentiment analysis and news impact: Machine learning algorithms can read news articles, social postings, and other written content to assess market sentiment and how possible influences on stock prices exist. If stock market forecasting model incorporates investor sentiment, an investor can get a sense of how positive or negative news will likely impact market behavior. This information could help an investor change their position, or to take a market sentiment used to seize a profit opportunity through trading.
- Automated trading and algorithmic strategies: Machine learning represents a new leap in algorithmic trading. Traders can automate their trading activities with machine learning algorithm and execute trades on their parameters. In addition, the algorithms are continuously evaluating now changes incremental to markets and identifying trading opportunities and executing trades very accurately and quickly. By automating trading activity, removes all emotional and bias out of the process and allows orderly and consistent trading decisions.
- Back testing and evaluation of performance: Machine learning algorithms can undergo back testing with historical data files to observe their performance and accuracy. By simulating a trade with historical data, an investor is able to review the performance of their different machine learning systems and modify each system as he/she wants. Back testing could be utilized to identify potential flaws, enhance models, and build better trading systems.

Machine learning has a lot of potential for figuring out the direction of the stock market, but it's important to remember that stock markets are inherently unclear, which means that no model will ever be 100% correct. How well machine learning models work will depend on the quality of the data, the features, the design of the model, and how the financial markets change over time. Thus, it is important to use machine learning as a means to augment human decisions and approach investment decisions with caution.

1.5.2 Role of data preprocessing and feature engineering in stock market prediction

For reliable stock market data predictions, data preparation and feature engineering are needed. This is because there is a huge amount of data that can be used as new information. The stock market is a very complicated and unstable system that is affected by many things, such as a company's financial records, market conditions, global signs, and even the way investors think.

Preprocessing typically includes the identification and removal pertinent and unusable data, data normalization, and treating missing values. Data cleaning is the actions taken to remove discrepancies, incorrect values, the presence of outliers that can distort the interpretation of the analysis or prediction. Normalization means data is taken from specific scales and mapped to facilitate analysis by removing possible biases from measurement units. Treating missing values means identifying and removing or replacing lost values through impute mechanisms. Each of these actions may be done repeatedly where needed, to eliminate all disclaimers from affecting data that could obstruct analysis. Feature engineering involves generating new features using existing datasets in order to present information in a new way to address the most pertinent features and minimize extraneous data. Feature engineering is an important in stock market forecasting since it generates new features inciting meaningful data from the surplus of data. Statistical tools like moving averages, volatility indicators, and technical tools like the Relative Strength Index (RSI) and Moving Average Convergence Divergence (MACD) can be used to do this. When the features from the feature creation are added to the analysis model, it gives the model a better idea of the stock market data's structure and trends. Furthermore, the use of domain knowledge during feature engineering is particularly useful for identifying and adding relevant external variables that might impact stock prices. External variables could involve news sentiment scores, economic indicators, social media activity, and even political events, so embedding those features means the predictive model can account for some of the reasons for stock price movements and improve accuracy. The quality of data preparation and feature engineering can have a big effect on how well and reliably stock market prediction models work. Once the data has been cleaned, processed, and prepared, and the right features have been chosen and added, the model is better able to find patterns and relationships in it. This helps investors, traders, and other financial institutions working in the stock market make better predictions and decisions.

When trying to figure out how the stock market will behave, there are important things to think about when preparing data and creating features. There are a lot of things that make the stock market difficult and unstable. These include public information about companies' financial success, current market trends, economic data, and investor mood. Using preparation and feature engineering steps to get the most useful information from the data is important for

making correct guesses. During data preparation, raw stock market data is organised and changed into a file that can be used for analysis. Data preprocessing is necessary, because the raw data sources may be incomplete, flawed, and have different representations of the same measurement. A part of the data preprocessing process is data cleaning, which removes inconsistencies, flaws, and outliers. At this point, the data is properly organized, cleaned and represents the underlying trends in market. The next part of data preprocessing is normalization. This is particularly important for stock market data, because the data may come from different sources that have different measurement units and scales. Data normalization puts the data on a standard scale that can eliminate biases from the differences in units, and allow for fair comparison. On the other hand, if there are many missing values, it may be reasonable to eliminate those missing values to keep from causing bias in the analyses. The approach to missing values is really about how the missingness and dataset are structured. Feature engineering is a way to use data to make new features that hold useful information for stock market predictions. This is done to improve the model's ability to make predictions by finding trends and connections that might not be obvious from the raw data. You can use moving averages, volatility indicators, and technical indicators like the Relative Strength Index (RSI) or the Moving Average Convergence Divergence (MACD) to do this. For future stock moves, these new features show how prices change, how fast prices move, and how volatile prices are.

Domain-specific knowledge, in addition to statistical and technical indicators, is frequently used during feature engineering to aid feature identification and the introduction of external factors that could have an impact on IMOX stock prices. For example, sentiment analysis of news can capture the market's relative response to positive or negative news regarding a company. Macroeconomic indicators, such as growth rates of GDP, interest rates, or inflation, could provide analytical insights regarding the overall market by introducing macroeconomic factors affecting the stock price. Analyzing social media trends can provide public sentiment and can affect stock prices. Geopolitical incidents could also be a viable external factor to consider. If these outside variables are added to the prediction model, it can also show how much these factors affected the movement of the stock market. The prediction model is now more accurate because of this. How well and reliably stock price predicting models work is

directly related to how well the data is preprocessed and how well the features are designed. Models can find important data trends and connections in the stock market by carefully planning the data, engineering features, and adding relevant features. In turn, this helps buyers, traders, and other financial companies that work with the stock market make more accurate predictions and better decisions. Correct preparation of data and feature engineering.

1.6 Machine Learning and Its Significance in Stock Market Prediction

A branch of artificial intelligence (AI) is machine learning. It helps computers learn from information and make choices or guesses without being told to do so directly. It does this by using algorithms and statistical methods. People in the financial world love this technology. It's especially useful for making predictions about the stock market because it can be used to look at old data sets and guess what trends will happen in the future. Stock market predicting is naturally difficult because it requires looking at a lot of different factors, such as market trends, company success, and events happening around the world, to guess how much a stock will be worth in the future. Traders and buyers can make smart decisions with the help of machine learning programs that can explain changes in stock prices, spot trends, and spot outliers in the market. A lot of different Machine Learning techniques are used to make predictions about the stock market. These include regression, classification, grouping, and deep learning. One way to guess how much a stock will be worth in the future is to look at how well it has done in the past. Classification techniques are employed to ascertain the likelihood of a stock's appreciation or depreciation in value. Clustering techniques allow for distinct stocks with similar attributes to be grouped together based, perhaps, on asset-related classifications such as, industry classifications or market capitalizations. Like neural networks, deep learning systems can look at big, complicated collections to find useful information. Machine Learning can handle a lot of data quickly and properly, which is one of the best things about using it to guess what will happen in the stock market. Machine learning programs can look at news stories, social media posts, old data, and other types of data to help traders and buyers guess what the market will do next. One more benefit is that machine learning systems can learn from their mistakes and get better at predicting the future over time. This means that adding more data to the programs will make them more accurate and reliable. This will help people make better spending decisions. While machine learning systems can help with some things,

it's important to remember that predicting the stock market is hard. Predictions aren't always accurate because of things like unplanned market events, quick changes in how well a company is doing, and bad or missing data. Overall, Machine Learning is a very useful tool for predicting the stock market. It gives traders and buyers a lot of useful information about market trends and predicted price changes. It is by no means a perfect solution, but could one day change how investment decisions are made in finance.

1.6.1 Machine Learning has several applications in stock market prediction, including:

1. As part of Machine Learning, computers can look at news stories, social media posts, and other things to figure out how other people in the market feel. Understanding how people feel about something can help you guess how the market will move and pick out the best choices.
2. Pattern recognition: Algorithms in Machine Learning can identify patterns in changes in stock prices, trend lines, price levels of support and resistance, and various technical indicators. Such analysis can help traders identify potential opportunities for buying or selling.
3. Portfolio optimization: Machine Learning algorithms can improve investment portfolios by using historical data to identify combinations of assets which produce the best returns while minimizing risks.
4. Fraud detection: Machine Learning algorithms can recognize fraud behavior (insider trading, market manipulation, etc.) and investigate the patterns in transaction data.
5. Algorithmic trading: Machine Learning algorithms can formulate trading strategies to trade automatically based upon market conditions and expected price changes.
6. Linear regression: Linear regression is a way to use statistics to show how one or more independent factors (like company success and market trends) affect a dependent variable (like stock price).
7. Decision trees: Different kinds of problems can be solved with decision trees, which are a type of method. They use a set of rules to divide the data into smaller sets that they can work with. Once they have these sets of data, they can use them to.

8. Random forest: We use a group method called "random forest" and lots of decision trees to make more accurate estimates. Random forests make a lot of decision trees and then figure out what will happen by averaging all of their guesses.
9. Support vector machines (SVMs): Help out SVMs are a type of technique that can be used to fix issues with both regression and classification. They figure out the best way to split the data into groups or guess how much a continuous variable will be worth.
10. Neural networks: Neural networks are a type of deep learning that is used to look at very big and difficult information. They get ideas from the way the brain works and use a similar structure to find connections.

In closing, Machine Learning is a sophisticated method of forecasting stock market behaviors that can provide valuable insights in regard to market movements, and allow investors to make educated decisions. While it doesn't provide a perfect solution, it can change how we think about making investment decisions in the financial space.

1.7 Significance of Stock Market Prediction:

- Investment Decision-Making: The stock market predicting models offer essential help to investors in making an informed decision when buying, selling or keeping a stock. These models can offer impressive insights into potential price movements, by analyzing past data and finding patterns.
- Risk Management: Accurate predictions of the stock market, will assist investors in managing risk, to better adjust his/her portfolio based on his/her expectations of change within the markets to minimize losses and to maximize profits.
- Algorithmic Trading: Models of predicting stock markets using machine learning techniques are often used as part of algorithmic trading systems, whereby trades are executed automatically based upon criteria determined by the user. These systems can react quickly to changing market conditions, thereby improving trading outcomes.

1.7.1 Machine Learning Techniques for Stock Market Prediction:

- Linear regression, support vector regression (SVR), and random forest regression are some of the regression methods that can be used to predict continuous factors, like stock rates.

These models use past price information and other important factors to guess what prices will be in the future.

- A time series analysis looks at how data from the stock market changes over time. Because of this, time series analysis can help you find trends and guess what prices will be in the future. If you want to look at a time series, you could use recurrent neural networks (RNN), long short-term memory (LSTM) networks, or autoregressive integrated moving average (ARIMA).
- Feelings: You can find things on social media, in news stories, and in text data that could change the price of stocks. So, natural language processing (NLP) and mood classification algorithms can be used on text data to learn how people feel about the market. This should help people make stock market predictions.

1.7.2 Data Sources and Preprocessing

- Historical stock data: Historical stock price data, consisting of open, close, high, low prices, volume, and other financial indicators, are a required data set to train or test a stock market prediction model. There are multiple financial data providers that provide complete historical data sets.
- Fundamental data: Financial records, profit reports, company news, and general data all show how well a business is doing financially and how the market is doing overall. When you use this basic data along with past stock data, you can make more accurate estimates.
- Data Cleaning and Feature Engineering: Although the data sets provided by financial data providers may be clean, it is important to remember that raw data must always be preprocessed for any missing values, outliers, and unmatched values. Data preprocessing also includes data cleaning. Feature engineering techniques can be used to help preprocess, i.e., scale and normalize, select features and extract relevant information in order to improve how well the model predicts.

1.7.3 Evaluation and Model Selection:

- There are several ways to measure how well a stock market prediction model works. These include accuracy, mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE). These measurements check how far off the expected stock prices are from the real prices.

- Cross-Validation: When evaluating models for most machine learning or deep learning approaches, cross-validation (like k-fold cross-validation) is a common step added to your modeling process. This technique is used because it ensures generalizability by training and testing a model using subsets.
- Model Selection: After developing a model or models for stock market predictions, you will compare the performance metrics and cross-validation results, and to determine which models would be the best stock prediction.

Using machine learning to guess what will happen in the stock market is very helpful for buyers and sellers because it lets them learn more about the market and choose better options. There are many strong machine learning methods that can be used to make good guesses about stock returns. These include regression models, time series analysis, and mood analysis. This can help you choose investments, look at risks, and set up high-frequency automatic trade options. Data requirements for the purposes of using machine learning to predict stock market directions should be obtained from different sources for maximum accuracy and utility regarding investment options. When obtaining historical stock data for training and testing forecast models, internet resources or financial data vendors can be beneficial. Along with fundamental variables including; financial statements, earnings announcements, and macro-economic indicators enhances analysis so that stock returns can be placed in context with the overall business health. There may be variations in the data collection process; incomplete figures, incorrect values, or inconsistent statements often need to be repaired before it is useful. This will help you get the most accurate results. You can clean up the data and pick out the traits that are most vital to the machine learning stock return forecast model. To find out how far off expected and real stock prices are, performance factors like precision, mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE) are often used. You can find out how general the models are with k-fold cross-validation and other cross-validation methods that train and test the models on different groups of data. You can choose the best machine learning model for telling the stock market what will happen after reading the study. You can use regression models, time series analysis methods like ARIMA and LSTM networks, and mood analysis tools. It all depends on the data and what it needs to be used for. Finally, it is very important in today's business world to use AI to guess what will happen in

the stock market. Traders and buyers can use advanced methods and techniques to learn more about market trends, deal with risk well, and get the most out of their financial plans. To make sure accurate and reliable stock market forecasts, it is important to get good data, process it correctly before using it, test different prediction models, and pick the best one.

1.8 Objectives

- Make a deep learning model using LSTM and GRU networks to accurately predict stock prices by studying past market data and finding complex trends in how prices change.
- Using time series cross-validation to check how well the model works on different time periods is important to make sure the model is stable and can work with unknown data.
- Check how accurate and reliable the model is at predicting stock prices by finding the MSE, RMSE, and MAE. This will show how well the model is at making forecasts and reducing mistakes.
- - Use risk management tools like stop-loss limits to avoid big losses and portfolio variety to spread risk. This will reduce the damage that could come from wrong stock price predictions and help you make better decisions when the market is volatile.

CHAPTER 2

LITERATURE REVIEW

LITERATURE REVIEW

2.1 General

Review of literature is a way to look at many books, papers, and other types of materials that are connected to the topic of the research. This is done to give a full account of the study, a summary of the study, a critical assessment of the study, and a thorough look into the problem. A literature study gives an overview of the research that has already been done by different authors that is connected to the current research. When you do a literature review, you organise the patterns of the articles you read into groups of related ideas through summary and analysis. This is done to make sure that the old studies are still useful today and to connect the old studies with new research. Plus, it helps you figure out what intellectual changes have happened in the last few hundred years. Also, it helps find the holes in the study.

2.2 Literature Map

In this study survey has been taken to Study Stock Market Prediction Using Machine Learning.

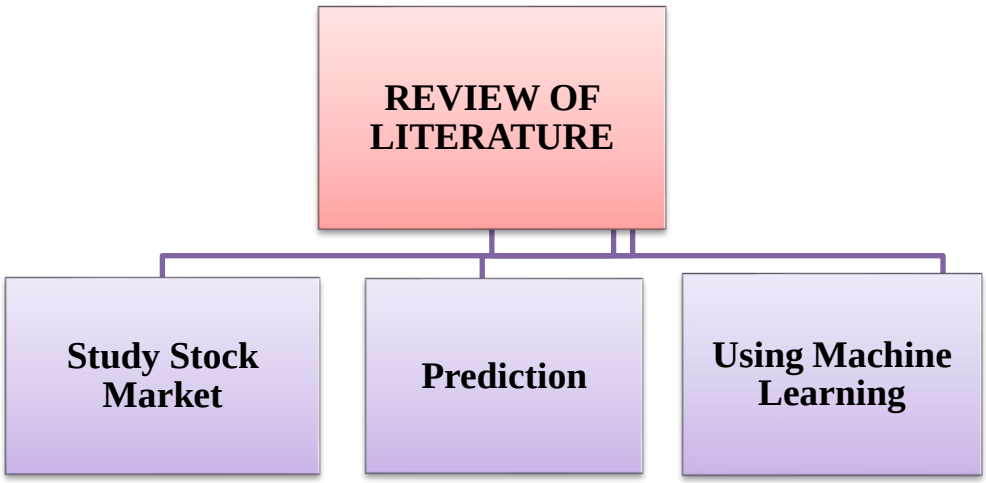


Figure 2.1 Literature Map

Stock Market Prediction Using Machine Learning

Publication Year: (2018)

Author: Lokesh Chouhan, Navanshu Agarwal

Summary: You have to guess how much a company's stock will be worth in the future to win Stock Market Prediction. A new trend is to use machine learning to try to guess what will happen in the stock market. Figures out what things in the stock market are worth now based on what they were worth in the past. In machine learning, various models are used to help people guess what will happen more accurately and more easily. LSTM-based machine learning and regression are used to predict how much stocks will be worth. This is what the research is mostly about. Among other things, it looks at open, close, low, high, and number. The Yahoo Finance dataset had been utilised for both LSTM and Regression in this work. Both of these ways have helped make estimates more correct, which is a good sign. New machine learning methods have been used to make accurate guesses about stocks. These predictions are now used in trade plans that make money.

Stock Market Prediction Using Machine Learning

Publication Year: (2018)

Author: V Kranthi Sai Reddy

Summary: To run a business, you need to trade stocks. On a stock market, people buy and sell stocks and other financial items. You can try to guess how much these items will be worth in the future. The name for this is stock market forecasts. Machine Learning can be used to guess what will happen with a stock in this study. Some of the ways that stockbrokers try to guess what will happen with stocks are through simple analysis, time series analysis, and statistical analysis. People use machine learning to try to guess what will happen on the stock market. Python code is used to make this happen. For this work, a Machine Learning (ML) technique is being put forward. It will learn from stock data that already exists, get better, and then use what it has learnt to make a good guess. Help Vector Machine (SVM), a type of machine learning, is used to guess stock prices for different markets and companies with big and small market caps. The prices change every day and every minute.

Stock Market Prediction Using Machine Learning Techniques

Publication Year: (2016)

Author: Mehak Usmani, Syed Hasan Adil, Kamran Raza Iqra

Summary: The main point of this study is to guess what will happen on the Karachi Stock Exchange (KSE) market at the end of the trade day using various machine learning techniques. The model that makes the predictions looks at different factors and says that the market will either be up or down. Aside from interest rates and news feeds, the model also looks at the prices of gold and silver, oil, and foreign exchange (FEX) rates. If you want to use an older statistical method, you can use a SMA or an ARIMA, which stands for "Simple Moving Average." There is a competition between the Single Layer Perceptron (SLP), Multi-Layer Perceptron (MLP), Radial Basis Function (RBF), and Support Vector Machine (SVM) machine learning methods. Each of these traits is also looked at by itself. This was the best way for the MLP algorithm to work. The most important thing for market success was found to be the oil rate. We can use machine learning to guess how the KSE-100 index will do based on the results.

Stock Market Forecasting Using Machine Learning Algorithms

Publication Year: (2015)

Author: Shunrong Shen, Haomiao Jiang, Tongda Zhang

Summary: It's been a long time since people who study many things tried to guess what would happen in the stock market. You can use two types of machine learning to try to guess how the stock market will move. These are help vector machine (SVM) and reinforcement learning. To guess what the next day's stock trend will be, we're working on a new project that looks at the time link between global stock markets and other financial goods. When it came to NASDAQ, the guesses were right 74.4% of the time, 76% of the time for S&P500, and 77.6% of the time for DJIA. Along with the same method, different regression techniques are also used to find the real market rise. That's it! Now you just need to make a simple trade model to see how well the suggested prediction system works against other rules.

Stock market prediction using machine learning classifiers and social media, news

Publication Year: (2022)

Author: Wasiat Khan, Mustansar Ali Ghazanfar, Muhammad Awais Azam, Amin Karami, Khaled H. Alyoubi

Summary: There are people who want to know how to correctly guess the stock market. But uncertain things like news and microblogs may affect the stock market. It's hard to predict the stock market index's future based on its history. It's hard to predict the stock market, therefore it's crucial to understand how other factors impact it. AI systems can predict stock market outcomes using social media and financial news. Buyers might respond differently when they know what will happen. This research examines how social media and financial news algorithms affected stock market projections over 10 days. Features are selected and garbage tweets are reduced in data sets to improve predictions. We also examine which stock markets are tougher to anticipate and which are more swayed by money and social media news. If we want to find a filter that always does the right thing, we look at how different methods work. Last but not least, the most accurate guesses are made with deep learning, a few models are ensemble.

NSE Stock Market Prediction Using Deep-Learning Models

Publication Year: (2018)

Author: Hiransha Ma, Gopalakrishnan E. Ab, Vijay Krishna Menonab, Soman K.P

Summary: Neural networks have been used by experts in many fields to gather smart data for the past 10 years. Right now, it's very important for business to make predictions and look into stock market info. Arc, GARch, and Neural Network are examples of nonlinear models that you can use to figure out what will happen. This study uses four different kinds of deep learning models to try to guess what a company's stock price will be based on what it has been in the past. These are the Convolutional Neural Network (CNN), the Multilayer Perceptron (MLP) and the Recurrent Neural Networks (RNN). In this case, we use the last prices of the New York Stock Exchange (NYSE) and the National Stock Exchange (NSE) of India. It knew what would happen with NYSE even though it had learnt from NSE data. Some of the same things that made this possible also work in the stock market. The results were compared

using ARIMA, and it was discovered that neural networks work better than the usual linear model (ARIMA).

Stock Market Prediction Using Machine Learning (ML) Algorithms

Publication Year: (2019)

Author: M Umer Ghania, M Awaisa and Muhammad Muzammu

Summary: When it comes to trading, stocks are the most important thing to keep in mind. They may be the most well-known way to get rich. Thanks to better trade technology, the markets are now more open, and almost anyone can own stocks. In the last few decades, most people's interest in the stock market has grown a lot. When you're in a market that moves quickly, like the stock market, you need to know something about what will happen next. Because of the bad market and the need to make money, it's important to be able to guess how much stocks will be worth right. To guess a signal that doesn't follow a straight line, you need advanced machine learning and artificial intelligence (AI). We will use the Machine Learning Algorithm to look at Time Series Forecasting, Three Month Moving Average (3MMA), Linear Regression (LR), and Exponential Smoothing (ES). Since we are going to use MS Excel to make the estimates, graphs and tables are the best ways to show what the results will be.

Stock Market Prediction using Machine Learning: A Systematic Literature Review
Publication Year: (2017)

Author: Siddhartha Vadlamudi

Summary: Several types of machine learning are talked about in this book study. You can guess what will happen in the stock market with these tools. You have to do some hard things, like guess what will happen in the stock market. We talk about how machine learning might be utilised to guess how much a stock will be worth in this study. There are a number of different characteristics that can be used to teach the program how to do this. People also talk about other things that might make the stock worth more or less. People who invest need to be able to guess how much a stock will be worth in the future. Forex traders can make more and more money if they can guess the right things. They have to make a tough choice about

when to buy and when to sell. The tool that uses machine learning techniques can help you make this choice. People are more likely to take chances when they know how much they think the stocks will be worth in the future. Someone smart has tried to guess how much the stock will be worth in the future. This paper talks about this work so the reader doesn't have to read all of them.

Stock Closing Price Prediction using Machine Learning Techniques

Publication Year: (2020)

Author: Mehar Vijha, Deeksha Chandolab, Vinay Anand Tikkiwalb, Arun Kumar

Summary: A straight line doesn't show how the stock market will move, so it's hard to tell what the prices of stocks will do. But as AI and faster computers have become more common, automatic ways of making predictions have become more accurate. Random Forest and Artificial Neural Network were utilised to guess what five companies in different fields would trade for the next day at the end of the day. The Open, High, Low, and Close prices of stocks were used to figure out.

Stock Market Prediction based on Social Sentiments using Machine Learning

Publication Year: (2018)

Author: Tejas Mankar, Tushar Hotchandani, Manish Madhwani, Akshay Chidrawar, Lifna C.S

Summary: Many problems in the real world are being solved with the help of data mining and machine learning and artificial intelligence. It has been shown that these methods work very well, giving the best results with the least amount of money and time spent. People these days are looking at stock purchases as a way to make extra money every year. We can almost double our yearly income from stock gains if we get good advice and plan ahead. Still, investing in stocks is still a risky thing to do for people who have never done it before. The huge salaries of investment experts and the fact that most people don't know much about money keep many people from buying stocks. Many people also don't do it because they are afraid of losing. Because of these facts, we decided to use machine learning to guess how stocks would move.

Using the Twitter API to collect mood-based tweets and stock prices at the end of the day, we want to make a system that can guess how the stock prices of different companies will change.

Stock market prediction using machine learning methods

Publication Year: (2020)

Author: Subhadra Kompella

Summary: Most people know that predicting stock prices is a big deal in business and many other areas of study as well. It is not a good idea to try to guess or predict the price of a share on the stock market. It is too risky. On the stock market, people try to guess the price in a number of ways, such as through basic analysis, time series analysis, statistical analysis, and more. But it has been shown over and over that none of these methods are good ways to guess the price. In this study, we used a Random Forest technique to try to guess what the stock market prices would be. Forecasting stock prices, returns, and stock modelling are all very good uses for Random Forests. We describe how the Random Forest is put together, including its most important aspects and factors that can be changed. We pay attention to a small group of factors that have a big effect on a company's share price. We came up with the polarity score of the new piece with the help of mood analysis, which helped us make correct predictions about what would happen.

Stock Market Prediction Using Machine Learning Techniques

Publication Year: (2019)

Author: Naadun Sirimevan, I.G. U. H. Mamalgaha, Chandira Jayasekara, Y. S. Mayuran1, and Chandimal Jayawardena

Summary: Right now, it's very important to learn how to guess stock market prices. So, experts are trying to figure out new ways to guess what will happen in the stock market. It is possible to guess what stock prices will be in the future by looking at past stock prices, search engine questions, and the social mood from user-generated content on sites like Twitter and Web News. Not enough was known about things like social mood in previous studies. We talk about a good way to use a lot of various sources of data to fill in the blanks and make

correct guesses about the future in this study. This was done with LSTM–RNN models that looked at different sources. To make more accurate stock price forecasts, the Ensembled method with Weighted Average and Differential Evolution technique was used. And very accurate guesses were made for the next one, seven, fifteen, and thirty days. Buyers need to know what the companies are making, and the companies need to know how well they will do in the stock market.

Stock Market Prediction Using Machine Learning Techniques

Publication Year: (2021)

Author: Reshma R, Usha Naidu S, Sathiyavathi V and Sai Ramesh.

Summary: Using machine learning to guess the future in every area was the latest study in the present situation. The stock market is one of them. It requires knowing what will happen so it can buy shares in new businesses or sell them to make money. This needs a good way to guess what will happen in the future by looking at trades that have already happened on the stock market before. If you look at how the stock market used to work, this story says you can guess how much stocks will cost on the market. Based on how the market price of a good changes over a certain amount of time, this method can tell you if the price will go up or down in the future. When we talked about the different things and methods that should be used, the regression algorithm, random forest, and support vector machine came up as some of the most used ones. A good and useful way to learn how to understand a sample of a time collection dataset is to use support vector regression. The information from four years was put together to figure out what the share prices of the company were likely to be. It can make good guesses if the fees for the important factors are set properly. It has been discovered that the guide vector regression model with the RBF kernel works better than other models in general.

Stock Market Data Prediction Using Machine Learning Techniques

Publication Year: (2021)

Author: Edgar P. Torres P, Myriam Hernández-Álvarez, Edgar A. Torres Hernández and Sang Guun Yoo.

Summary: It looks at how past data and machine learning systems can be used to guess what prices will be on the stock market. Random trees and stacked perceptron algorithms were used to play around with Apple Inc. stock market data and try to guess what the closing prices would be. Some guided machine learning techniques were also put to the test to see how useful they might be in the financial world. Data from the Bolsa de Valores de Quito (BVQ) and Bolsa de Valores de Guayaquil (BVG) stock exchanges in Ecuador could also be used to see how well the algorithms work in markets that don't trade as often. This could help people in the market and other interested parties save money on the costs of not being efficient.

Stock Market Prediction using Supervised Machine Learning Techniques: An Overview

Publication Year: (2021)

Author: Zaharaddeen Karami Lawal, Hayati Yassin and Rufai Yusuf Zakari.

Summary: One of the most studied and hard topics is predicting stock prices. Experts from business, economics, mathematics, and computer science, among other fields, are working on it with help from academics and business people. A big reason why it's hard to tell what will happen in the stock market is that stock time series act a lot like random walks. Everyone in the world puts money into the stock market every day. People who buy, handle, and make decisions will be able to make smart choices if they have a good model for predicting stock prices. In this work, we look at research that used supervised machine learning models to guess what would happen in the stock market. The research talked about how guided machine learning can be used to get better predictions of the stock market. A lot of people use Support Vector Machine (SVM) to guess stock prices because it works well and is accurate. Artificial Neural Network (ANN), K-Nearest Neighbour (KNN), Naïve Bayes, Random Forest, Linear Regression, and Support Vector Regression (SVR) were some of the other methods that did a good job of making predictions.

Stock Market Prediction Using Machine Learning Techniques

Publication Year: (2023)

Author: Naadun Sirimevan, I.G. U. H. Mamalgaha, Chandira Jayasekara, Y. S. Mayuran, and Chandimal Jayawardena.

Summary: Right now, it's very important to learn how to guess stock market prices. So, experts are trying to figure out new ways to guess what will happen in the stock market. It is possible to guess what stock prices will be in the future by looking at past stock prices, search engine questions, and the social mood from user-generated content on sites like Twitter and Web News. Not enough was known about things like social mood in previous studies. We talk about a good way to use a lot of various forms of data to fill in the blanks and make correct guesses about the future in this study. This was done with LSTM–RNN models that looked at different sources. To make more accurate stock price forecasts, the Ensembled method with Weighted Average and Differential Evolution technique was used. And very accurate guesses were made for the next one, seven, fifteen, and thirty days. Business can keep track of how well they will do and buyers can understand what they are coming up with.

Prediction of Stock Market performance by using Machine Learning Techniques

Publication Year: (2017)

Author: Prof. Dr. Kamran Raza

Summary: One choice in the stock market can change the course of an investment's life in a big way. It's hard to think about all the things that might affect a choice before making one because the stock market is hard to understand and often kept secret. There were a lot of different things that went into this study as we tried to make a model for predicting the stock market. It was asked to guess how the KSE-100 index would do. Different traits in the forecast model help us figure out whether the market will go up or down. Among these are changes in the prices of goods, interest rates, foreign exchange, and fuel; the public's mood; news stories about the subject; and forecasts made by Auto-Regressive Integrated Moving Average (ARIMA) and Simple Moving Average (SMA) using market past. AI neural networks come in four different types: Single Layer Perceptron (SLP) and Multi-layer Perceptron (MLP); Radial Basis Function (RBF) and Deep Belief Network (DBN). All of these are used to make predictions. There are also the Support Vector Machine (SVM), the Decision

Tree, and the Naïve Bayes method. To find the best model for making predictions, all of these methods had to be put up against each other. 77% of the time, MLP got the market predictions right, which meant they did the best. The link between each factor and market growth was looked at on its own. The most important link between market growth and changes in petrol prices was found. Machine learning techniques can be used to guess what will happen in the market, as shown by the results.

Stock Market Prediction Based on Machine Learning Approaches

Publication Year: (2019)

Author: V. Lalithendra Nadh and G. Syam Prasad

Summary: Based on the knowledge we have, it's hard to make correct predictions about the stock market because it is rarely stable. You can guess what will happen with the stock market in a lot of different ways. When it comes to what people expect, machine learning systems are better than other ways of doing things. Many experts have studied a wide range of topics over the years using various machine learning techniques. In this study, computer tools like genetic algorithms (GAs), support vector machines (SVMs), and artificial neural networks (ANNs) are looked at. They are used to guess what will happen with the stock market. The stock market outlook can help traders and buyers opt for the best stocks and know what to expect when they buy and sell them. This study looks at how genetic algorithms (GAs), support vector machines (SVMs), and artificial neural networks (ANNs) can be used to predict the stock market. What the stock does and how it moves are explained. The ending price for the next day is used as a guide, and there are notes for five days. Getting the right answers is important. ANN was mostly concerned with making the learning algorithm and link weights better.

Stock Market Prediction Using Data Mining Techniques Archana

Publication Year: (2019)

Author: Archana Gupta, Pranay Bhatia, Kashyap Dave and Pritesh Jain.

Summary: People buy and sell stocks, which are also called shares, at a stock market. You can buy and sell stocks on a public stock market or on your own. Stocks are claims on companies. Over the years, we've seen people lose a lot of money, which has ruined their lives. Because of this, we need a prediction system that people can trust and that works the same way throughout a person's life. Predicting stock prices is another important part of financial time series forecasting. Stock buyers, stock traders, and applied experts are all very interested in this field. For buyers to make huge profits, they need to be able to accurately predict stocks. It's hard to make predictions like this though because the market is so unstable. We show a client how Data Mining and Machine Learning can help them decide what to do. To make a model that can guess how much a company's stock will be worth, we need to use Data Mining techniques like KNN for sorting and regression, Machine Learning techniques like Genetic algorithm and SVR, and text from social media that has been analysed for mood. If everything is set up right, the system can help sellers and new users start spending and give them a lot of benefits. If you look at the input choices and the data that has been gathered over time, you can make the system better.

A systematic review of stock market prediction using machine learning and statistical techniques

Publication Year: (2020)

Author: Deepak Kumar, Pradeepta Kumar Sarangi, Rajit Verma

Summary: Patterns for guessing what will happen in the stock market are seen as very important, and they do work better. So, making smart choices about stock prices will help you make a lot of money. It's hard for investors to guess what will happen in the stock market because data is often dull and noisy. People who want to make more money need to be able to guess what will happen in the stock market. Math skills and learning tools are used to make guesses about the stock market. Along with measurement methods, machine learning algorithms, success factors, and top journals, this article takes a close look at 30 research papers that suggest various ways to do things. There are research questions that are used to pick the works. So, these studies were picked because they help us find the machine learning techniques and data sets that can be used to guess what will happen in the stock market. Most

of the time, ANN and NN techniques are used to make good predictions about the stock market. There is a lot of work that goes into the new way to guess what will happen in the stock market, but it is not very good. It is thought in this study that guessing the stock market is a process that uses more than one factor to find the most accurate predictions.

Share Price Prediction using Machine Learning Technique

Publication Year: (2018)

Author: Jeevan B (jeev049@gmail.com), Naresh E (naresh@msrit.edu), Vijaya kumar B P (hod_is@msrit.edu), Prashanth Kambli (prash.kambli@gmail.com)

Summary: There are more scholars and businesspeople interested in the stock market now than ever before. Basically, this study is mostly about how to guess the share price using LSTM and RNN for Long Short Term Memory. Based on NSE data, the goal is to guess the stock price by looking at things like the base value, the current market price, the price-earnings ratio, and other events. We use both real data and suggested data for our business. The suggested data comes from a joint solution based on content. There is a guidance system and a machine learning model based on the LSTM and RNN procedures in this study that talk about how to pick a company. Compare the company data to the expected data in an RNN graph to see how well the model works.

A Survey on Machine Learning for Stock Price Prediction: Algorithms and Techniques

Publication Year: (2020)

Author: Mehtabhorn Obthong1, Nongnuch Tantisantiwong2, Watthanasak Jeamwatthanachai1 c, and Gary Wills

Summary: When buyers trade stocks on the stock market, they need quick and correct information to make good choices. A stock market trades a lot of different stocks, so there are a lot of things that go into making decisions. It's also tough to guess what will happen with stock prices in the future. This is why it's important and hard to guess what stock prices will do. So, experts are looking for the best prediction model that can guess the right things most of the time with the least amount of error. It talks about studies that have been done on machine

learning techniques and methods that are used to get better at predicting stock prices. ML algorithms and methods used in finance, especially to guess stock prices, were looked at and compared in this study. A lot of talk has been made about the different kinds of data, goals, pros and cons, and several hundred different machine learning techniques and methods. Choose some machine learning techniques and methods based on how well they guess stock prices and how few mistakes they make. Stock prices are affected by more than just past prices. Politics, economic growth, financial news, and the mood on social media can all happen. There are a lot of studies that show that emotion (or mood) research has a big effect on future prices. So, a good prediction can be made by using both technical and basic studies together.

Machine Learning Approaches in Stock Price Prediction: A Systematic Review

Publication Year: (2022)

Author: Payal Soni, Yogya Tewari and Prof. Deepa Krishnan

Summary: Predicting stock prices is something that both experts and people who work in the stock market are very interested in. Since artificial intelligence (AI) has grown, different systems have been used to try to guess how the stock market will move. Statistics and machine learning programs have been put together to try to guess what the stock price will be the next day when the market opens or to figure out how the market will work in the long term. Some ways to try to guess what a stock's price will be in the future are neural networks, normal machine learning, and deep learning. This article talks about these all. It goes into detail about the ways people try to guess stock prices and talks about the problems that come with them. It also points out areas where more study needs to be done in the future.

CHAPTER 3

METHODOLOGY

METHODOLOGY

3.1 Introduction

1. Give an outline of the study topic, focusing on how important it is to predict the stock market and what role machine learning plays in this area.
2. Write down your study's goals and research questions.

3.2 Overview of Existing Work

In recent years, it has become more common to use machine learning methods like LSTM to guess what will happen in the stock market. The goal of this method is to get the most accurate predictions about how much stocks will be worth in the future. LSTM can spot trends that don't always happen in the same way by looking at how the stock market has changed over a certain time period. A sample from Yahoo Finance was chosen to show how machine learning can be used to normalize stock market predictions. There are about 9 lakh records in the collection that show information about the stock market. These records include date, sign, open market, close market, low market, high market, and volume. The data was turned into a data frame via the Pandas tool in Python, and only data from one company was looked at. The sklearn tool was used to normalize the data, which was then split into training and testing sets. The experiment set made up 20% of the dataset.

The Regression-based Model and the LSTM are the two design models. A linear function is used with the regression-based model to predict continuous values based on given independent values. LSTM design, on the other hand, can spot changes and trends that don't always happen in the same way. The results show that LSTM is the best model for the suggested method. This means that the system can find connections between the data. Trends and shifts in the stock market don't always follow the same pattern. They can be different for different companies and industries. So, using networks like LSTM to look at trends and cycles can help buyers make more money. These networks use up-to-date information to look at different sets of data and make predictions based on that. Overall, machine learning methods such as LSTM have a lot of promise to help predict the stock market and can be used to get better results.

3.3 Literature Review

1. Read everything that has already been written about predicting the stock market and using machine learning methods in this field.
2. List the good and bad points of earlier research.
3. Point out the holes in the existing research that your study wants to fill.

3.4 Proposed Work

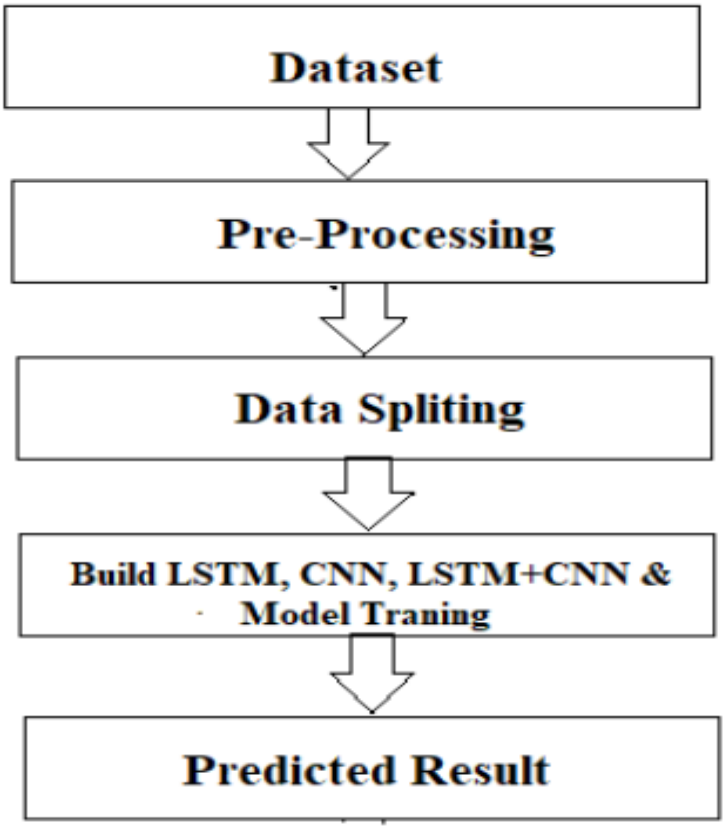


Figure 3.1 Proposed Workflow

The system presented here composes of five modules: -

1. Input as Dataset

2. Preprocessing

3. Data splitting

4. Build & Model train Lstm, CNN and Hybrid approach of LSTM+CNN

5. Output as Predicted Result

Market open, high, low, close, and modified close prices from a very large dataset are fed into the models to train them. To prepare the dataset, methods like normalisation and one-hot encoding are used. Following this, the data is split into two sets, training and testing, with an 80:20 split. Then, this set can be utilised to train a model in three different ways: LSTM, CNN, and a hybrid method that combines LSTM and CNN. Lastly, Root mean square error is used to grade all of these courses.

3.5 Data Collection

- Describe the data sources you utilized for your research, such as historical stock market data, financial indicators, or news sentiment data.
- Explain the criteria used for selecting the data sources and the period of data collection.
- Discuss any preprocessing steps applied to the data, such as data cleaning, normalization, or feature engineering.

3.6 Methodology

1. Specify the machine learning algorithms or models you employed for stock market prediction. This could include regression models, time series analysis, or deep learning architectures like LSTM.
2. Justify the selection of these models based on their suitability for the problem and the available data.
3. Describe the parameters and hyper parameters of the models and how they were determined (e.g., cross-validation, grid search).

3.7 Model Training and Evaluation

1. Explain how you divided the data into training, validation, and test sets.
2. Detail the process of training the models on the training data, including any techniques used for regularization or optimization.
3. Outline the evaluation metrics used to assess the performance of the models, such as accuracy, mean squared error, or precision-recall curves.

3.7.1 Working of LSTM model

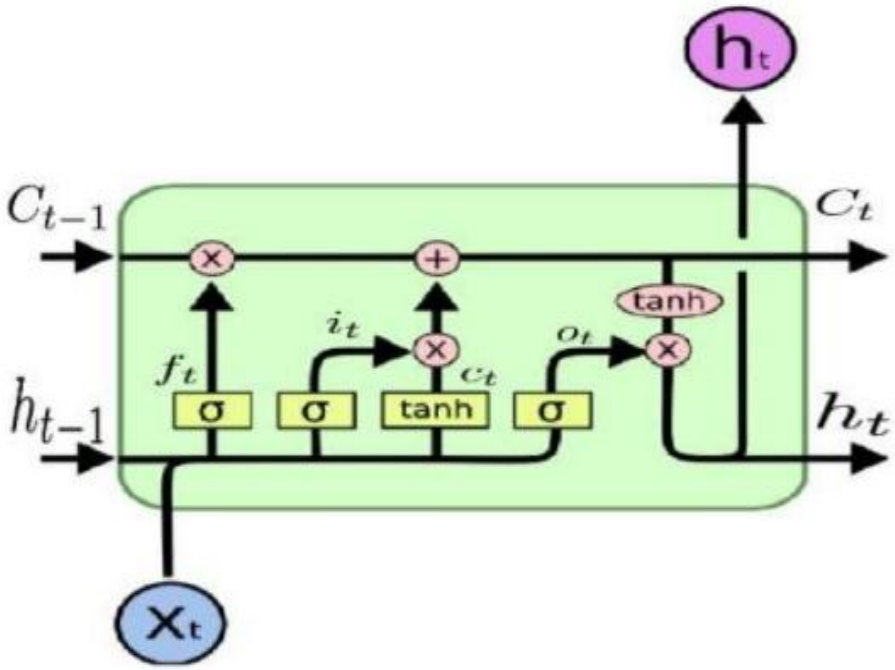


Figure 3.2 LSTM Architecture

A recurrent neural network (RNN) called Long Short Term Memory (LSTM) solves the problem of RNNs being dependent on each other for a long time. LSTM can correctly predict what will happen in the future because it stores important information for a longer time. This is different from regular RNNs, which only use output from the previous step as input in the current step. LSTM can make more accurate predictions based on recent data, while RNNs may have trouble guessing what will happen based on long-term memory. RNNs may not work as well as they used to as the gap length gets longer. However, LSTM can effectively retain knowledge for extended periods, making it an ideal tool for processing, predicting, and classifying time-series data.

3.7.1.1 Structure of LSTM

- LSTM is set up in a chain, with four neural networks and various memory blocks called cells.
- The new part of LSTM is called a memory cell. What knowledge to keep and when to let the brain read, write, and forget are all choices made by the memory cell.

- There are three main gates in a memory cell:
  - I/O gate: a new value is sent into the memory cell.
  - Forget gate: The memory cell still holds a value.
  - The memory cell's output gate number is used to figure out the output.

3.7.1.2 Applications of LSTM

- Language Modelling
- Machine Translation
- Image Captioning
- Handwriting generation
- Question Answering Chatbot

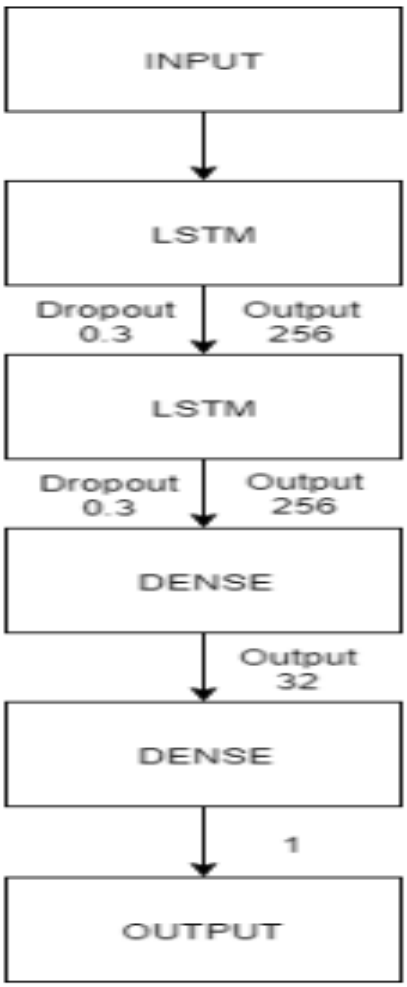


Figure 3.3 LSTM layers

3.7.2. Working of CNN model

It is a type of deep learning models called convolutional neural networks (CNNs) that has changed the way computer vision and picture recognition are done. They are made to instantly learn and pull out useful information from visual data. This makes them very good at jobs like picture segmentation, object recognition, and image classification. In this passage, I will explain the working of CNN models in detail.

CNN Architecture: A normal CNN design has many levels, and each layer does a certain job. A CNN is made up of three main parts: fully linked layers, convolutional layers, and pooling layers.

Convolutional Layers: The most important part of a CNN is the convolutional layer. The raw picture is run through a set of learnable filters, which are also called kernels or feature analysers. Each filter moves across the picture, multiplying each element one at a time and adding up the results to make a feature map. These filters are meant to pick up lines, textures, and patterns, among other neighbourhood traits. The filters' weights are learnt while they are being trained.

- Pooling Layers: These layers are used to make the feature maps that the convolutional layers make smaller in size (width and height). Max pooling and average pooling are two common pooling techniques. By pooling, the models become smaller and more efficient to compute. It also makes the translation less sensitive to small changes in the input.
- Fully Connected Layers: These layers make the final predictions using the information gathered by the layers that came before them. These layers link every neurone in the present layer to every neurone in the layer below it. A softmax activation function is often used to turn the output of the fully linked layers into a chance distribution over the different classes for classification tasks.
- How It Works: A CNN uses both the forward pass (for reasoning) and the backward pass (for training) to do its work. In general, this is how it works:
- Forward Pass: A picture is sent to the CNN during the forward pass. The picture goes through the convolutional layers. At each layer, the data is filtered, which makes feature maps. After that, the feature maps are run through the pooling layers to make them smaller

in space. This process is continued for many layers, which helps them learn more complicated traits over time.

- When the convolutional and pooling layers are done, the feature maps are smoothed into a 1D vector. To connect the features to the fully connected layers, this lowering step has to be taken.
- Fully Connected Layers: After the features have been smoothed, they are sent through one or more fully connected layers. To make the end product, these layers use non-linear activation functions and weighted summations. A softmax activation function is often used to turn results into class odds in classification tasks.
- Loss Computation: Once the predictions are made, a loss function (like cross-entropy) is used to find the difference in likelihoods between the forecasts and the real labels. The training process is meant to cut down on this loss as much as possible.
- Backward Pass (Training): In the backward pass, the CNN's weights are changed to reduce the loss as much as possible. A method for improving things like stochastic gradient descent (SGD) or its variations can be used to do this. Backpropagation is used to find the gradients of the loss with respect to the model parameters. After that, these gradients are used to change the weights in the fully linked and convolutional layers over and over again.
- Training Iterations: The training process has several steps, which are called epochs. Every time an epoch starts, the CNN is fed the whole training dataset, and the weights are changed based on the gradients that were found. This process will keep going until the model converges or a set end point is reached.

By doing the forward pass and the backward pass over and over, a CNN learns to find important details in the images it is given and make correct guesses over time. In the early layers, the convolutional layers learn to find low-level features like lines and patterns. In the later layers, they move on to higher-level features and abstract models. In order to train a CNN, the determined slopes are used to change the weights of the filters and fully linked layers. An optimisation method is used to do this, and its goal is to make the loss function as small as possible. The optimisation method you choose and its settings can affect how fast you train and how well the features you learn are. During training, data enrichment is often used to make the training set more diverse. Random shifts like flips, rotations, translations, and changes in

colour or contrast can be part of this. Adding more data to an existing model stops it from overfitting and makes it better at generalising. After CNN is taught, it can be used to make guesses based on data it has never seen before. The model gives out the expected odds for each class after the forward pass is done on the new data. Most of the time, the end forecast is the class with the best chance. It's important to remember that CNNs can use more than just picture data. In other fields, like natural language processing and speech recognition, they have been used successfully. When this happens, the raw data is usually changed into a shape that the CNN design can understand, like word embeddings for text data. To sum up, CNNs are strong deep learning models for analysing visual data. They use pooling layers to lower the size of the space they work in, convolutional layers to pull out features, and fully connected layers to make predictions. Through repeated forward and backward passes, the model learns to spot patterns in pictures and put them into groups. CNNs have made a big difference in the field of computer vision, and they are still being used in many different areas.

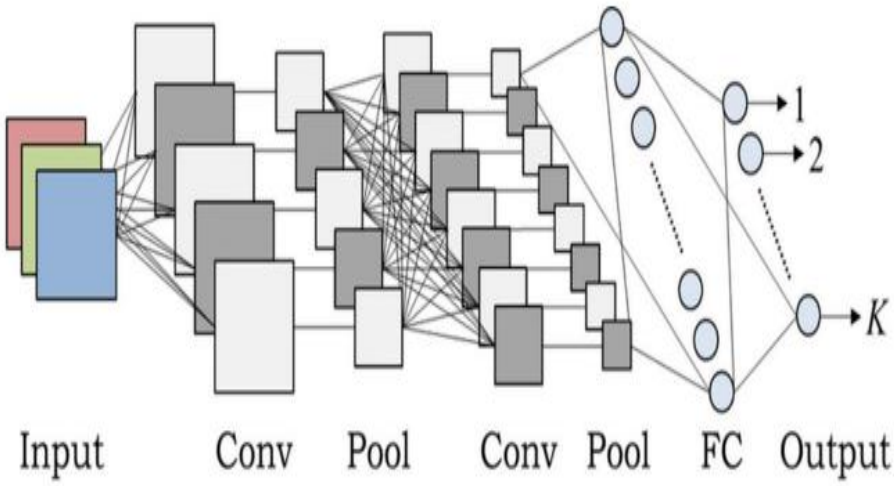


Figure 3.4 CNN Architecture

- **Layer of CNN model:**
  - Convolution
  - MAX Pooling
  - Dropout

- Flatten
- Dense
- Activation

- **Convolution:** In the Convolution extract the featured from the input image. It given the output in matrix form.
- **MAX Pooling:** In the MAX polling it takes the largest element from a rectified feature map.
- **Dropout:** Dropout is randomly selected neurons are ignored during training.
- **Flatten:** Flatten feed output into a fully connected layer. It gives data in list form.
- **Dense:** A Linear operation in which every input is connected to every output by weight. It followed by a nonlinear activation function.
- **Activation:** It used sigmoid function and predict the probability 0 and 1.

Applications of CNN includes:

1. Decoding Facial Recognition

2. Analyzing Document

3.7.3. Hybrid Approach of LSTM + CNN

The mixed method uses both Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs). Each has its own perks when learning straight data. CNNs can use convolutional filters on input layers to get to easy choices. This has made computer vision, natural language processing, and other jobs better. CNNs can also choose traits that make predictions more accurate. On the other hand, LSTMs are good at figuring out how feature sequences depend on each other over time by looking at the results of earlier steps.

A feature map is made by sliding several convolutional filters over a grid. Filters of different sizes make different features. The Max-pooling layer then figures out what the highest number of a related trait is for a certain filter. The Max-pooling layer's output vectors are then sent to LSTM networks as inputs to find out how feature sequences depend on each other over time. The LSTM layer has a set number of units, or cells. The output from the previous dropout layer goes into each cell. This layer's end output has the same number of units as the network itself.

The LSTM outputs are mixed and put together in a grid before being sent to a fully linked layer. The fully linked layer then turns the array into a single result that is between 0 and 1, and the sigmoid function is used to classify it. This approach has been reported to produce improved results, leveraging the strengths of both CNNs and LSTMs.

3.8 Jupyter and Python

In the fields of data science and machine learning, like predicting the stock market, Jupyter and Python are two useful tools that are often used together. Let's look at each one more closely.

Jupyter: Jupyter is a free web app that lets you make and share papers with live code, math, visualizations, and text that explains things. It works with more than 40 computer languages, such as Python, R, Julia, and more. Jupyter is a dynamic computer system that makes it easy to work together, explore, and look at data.

Jupyter Notebooks: Jupyter Notebooks, often referred to simply as Jupyter, are the most common type of document used within the Jupyter ecosystem. Notebooks allow you to create and run code cells interactively, making it easy to experiment with different algorithms, visualize data, and analyze results. A Jupyter Notebook consists of a collection of cells, which can be either code cells or markdown cells (used for text explanations). You can execute code cells individually or all at once, facilitating step-by-step analysis and exploration.

3.9 Advantages of Jupiter

Interactive Environment: You can run and change code in real time in Jupyter's dynamic environment. This tool is especially helpful for exploring data and building machine learning models over and over again.

Collaboration: Sharing and working on Jupyter notebooks with other people is easy. Data scientists and experts who want to work on projects together, share their results, and trade ideas like this one a lot.

Data Visualization: Jupyter integrates well with popular data visualization libraries such as Matplotlib and Seaborn. This allows you to create rich visualizations directly within your notebooks, making it easier to understand and communicate insights from your data.

Documentation: Popular tools for visualizing data, like Matplotlib and Seaborn, work well with Jupyter. This lets you make complex visualizations right in your notes, which makes it easier to understand and share what your data means.

Documentation: Jupyter notebooks are a great way to keep track of how you do your research. You can use both code cells and markdown cells together to give reasons, insights, and conclusions throughout your analysis. This makes it easier for others (and yourself in the future) to understand and copy your work.

3.10 Python

- **Python:** Python is a general-purpose, high-level computer language that is used a lot in many fields, such as data science and machine learning. It has a lot of libraries and systems that work together to make useful tools for working with data, creating machine learning models, and seeing the results. Python is a popular choice among both new and experienced coders because it is easy to understand and use.
- **Machine Learning with Python:** Python has a lot of tools and systems that can help you use machine learning methods correctly. These Python tools are some of the most popular ones for machine learning:
- **NumPy:** The most important tool for doing number computing in Python. It supports big, multidimensional arrays and matrices and comes with a set of mathematical tools that make it easy to work with these arrays.
- **Pandas:** A tool that lets you work with and analyse data. Pandas gives you data structures, like Data Frames, that let you keep and change organised data. It also has strong tools for cleaning and preparing data.
- **Scikit-learn:** A well-known library for machine learning that has many methods for sorting, regression, grouping, and other tasks. Scikit-learn is a complete set of tools for machine learning tasks because it has services for preparing data, evaluating models, and choosing features.
- **TensorFlow and Keras:** TensorFlow is a free tool for numbers and machine learning. Keras is an API for high-level neural networks that runs on top of TensorFlow. A lot of people use these tools to make and train deep learning models.

3.11 working with Jupiter and python for stock market prediction:

You can use the many libraries and tools that are available in Jupyter and Python to prepare and analyse past stock market data, pick out features that are important, and create machine learning models when you use them to predict the stock market. Here's a quick look at how things usually work:

- **Data Collection:** Get past details on the stock market. This usually includes things like company prices, trade rates, and financial factors. You can get data from a number of places, including public financial databases, specialised financial data companies, and financial APIs.
- **Data Preprocessing:** To make sure the quality of the received data and get it ready for analysis, clean and preprocess it. In this step, you may have to deal with missing values, scale or normalize features, deal with outliers, and do other jobs that clean up and change the data. Python data preparation is often done with the Pandas and NumPy tools.
- **Exploratory Data Analysis (EDA):** You can learn more about the data and its patterns, trends, and connections by doing EDA. Using tools like Matplotlib and Seaborn to make visualizations of your data can help you find any trends, connections, or strange things that are happening in it. The use of EDA can help with choosing features and building features.
- **Feature Selection:** Select the most relevant features that are likely to have a significant impact on predicting stock market performance. Feature selection techniques, such as statistical tests, correlation analysis, or domain expertise, can be employed to identify the most informative features for the prediction model.
- **Model Selection and Training:** Pick the right machine learning method for predicting the stock market based on the type of problem and the data that is available. Linear regression, support vector machines (SVM), random forests, and neural networks are some methods that are often used. Scikit-learn has a lot of methods and APIs that are easy to use for training and testing models.
- **Model Evaluation:** Use the right evaluation metrics, like mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), or accuracy measures for classification problems, to check how well the learnt model is doing. Cross-validation methods can help figure out how well the model generalises and find cases of overfitting.

- **Hyperparameter Tuning:** Adjust the model's hyperparameters to make it work better. The hyperparameters determine how the machine learning method works, and by making them better, you can improve the model's ability to predict the future. To get to the best Hyperparameter setup, methods like grid search and randomised search can be used.
- **Model Deployment and Prediction:** Once you have a model that you know will work, use it to make guesses based on new data that you haven't seen before. You can use the training model to guess how a certain stock will do and make guesses based on the traits you chose.
- **Iteration and Improvement:** Stock market prediction is an ongoing process. Continuously monitor and evaluate the performance of your model, and iterate on your approach as new data becomes available or market conditions change. Regularly update the model by retraining it with fresh data to ensure its accuracy and relevance.

Remember that predicting the stock market is hard and complicated, and you should think about more than just machine learning algorithms. For example, you should look at economic data, market trends, and global events. Instead of depending only on their predictions, machine learning models should be utilised to help people make choices.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

DATA ANALYSIS AND INTERPRETATION

4.1 Software Used

Python is the pioneer, it is a generally utilized language, and however it is the essential decision for the vast majority of its clients because of the wide source of different libraries. Python is the best decision for novices in the field. There are a lot of python libraries accessible for machine learning and artificial intelligence. Numpy is its library that assists with illuminating numerous calculations. Python is the best well-known, broadly useful programming language reasonable for an assortment of procedures in machine learning. R is utilized for information investigation and factual calculations. The best language for machine learning relies upon the territory on which it will be applied. Another explanation behind its acceptance is that its language structures are basic and can be effectively realized which makes calculations handily applied. It provides straight entree to its clients for predictive investigation. It is a favoured language for designers who are hoping to outline better inquiries and extend their capacities of existing machine learning frameworks. Artificial intelligence endeavours divergence by customary programmes endeavours. a dissimilarities lie in the modernization stack, the abilities required for an artificial intelligence grounded mission and the necessity of insightful research. To actualize your Artificial intelligence goals, you must consume a programming dialectal that is firm, adjustable and has apparatuses accessible. Python compromises the completeness of this features that is the motive to see heaps of Artificial intelligence projects are using python today. Stating from advancement to organization, python assist engineer by being beneficial and sure about the product they are constructing.

Compensations of brand Python a finest suitable for artificial intelligence & machine learning based activities incorporate effortlessness & uniformity, entree to unbelievable libraries & systems for AI & AI (ML), adaptableness, stage liberty, and an extensive network. These add to the general acceptance of the language. Another purpose behind its acceptance is that its sentence structures are exceptionally straightforward. The IT business has been helped by machine learning. There is now artificial intelligence that helps with tasks that are hard to do by hand. One of the main ways that artificial intelligence can see and explain the affects of huge amounts of data is through machine learning. In the business world, this new idea has become necessary. For creators to carry out their goals in AI and machine learning, they need to use a good computer language. Python is utilized for machine learning by practically all

software engineers for their work. There are numerous advantages of utilizing Python in Machine learning. These advantages are expanding the fame of Python. That is the reason you see Python generally in Machine Learning Software. Python is steady, adaptable and gives different tools to engineers. Every one of these properties of Python settles on it the principal decision for Machine learning. From advancement to usage and support, Python is helping designers to be profitable and sure about the product they are creating. Reasons of selecting python language for machine learning project.

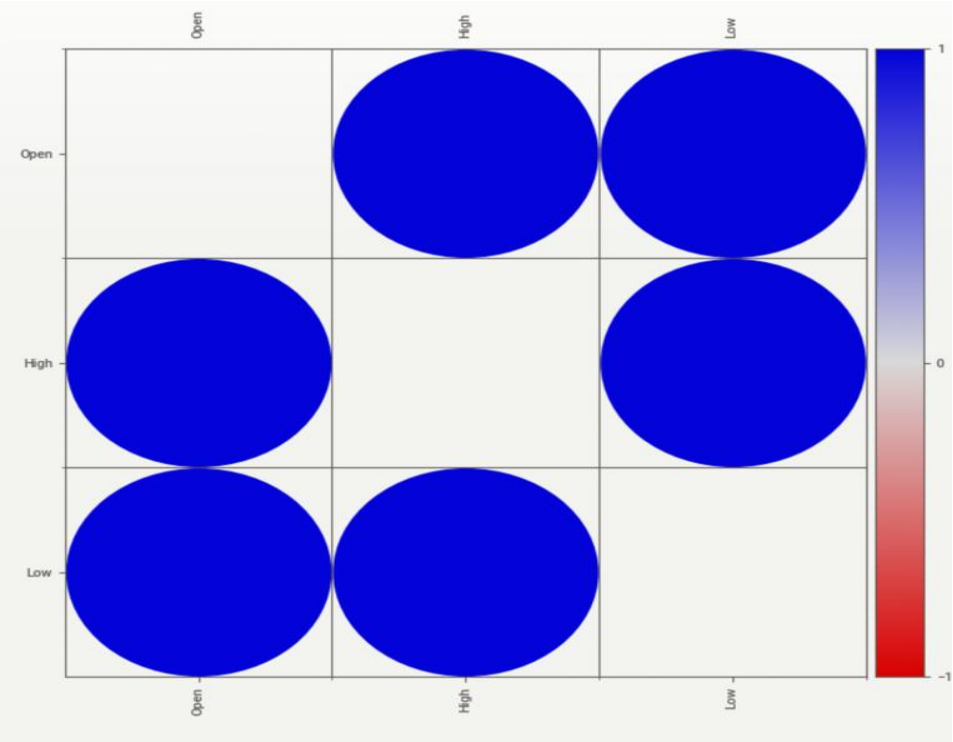
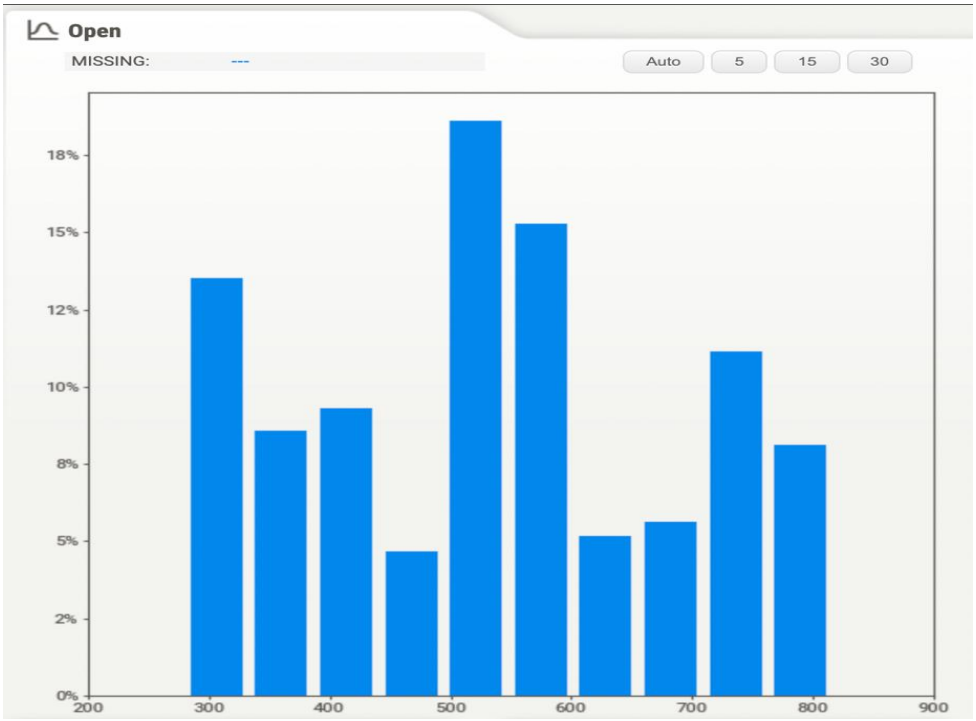
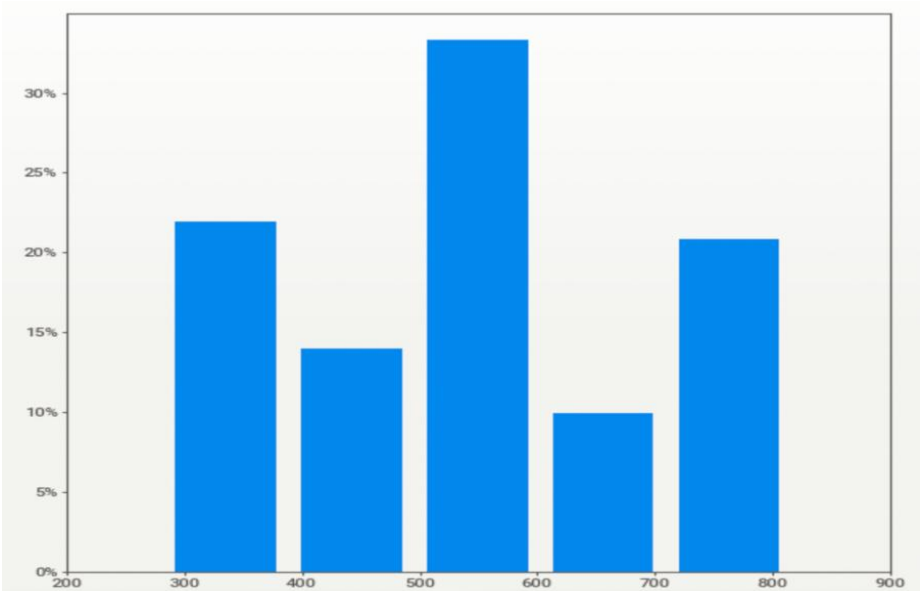


Figure 4.1 Correlation Matrix of Stock Market Features (Open, High, Low)

This image shows the relationship between different features. Blue circle indicates the correlations between features. Pearson correlation is used here.



Graph 4.1 Histogram Showing Distribution of 'Open' Stock Prices



Graph 4.2 Histogram Showing Distribution of Stock Price Values

These are the histograms of different features used. The histograms show the distribution of data throughout the bins.

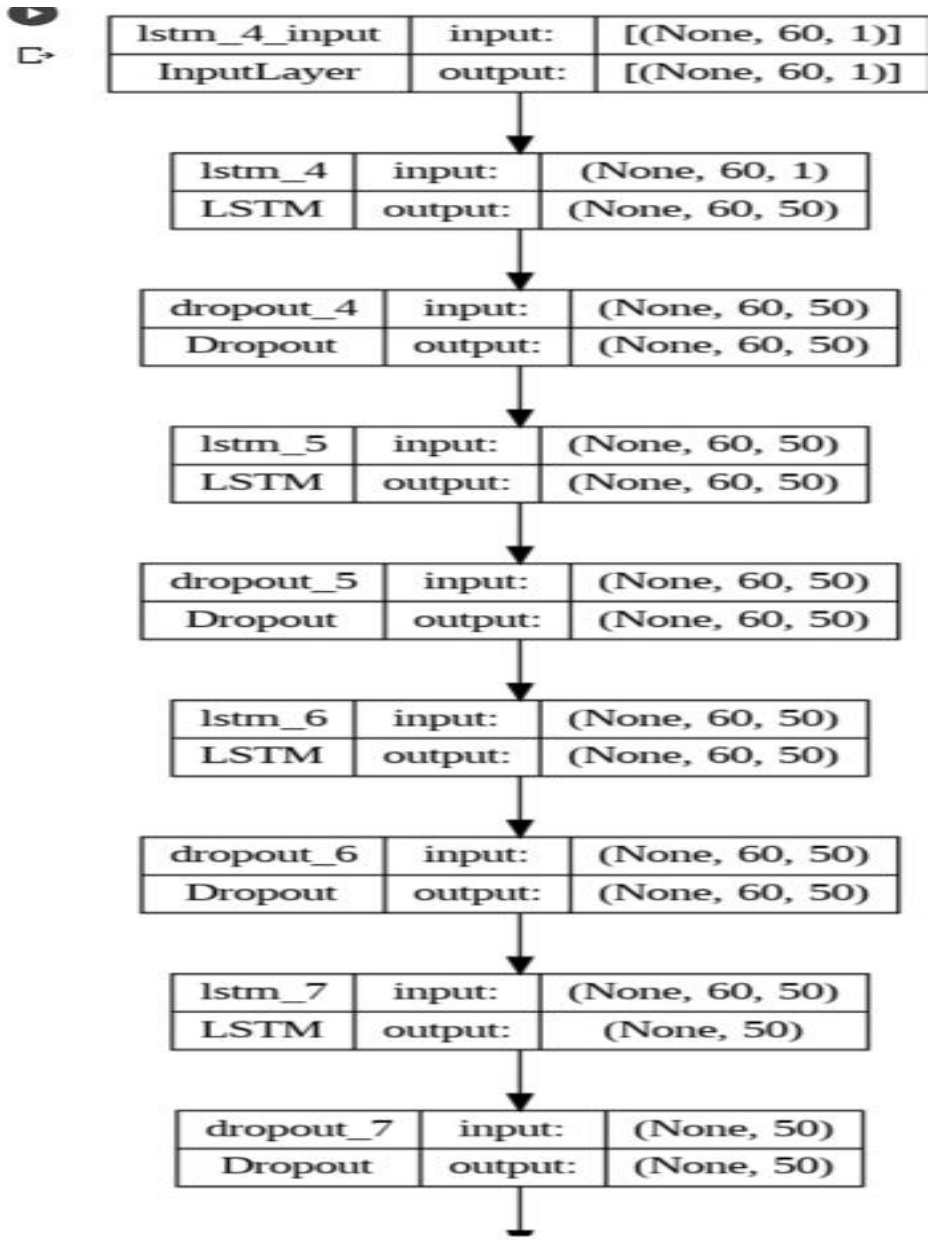


Figure 4.2 Layer-wise Architecture of LSTM Model for Stock Market Prediction

This is the layered architecture of LSTMS used throughout the objectives. Each layer resonates with the KERAS code done in python.

4.2 Coding used

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

dataset_train = pd.read_csv('https://raw.githubusercontent.com/AMoazeni/Machine-Learning-Stock-Market-Prediction/master/Data/Google_Stock_Price_Train.csv')

training_set = dataset_train.iloc[:, 1:2].values

from sklearn.preprocessing import MinMaxScaler

sc = MinMaxScaler(feature_range = (0, 1))
training_set_scaled = sc.fit_transform(training_set)


X_train = []

y_train = []
for i in range(60, 1258):
    X_train.append(training_set_scaled[i-60:i, 0])
    y_train.append(training_set_scaled[i, 0])
X_train, y_train = np.array(X_train), np.array(y_train)


X_train = np.reshape(X_train, (X_train.shape[0], X_train.shape[1], 1))


display(dataset_train)
```

Figure 4.3 Loads, scales data, creates sequences, prepares for stock prediction.

Things done here:

1) Importing libraries.

2) Read the dataset.

3) Basic preprocessing done.

```
# Importing the Keras libraries and packages
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import LSTM
from keras.layers import Dropout

# Initialising the RNN
# Regression is when you predict a continuous value
regressor = Sequential()

# Adding the first LSTM layer and some Dropout regularisation
# 'units' is the number of LSTM Memory Cells (Neurons) for higher dimensionality
# 'return_sequences = True' because we will add more stacked LSTM Layers
# 'input_shape' of x_train
regressor.add(LSTM(units = 50, return_sequences = True, input_shape = (X_train.shape[1], 1)))
# 20% of Neurons will be ignored (10 out of 50 Neurons) to prevent Overfitting
regressor.add(Dropout(0.2))

# Adding a second LSTM layer and some Dropout regularisation
# Not need to specify input_shape for second Layer, it knows that we have 50 Neurons from the previous layer
regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))

# Adding a third LSTM layer and some Dropout regularisation
regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))
```

Figure 4.4 Stacked LSTM Layers with Dropout for Regression Model Initialization

Here we are designing the LSTM architecture for our stock market prediction. Each layer signifies different neural features to be added.

```
# Adding a third LSTM layer and some Dropout regularisation
regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))

# Adding a fourth LSTM layer and some Dropout regularisation
# This is the last LSTM Layer. 'return_sequences = false' by default so we leave it out.
regressor.add(LSTM(units = 50))
regressor.add(Dropout(0.2))

# Adding the output layer
# 'units = 1' because Output layer has one dimension
regressor.add(Dense(units = 1))

# Compiling the RNN
# Keras documentation recommends 'RMSprop' as a good optimizer for RNNs
# Trial and error suggests that 'adam' optimizer is a good choice
# loss = 'mean_squared_error' which is good for Regression vs. 'Binary Cross Entropy' previously used for Classification
regressor.compile(optimizer = 'adam', loss = 'mean_squared_error')

# Fitting the RNN to the Training set
# 'X_train' Independent variables
# 'y_train' Output Truths that we compare X_train to.
regressor.fit(X_train, y_train, epochs = 10, batch_size = 32)
```

Figure 4.5 LSTM model architecture with dropout and Adam optimizer compilation

LSTM layer is LSTM based architecture, please add details of LSTM architectures in the docs. Dropout is used for controlling overfitting.

```
Epoch 1/10
38/38 [==============================] - 15s 136ms/step - loss: 0.0395
Epoch 2/10
38/38 [==============================] - 6s 170ms/step - loss: 0.0064
Epoch 3/10
38/38 [==============================] - 5s 134ms/step - loss: 0.0058
Epoch 4/10
38/38 [==============================] - 6s 167ms/step - loss: 0.0053
Epoch 5/10
38/38 [==============================] - 5s 137ms/step - loss: 0.0062
Epoch 6/10
38/38 [==============================] - 6s 148ms/step - loss: 0.0057
Epoch 7/10
38/38 [==============================] - 6s 160ms/step - loss: 0.0047
Epoch 8/10
38/38 [==============================] - 5s 138ms/step - loss: 0.0051
Epoch 9/10
38/38 [==============================] - 6s 170ms/step - loss: 0.0048
Epoch 10/10
38/38 [==============================] - 5s 135ms/step - loss: 0.0044

<keras.callbacks.History at 0x7f31289c93d0>
```

Figure 4.6 Training Loss Decreasing Over Ten Epochs in Neural Network Model

Training is happening here.

```
# Getting the real stock price of 2017
dataset_test = pd.read_csv('https://raw.githubusercontent.com/AMoazeni/Machine-Learning-Stock-Market-Prediction/master/Data/Google_Stock_Price_Test.csv')
real_stock_price = dataset_test.iloc[:, 1:2].values

# Getting the predicted stock price of 2017
# We need 60 previous inputs for each day of the Test_set in 2017
# Combine 'dataset_train' and 'dataset_test'
# 'axis = 0' for Vertical Concatenation to add rows to the bottom
dataset_total = pd.concat((dataset_train['Open'], dataset_test['Open']), axis = 0)
# Extract Stock Prices for Test time period, plus 60 days previous
inputs = dataset_total[len(dataset_total) - len(dataset_test) - 60:].values
# 'reshape' function to get it into a NumPy format
inputs = inputs.reshape(-1,1)
# Inputs need to be scaled to match the model trained on Scaled Feature
inputs = sc.transform(inputs)
# The following is pasted from above and modified for Testing, romove all 'Ys'
X_test = []

for i in range(60, 80):
    X_test.append(inputs[i-60:i, 0])
```

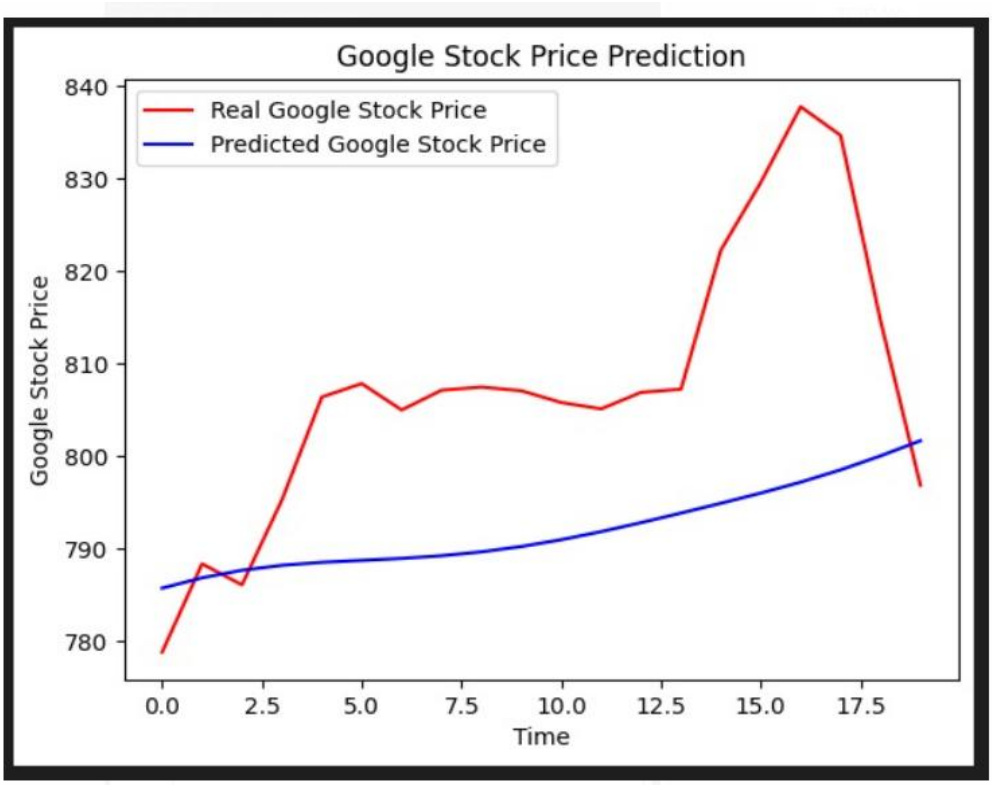
Figure 4.7 Preparing stock price data for 2017 prediction model input.

Testting on new stock prices and comparing the results via graphs.

```
X_test = np.array(X_test)
# We need a 3D input so add another dimension
X_test = np.reshape(X_test, (X_test.shape[0], X_test.shape[1], 1))
# Predict the Stock Price
predicted_stock_price = regressor.predict(X_test)
# We need to inverse the scaling of our prediction to get a Dollar amount
predicted_stock_price = sc.inverse_transform(predicted_stock_price)

# Visualising the results
plt.plot(real_stock_price, color = 'red', label = 'Real Google Stock Price')
plt.plot(predicted_stock_price, color = 'blue', label = 'Predicted Google Stock Price')
plt.title('Google Stock Price Prediction')
plt.xlabel('Time')
plt.ylabel('Google Stock Price')
plt.legend()
plt.show()
```

Figure 4.8 Comparison of Real vs Predicted Google Stock Price Over Time



Graph 4.3 Layer-wise Architecture of LSTM Model for Stock Market Prediction

This graph shows the comparison between predicted stock prices and real stock prices. With time on the X-axis you can see the stock prices changes and fluctuations of it.

CHAPTER 5

SUMMARY AND CONCLUSION

SUMMARY AND CONCLUSION

5.1 Summary

Using artificial intelligence to guess how the stock market will do in the future is called stock market prediction using machine learning. There are many economic, social, and political forces that affect the stock market, which changes all the time. It's hard to tell what the next price of a stock will be, but machine learning systems can help find patterns and trends that can help. Regression, time series analysis, artificial neural networks, and deep learning are some of the machine learning methods that are used to guess what will happen in the stock market. Time series analysis is used to guess what values will be in the future based on data from the past, while regression models are used to find the link between two factors. It is possible to model complicated patterns and connections with artificial neural networks, and it is possible to get features from big datasets with deep learning models. Predicting the stock market with machine learning can be used in many areas of finance, such as trade methods, optimising portfolios, and managing risk. It can also help buyers make smart choices by giving them correct information about how much stocks will cost in the future. It's important to keep in mind, though, that using machine learning to guess the stock market is not a foolproof way. How well the plans work depends on how good the data is, what traits are utilised in the model, and how hard the method is to understand. The stock market is very unstable, and prices can change quickly when something unexpected happens. This makes it hard to make accurate predictions. In conclusion, using machine learning to guess the stock market is an exciting area that could have many uses. But you need to know a lot about both economics and machine learning for it to work, and how well it works rests on the data and the method you choose.

5.2 Conclusion

In conclusion, using machine learning models to guess what will happen in the stock market has been very successful in recent years. Traditional ways of analysis are still useful, but machine learning models' ability to handle and analyse huge amounts of data in real time has made forecasts more accurate and useful. But it's important to remember that stock market predictions are hard to make and can't always be accurate. Machine learning models can give us useful information, but they're not perfect, so we shouldn't rely on them alone to make financial choices. It is also important to think about what using machine learning to guess the

stock market means in terms of ethics. To make sure that everything is fair and clear, the use of algorithms like LSTM design that can change market trends and even change the results of markets needs to be closely watched and controlled. In conclusion, the picture shows how different parts of the image relate to each other and how they change over time. Pearson correlation was used to figure out how strong the link is between the traits. The blue circle shows how much they are related. The different feature histograms showed how the data was spread out, showing how dense and spread out the numbers were within each bin. You can use this knowledge to get a better sense of the trends and connections that lie beneath the collection. The LSTMS design used in this study is also based on layers, which is similar to KERAS code written in Python. LSTMS are a type of recurrent neural network that work well with time series data and other sequential data. This architecture's stacked method makes it possible to build a more complicated model that might be able to show more complex connections between the dataset's traits. As a whole, using machine learning to analyse the stock market could completely change the field and give buyers very useful information. But you should be careful when using these tools and only use them as part of a bigger, more complete financial plan.

5.3 Future Scope of the Study

Predicting the stock market with machine learning is a field of study that is changing quickly and has a lot of room for growth in the future. If machine learning systems keep getting better, it's possible that predictions about the stock market will keep getting better too. Adding more types of material from different sources is one way that this area can grow in the future. At the moment, most models that try to predict the stock market use past stock prices and financial records to do so. Predictions might be more correct, though, if data from news stories, social media, and other sources is used. The creation of more complex programs that can recognise non-linear connections between different factors is another possible area of study. Linear regression is used in a lot of stock market forecast models right now. It works by assuming that there is a straight line between the input variables and the output variables. But there are a lot of non-linear connections in financial data. Making programs that can find these relationships could help make estimates that are more correct. More and more people are also interested in how deep learning methods, like neural networks, can be used to predict the stock market. Some other areas, like speech and picture recognition, have seen progress with these methods.

They might also work for predicting the stock market. Overall, the future of using machine learning to predict the stock market is bright, and there are many areas where study and development could go in the future. As machine learning systems keep getting better and more types of data are added, it's likely that estimates about how the stock market will do will become more accurate.

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ANNEXURE

Date	Open	High	Low	Close*	Adj Close**	Volume
Apr 28, 2023	33,797.43	34,104.56	33,728.40	34,098.16	34,098.16	354,310,000
Apr 27, 2023	33,381.66	33,859.75	33,374.65	33,826.16	33,826.16	343,240,000
Apr 26, 2023	33,596.34	33,645.83	33,235.85	33,301.87	33,301.87	321,170,000
Apr 25, 2023	33,828.34	33,875.49	33,525.39	33,530.83	33,530.83	297,880,000
Apr 24, 2023	33,805.04	33,891.15	33,726.09	33,875.40	33,875.40	252,020,000
Apr 21, 2023	33,793.60	33,858.83	33,688.57	33,808.96	33,808.96	291,080,000
Apr 20, 2023	33,740.60	33,875.39	33,677.74	33,786.62	33,786.62	307,910,000
Apr 19, 2023	33,889.83	33,957.98	33,814.65	33,897.01	33,897.01	251,490,000
Apr 18, 2023	33,965.16	34,018.62	33,791.89	33,976.63	33,976.63	253,570,000
Apr 17, 2023	33,930.46	33,991.23	33,796.91	33,987.18	33,987.18	234,530,000
Apr 14, 2023	33,981.71	34,082.94	33,730.85	33,886.47	33,886.47	276,720,000
Apr 13, 2023	33,668.97	34,054.99	33,605.17	34,029.69	34,029.69	277,230,000
Apr 12, 2023	33,764.21	33,895.43	33,593.32	33,646.50	33,646.50	260,300,000
Apr 11, 2023	33,586.75	33,781.29	33,586.75	33,684.79	33,684.79	241,430,000
Apr 10, 2023	33,425.25	33,590.24	33,343.43	33,586.52	33,586.52	232,920,000
Apr 06, 2023	33,420.96	33,525.96	33,325.37	33,485.29	33,485.29	258,680,000
Apr 05, 2023	33,394.60	33,543.84	33,376.27	33,482.72	33,482.72	286,660,000
Apr 04, 2023	33,594.79	33,634.72	33,275.62	33,402.38	33,402.38	283,360,000
Apr 03, 2023	33,245.78	33,632.90	33,245.78	33,601.15	33,601.15	311,890,000
Mar 31, 2023	32,901.96	33,291.00	32,901.96	33,274.15	33,274.15	353,740,000
Mar 30, 2023	32,807.43	32,905.80	32,682.87	32,859.03	32,859.03	283,380,000
Mar 29, 2023	32,566.54	32,728.04	32,539.94	32,717.60	32,717.60	314,110,000
Mar 28, 2023	32,434.85	32,551.01	32,295.50	32,394.25	32,394.25	248,560,000
Mar 27, 2023	32,276.72	32,564.00	32,276.72	32,432.08	32,432.08	291,560,000
Mar 24, 2023	32,038.22	32,257.24	31,805.18	32,237.53	32,237.53	317,070,000
Mar 23, 2023	32,101.49	32,511.49	31,864.74	32,105.25	32,105.25	335,430,000
Mar 22, 2023	32,570.19	32,761.89	32,020.46	32,030.11	32,030.11	325,990,000
Mar 21, 2023	32,420.71	32,593.68	32,346.12	32,560.60	32,560.60	363,190,000
Mar 20, 2023	31,872.33	32,280.07	31,872.33	32,244.58	32,244.58	357,210,000
Mar 17, 2023	32,217.32	32,217.32	31,728.70	31,861.98	31,861.98	712,100,000
Mar 16, 2023	31,827.65	32,281.61	31,571.46	32,246.55	32,246.55	440,940,000
Mar 15, 2023	31,759.87	31,906.47	31,429.82	31,874.57	31,874.57	420,910,000
Mar 14, 2023	32,055.29	32,306.59	31,805.40	32,155.40	32,155.40	391,760,000
Mar 13, 2023	31,819.93	32,240.35	31,624.87	31,819.14	31,819.14	429,090,000
Mar 10, 2023	32,185.14	32,422.10	31,783.41	31,909.64	31,909.64	397,790,000
Mar 09, 2023	32,876.83	32,990.46	32,190.60	32,254.86	32,254.86	315,860,000
Mar 08, 2023	32,872.08	32,903.44	32,612.70	32,798.40	32,798.40	242,750,000

Mar 07, 2023	33,428.31	33,453.25	32,838.21	32,856.46	32,856.46	299,400,000
Mar 06, 2023	33,425.32	33,572.22	33,383.47	33,431.44	33,431.44	329,190,000
Mar 03, 2023	33,076.33	33,405.82	33,008.41	33,390.97	33,390.97	300,050,000
Mar 02, 2023	32,780.97	33,083.45	32,665.85	33,003.57	33,003.57	317,450,000
Mar 01, 2023	32,656.37	32,746.15	32,500.71	32,661.84	32,661.84	292,810,000
Feb 28, 2023	32,873.47	32,873.47	32,636.43	32,656.70	32,656.70	303,760,000
Feb 27, 2023	32,906.16	33,189.28	32,814.18	32,889.09	32,889.09	243,660,000
Feb 24, 2023	32,999.19	32,999.19	32,643.48	32,816.92	32,816.92	277,120,000
Feb 23, 2023	33,175.39	33,272.40	32,795.67	33,153.91	33,153.91	288,120,000
Feb 22, 2023	33,169.33	33,245.97	32,948.27	33,045.09	33,045.09	321,910,000
Feb 21, 2023	33,699.69	33,699.69	33,115.74	33,129.59	33,129.59	348,320,000
Feb 17, 2023	33,677.01	33,846.64	33,517.73	33,826.69	33,826.69	302,900,000
Feb 16, 2023	33,992.09	34,041.16	33,686.59	33,696.85	33,696.85	317,530,000
Feb 15, 2023	34,008.63	34,130.49	33,833.10	34,128.05	34,128.05	267,530,000
Feb 14, 2023	34,194.09	34,331.47	33,827.45	34,089.27	34,089.27	288,820,000
Feb 13, 2023	33,887.39	34,249.60	33,887.39	34,245.93	34,245.93	302,600,000
Feb 10, 2023	33,671.54	33,897.31	33,591.99	33,869.27	33,869.27	288,150,000
Feb 09, 2023	34,105.61	34,252.57	33,607.13	33,699.88	33,699.88	347,520,000
Feb 08, 2023	34,132.90	34,161.65	33,899.79	33,949.01	33,949.01	324,090,000
Feb 07, 2023	33,769.78	34,240.00	33,634.10	34,156.69	34,156.69	361,620,000
Feb 06, 2023	33,874.44	33,962.84	33,683.58	33,891.02	33,891.02	296,400,000
Feb 03, 2023	33,926.30	34,179.58	33,813.86	33,926.01	33,926.01	425,150,000
Feb 02, 2023	34,129.30	34,145.14	33,814.78	34,053.94	34,053.94	429,910,000
Feb 01, 2023	34,039.60	34,334.70	33,581.42	34,092.96	34,092.96	370,120,000
Jan 31, 2023	33,803.56	34,095.23	33,664.91	34,086.04	34,086.04	360,240,000
Jan 30, 2023	33,909.21	34,055.29	33,695.18	33,717.09	33,717.09	343,920,000
Jan 27, 2023	33,952.93	34,164.33	33,830.85	33,978.08	33,978.08	441,260,000
Jan 26, 2023	33,771.66	33,953.79	33,635.60	33,949.41	33,949.41	327,000,000
Jan 25, 2023	33,538.36	33,773.09	33,273.21	33,743.84	33,743.84	341,720,000
Jan 24, 2023	33,444.72	33,782.92	33,310.56	33,733.96	33,733.96	299,310,000
Jan 23, 2023	33,439.56	33,782.88	33,316.25	33,629.56	33,629.56	355,220,000
Jan 20, 2023	33,073.46	33,381.95	32,948.93	33,375.49	33,375.49	367,330,000
Jan 19, 2023	33,171.35	33,227.49	32,982.05	33,044.56	33,044.56	319,770,000
Jan 18, 2023	33,948.49	34,016.53	33,269.90	33,296.96	33,296.96	351,670,000
Jan 17, 2023	34,222.32	34,269.97	33,860.67	33,910.85	33,910.85	353,050,000
Jan 13, 2023	34,075.31	34,342.32	33,915.49	34,302.61	34,302.61	274,110,000
Jan 12, 2023	34,047.86	34,292.67	33,792.10	34,189.97	34,189.97	308,190,000
Jan 11, 2023	33,754.03	33,974.69	33,711.04	33,973.01	33,973.01	301,650,000
Jan 10, 2023	33,516.43	33,726.54	33,421.80	33,704.10	33,704.10	268,260,000
Jan 09, 2023	33,664.39	33,935.11	33,487.66	33,517.65	33,517.65	327,840,000
Jan 06, 2023	33,055.30	33,710.66	32,997.39	33,630.61	33,630.61	360,740,000

Jan 05, 2023	33,191.72	33,191.72	32,812.33	32,930.08	32,930.08	337,920,000
Jan 04, 2023	33,165.14	33,409.10	33,033.48	33,269.77	33,269.77	379,400,000
Jan 03, 2023	33,148.90	33,387.52	32,850.57	33,136.37	33,136.37	355,930,000
Dec 30, 2022	33,121.61	33,152.55	32,847.82	33,147.25	33,147.25	295,500,000
Dec 29, 2022	33,021.43	33,293.42	33,020.35	33,220.80	33,220.80	243,060,000
Dec 28, 2022	33,264.76	33,379.55	32,869.15	32,875.71	32,875.71	252,260,000
Dec 27, 2022	33,224.23	33,387.72	33,069.58	33,241.56	33,241.56	246,010,000
Dec 23, 2022	32,961.06	33,226.14	32,814.02	33,203.93	33,203.93	221,050,000
Dec 22, 2022	33,233.35	33,233.35	32,573.43	33,027.49	33,027.49	340,640,000
Dec 21, 2022	33,028.09	33,437.84	33,028.09	33,376.48	33,376.48	328,020,000
Dec 20, 2022	32,735.24	32,992.23	32,649.45	32,849.74	32,849.74	303,970,000
Dec 19, 2022	32,921.45	33,038.94	32,581.97	32,757.54	32,757.54	326,430,000
Dec 16, 2022	33,166.46	33,166.46	32,654.59	32,920.46	32,920.46	756,310,000
Dec 15, 2022	33,773.00	33,773.00	33,016.16	33,202.22	33,202.22	394,290,000
Dec 14, 2022	34,086.07	34,395.65	33,704.17	33,966.35	33,966.35	339,090,000
Dec 13, 2022	34,268.44	34,712.28	33,890.88	34,108.64	34,108.64	442,650,000
Dec 12, 2022	33,519.50	34,005.04	33,519.50	34,005.04	34,005.04	313,440,000
Dec 09, 2022	33,746.71	33,844.63	33,462.65	33,476.46	33,476.46	287,630,000
Dec 08, 2022	33,695.97	33,899.17	33,642.46	33,781.48	33,781.48	271,830,000
Dec 07, 2022	33,556.40	33,774.27	33,503.67	33,597.92	33,597.92	359,980,000
Dec 06, 2022	33,936.87	34,006.63	33,418.59	33,596.34	33,596.34	306,220,000
Dec 05, 2022	34,335.73	34,335.73	33,846.81	33,947.10	33,947.10	312,960,000
Dec 02, 2022	34,265.45	34,483.21	34,039.73	34,429.88	34,429.88	290,580,000
Dec 01, 2022	34,533.59	34,595.51	34,129.35	34,395.01	34,395.01	348,720,000
Nov 30, 2022	33,795.43	34,589.77	33,583.77	34,589.77	34,589.77	524,440,000
Nov 29, 2022	33,847.80	33,933.91	33,662.45	33,852.53	33,852.53	279,750,000
Nov 28, 2022	34,275.91	34,303.88	33,799.35	33,849.46	33,849.46	300,330,000
Nov 25, 2022	34,213.04	34,386.51	34,199.57	34,347.03	34,347.03	131,660,000
Nov 23, 2022	34,091.57	34,262.07	34,004.64	34,194.06	34,194.06	236,820,000
Nov 22, 2022	33,810.05	34,115.70	33,810.05	34,098.10	34,098.10	277,070,000
Nov 21, 2022	33,760.30	33,864.59	33,559.18	33,700.28	33,700.28	352,960,000
Nov 18, 2022	33,606.59	33,827.94	33,540.64	33,745.69	33,745.69	312,950,000
Nov 17, 2022	33,329.27	33,616.02	33,239.75	33,546.32	33,546.32	319,210,000
Nov 16, 2022	33,554.93	33,682.83	33,516.99	33,553.83	33,553.83	301,560,000
Nov 15, 2022	33,755.94	33,987.06	33,320.20	33,592.92	33,592.92	390,610,000
Nov 14, 2022	33,662.05	33,964.30	33,533.96	33,536.70	33,536.70	354,340,000
Nov 11, 2022	33,797.75	33,817.96	33,394.43	33,747.86	33,747.86	427,790,000
Nov 10, 2022	33,263.91	33,727.73	33,167.82	33,715.37	33,715.37	480,130,000
Nov 09, 2022	33,004.47	33,065.11	32,478.87	32,513.94	32,513.94	355,200,000
Nov 08, 2022	32,934.56	33,355.39	32,831.46	33,160.83	33,160.83	335,340,000
Nov 07, 2022	32,454.10	32,895.59	32,424.99	32,827.00	32,827.00	336,300,000

Nov 04, 2022	32,265.01	32,611.52	31,938.92	32,403.22	32,403.22	422,370,000
Nov 03, 2022	31,985.05	32,185.71	31,727.05	32,001.25	32,001.25	354,440,000
Nov 02, 2022	32,576.28	33,071.93	32,139.77	32,147.76	32,147.76	398,430,000
Nov 01, 2022	32,862.79	32,975.48	32,485.23	32,653.20	32,653.20	323,210,000
Oct 31, 2022	32,754.27	32,883.86	32,586.93	32,732.95	32,732.95	390,890,000
Oct 28, 2022	32,204.31	32,889.81	32,158.38	32,861.80	32,861.80	495,830,000
Oct 27, 2022	32,062.14	32,388.42	31,993.21	32,033.28	32,033.28	375,720,000
Oct 26, 2022	31,738.44	32,172.42	31,738.44	31,839.11	31,839.11	430,400,000
Oct 25, 2022	31,463.65	31,876.15	31,423.53	31,836.74	31,836.74	336,930,000
Oct 24, 2022	31,187.32	31,603.63	31,161.41	31,499.62	31,499.62	345,040,000
Oct 21, 2022	30,291.18	31,119.27	30,206.28	31,082.56	31,082.56	418,190,000
Oct 20, 2022	30,411.92	30,822.96	30,265.20	30,333.59	30,333.59	333,190,000
Oct 19, 2022	30,439.04	30,662.11	30,226.90	30,423.81	30,423.81	314,030,000
Oct 18, 2022	30,697.52	30,837.90	30,301.83	30,523.80	30,523.80	387,130,000
Oct 17, 2022	30,059.58	30,311.95	29,997.62	30,185.82	30,185.82	369,820,000
Oct 14, 2022	30,192.21	30,428.82	29,614.30	29,634.83	29,634.83	379,090,000
Oct 13, 2022	28,755.83	30,168.54	28,660.94	30,038.72	30,038.72	470,260,000
Oct 12, 2022	29,233.06	29,455.12	29,136.03	29,210.85	29,210.85	304,420,000
Oct 11, 2022	29,085.20	29,608.42	29,074.23	29,239.19	29,239.19	352,920,000
Oct 10, 2022	29,419.09	29,485.82	29,010.41	29,202.88	29,202.88	315,410,000
Oct 07, 2022	29,685.27	29,685.27	29,142.66	29,296.79	29,296.79	367,950,000
Oct 06, 2022	30,206.13	30,329.61	29,859.34	29,926.94	29,926.94	306,840,000
Oct 05, 2022	30,078.36	30,454.46	29,886.44	30,273.87	30,273.87	290,610,000
Oct 04, 2022	29,826.67	30,325.05	29,826.67	30,316.32	30,316.32	375,640,000
Oct 03, 2022	28,855.25	29,647.79	28,855.25	29,490.89	29,490.89	397,830,000
Sep 30, 2022	29,123.03	29,355.78	28,715.85	28,725.51	28,725.51	464,260,000
Sep 29, 2022	29,513.73	29,513.73	28,997.34	29,225.61	29,225.61	388,820,000
Sep 28, 2022	29,198.92	29,811.78	29,114.97	29,683.74	29,683.74	436,040,000
Sep 27, 2022	29,419.88	29,659.12	28,958.22	29,134.99	29,134.99	355,460,000
Sep 26, 2022	29,536.84	29,630.77	29,161.12	29,260.81	29,260.81	369,450,000
Sep 23, 2022	29,955.65	29,955.65	29,250.47	29,590.41	29,590.41	388,200,000
Sep 22, 2022	30,204.52	30,302.28	29,994.53	30,076.68	30,076.68	335,270,000
Sep 21, 2022	30,819.39	31,020.79	30,181.99	30,183.78	30,183.78	354,960,000
Sep 20, 2022	30,888.53	30,888.53	30,465.91	30,706.23	30,706.23	317,970,000
Sep 19, 2022	30,722.86	31,026.89	30,559.37	31,019.68	31,019.68	291,200,000
Sep 16, 2022	30,866.61	30,866.61	30,550.08	30,822.42	30,822.42	627,920,000
Sep 15, 2022	31,124.68	31,277.69	30,882.08	30,961.82	30,961.82	345,500,000
Sep 14, 2022	31,141.02	31,276.82	30,885.02	31,135.09	31,135.09	348,670,000
Sep 13, 2022	32,006.50	32,006.50	31,018.54	31,104.97	31,104.97	428,930,000
Sep 12, 2022	32,159.49	32,504.04	32,159.49	32,381.34	32,381.34	339,250,000
Sep 09, 2022	31,876.22	32,227.74	31,876.22	32,151.71	32,151.71	310,980,000

Sep 08, 2022	31,481.95	31,783.37	31,321.50	31,774.52	31,774.52	344,340,000
Sep 07, 2022	31,145.63	31,644.34	31,095.21	31,581.28	31,581.28	345,080,000
Sep 06, 2022	31,343.79	31,463.89	31,048.46	31,145.30	31,145.30	347,320,000
Sep 02, 2022	31,848.50	32,026.90	31,182.09	31,318.44	31,318.44	306,180,000
Sep 01, 2022	31,454.58	31,677.51	31,219.75	31,656.42	31,656.42	339,360,000
Aug 31, 2022	31,827.15	31,966.04	31,509.59	31,510.43	31,510.43	392,780,000
Aug 30, 2022	32,163.84	32,205.66	31,647.22	31,790.87	31,790.87	317,150,000
Aug 29, 2022	32,188.00	32,325.16	31,972.79	32,098.99	32,098.99	317,820,000
Aug 26, 2022	33,293.43	33,364.70	32,278.22	32,283.40	32,283.40	352,850,000
Aug 25, 2022	33,029.27	33,306.09	32,889.85	33,291.78	33,291.78	308,530,000
Aug 24, 2022	32,899.01	33,095.50	32,828.56	32,969.23	32,969.23	256,900,000
Aug 23, 2022	33,033.52	33,138.26	32,858.89	32,909.59	32,909.59	251,870,000
Aug 22, 2022	33,586.59	33,586.59	33,007.63	33,063.61	33,063.61	308,900,000
Aug 19, 2022	33,904.69	33,904.69	33,627.09	33,706.74	33,706.74	293,990,000
Aug 18, 2022	33,981.86	34,043.83	33,844.46	33,999.04	33,999.04	282,550,000
Aug 17, 2022	34,029.55	34,161.01	33,828.41	33,980.32	33,980.32	281,390,000
Aug 16, 2022	33,924.35	34,281.36	33,852.57	34,152.01	34,152.01	288,670,000
Aug 15, 2022	33,710.70	33,954.71	33,582.96	33,912.44	33,912.44	262,670,000
Aug 12, 2022	33,430.72	33,767.09	33,398.32	33,761.05	33,761.05	299,140,000
Aug 11, 2022	33,451.51	33,651.80	33,289.83	33,336.67	33,336.67	308,030,000
Aug 10, 2022	33,130.63	33,364.41	33,130.63	33,309.51	33,309.51	311,740,000
Aug 09, 2022	32,807.36	32,877.53	32,702.66	32,774.41	32,774.41	281,080,000
Aug 08, 2022	32,877.36	33,109.96	32,769.10	32,832.54	32,832.54	275,350,000
Aug 05, 2022	32,593.90	32,814.61	32,489.62	32,803.47	32,803.47	273,010,000
Aug 04, 2022	32,805.69	32,829.22	32,652.63	32,726.82	32,726.82	315,590,000
Aug 03, 2022	32,514.21	32,912.74	32,514.21	32,812.50	32,812.50	342,160,000
Aug 02, 2022	32,691.29	32,772.93	32,387.12	32,396.17	32,396.17	316,740,000
Aug 01, 2022	32,755.71	32,972.03	32,640.79	32,798.40	32,798.40	321,180,000
Jul 29, 2022	32,515.62	32,910.18	32,493.02	32,845.13	32,845.13	487,360,000
Jul 28, 2022	32,197.62	32,609.54	31,982.61	32,529.63	32,529.63	331,110,000
Jul 27, 2022	31,865.05	32,336.23	31,799.43	32,197.59	32,197.59	365,480,000
Jul 26, 2022	31,950.04	31,950.04	31,705.36	31,761.54	31,761.54	324,930,000
Jul 25, 2022	31,950.93	32,029.03	31,821.67	31,990.04	31,990.04	255,450,000
Jul 22, 2022	32,167.92	32,219.25	31,731.47	31,899.29	31,899.29	319,880,000
Jul 21, 2022	31,826.49	32,040.63	31,534.08	32,036.90	32,036.90	302,320,000
Jul 20, 2022	31,829.99	31,944.45	31,646.95	31,874.84	31,874.84	313,520,000
Jul 19, 2022	31,165.91	31,842.25	31,165.91	31,827.05	31,827.05	378,720,000
Jul 18, 2022	31,475.98	31,644.68	30,982.97	31,072.61	31,072.61	309,960,000
Jul 15, 2022	30,775.37	31,288.26	30,775.37	31,288.26	31,288.26	321,570,000
Jul 14, 2022	30,451.80	30,680.12	30,143.93	30,630.17	30,630.17	325,130,000
Jul 13, 2022	30,743.63	30,979.17	30,514.61	30,772.79	30,772.79	291,830,000

Jul 12, 2022	31,113.31	31,346.10	30,860.51	30,981.33	30,981.33	324,190,000
Jul 11, 2022	31,277.98	31,367.55	31,114.40	31,173.84	31,173.84	245,020,000
Jul 08, 2022	31,348.43	31,511.46	31,212.45	31,338.15	31,338.15	238,120,000
Jul 07, 2022	31,190.65	31,421.84	31,154.64	31,384.55	31,384.55	296,620,000
Jul 06, 2022	30,957.30	31,224.69	30,794.84	31,037.68	31,037.68	289,800,000
Jul 05, 2022	30,903.12	30,971.74	30,355.12	30,967.82	30,967.82	323,080,000
Jul 01, 2022	30,737.77	31,139.35	30,487.79	31,097.26	31,097.26	310,440,000
Jun 30, 2022	30,790.00	30,979.85	30,431.87	30,775.43	30,775.43	394,070,000
Jun 29, 2022	31,067.41	31,152.96	30,894.53	31,029.31	31,029.31	269,390,000
Jun 28, 2022	31,549.05	31,885.09	30,934.33	30,946.99	30,946.99	350,780,000
Jun 27, 2022	31,533.60	31,598.59	31,351.37	31,438.26	31,438.26	309,910,000
Jun 24, 2022	30,846.94	31,517.29	30,846.94	31,500.68	31,500.68	465,480,000
Jun 23, 2022	30,570.33	30,715.63	30,293.40	30,677.36	30,677.36	361,420,000
Jun 22, 2022	30,352.57	30,777.92	30,166.01	30,483.13	30,483.13	343,490,000
Jun 21, 2022	30,074.69	30,653.98	30,074.69	30,530.25	30,530.25	376,900,000
Jun 17, 2022	29,912.70	30,167.52	29,653.29	29,888.78	29,888.78	692,830,000
Jun 16, 2022	30,305.74	30,305.74	29,740.35	29,927.07	29,927.07	442,910,000
Jun 15, 2022	30,570.50	31,011.97	30,185.08	30,668.53	30,668.53	392,670,000
Jun 14, 2022	30,592.34	30,690.80	30,144.23	30,364.83	30,364.83	366,800,000
Jun 13, 2022	31,144.91	31,144.91	30,373.72	30,516.74	30,516.74	472,290,000
Jun 10, 2022	32,053.52	32,053.52	31,387.84	31,392.79	31,392.79	362,300,000
Jun 09, 2022	32,828.62	32,956.73	32,267.78	32,272.79	32,272.79	289,710,000
Jun 08, 2022	33,087.07	33,156.50	32,824.37	32,910.90	32,910.90	270,470,000
Jun 07, 2022	32,783.03	33,207.45	32,641.85	33,180.14	33,180.14	270,750,000
Jun 06, 2022	33,032.04	33,235.37	32,819.50	32,915.78	32,915.78	253,010,000
Jun 03, 2022	32,986.32	33,135.61	32,839.21	32,899.70	32,899.70	298,570,000
Jun 02, 2022	32,809.01	33,248.61	32,509.43	33,248.28	33,248.28	333,210,000
Jun 01, 2022	33,156.31	33,272.34	32,584.76	32,813.23	32,813.23	338,210,000
May 31, 2022	33,160.59	33,240.22	32,752.34	32,990.12	32,990.12	533,560,000
May 27, 2022	32,735.09	33,213.62	32,682.01	33,212.96	33,212.96	328,770,000
May 26, 2022	32,248.17	32,774.14	32,248.17	32,637.19	32,637.19	351,220,000
May 25, 2022	31,816.31	32,254.44	31,754.33	32,120.28	32,120.28	343,460,000
May 24, 2022	31,717.61	32,014.86	31,365.59	31,928.62	31,928.62	382,950,000
May 23, 2022	31,395.89	31,968.42	31,395.89	31,880.24	31,880.24	402,010,000
May 20, 2022	31,426.94	31,515.78	30,635.76	31,261.90	31,261.90	495,510,000
May 19, 2022	31,262.62	31,569.13	31,016.41	31,253.13	31,253.13	517,190,000
May 18, 2022	32,468.67	32,468.67	31,393.95	31,490.07	31,490.07	465,060,000
May 17, 2022	32,427.00	32,689.14	32,308.15	32,654.59	32,654.59	389,310,000
May 16, 2022	32,152.15	32,514.02	31,928.18	32,223.42	32,223.42	330,030,000
May 13, 2022	31,963.86	32,276.05	31,862.79	32,196.66	32,196.66	388,520,000
May 12, 2022	31,699.04	31,914.46	31,228.22	31,730.30	31,730.30	544,240,000

May 11, 2022	32,123.24	32,584.02	31,798.86	31,834.11	31,834.11	458,460,000
May 10, 2022	32,504.09	32,752.17	31,887.89	32,160.74	32,160.74	439,290,000
May 09, 2022	32,685.17	32,685.17	32,121.98	32,245.70	32,245.70	470,770,000
May 06, 2022	32,773.88	33,055.84	32,474.69	32,899.37	32,899.37	416,520,000
May 05, 2022	33,854.17	33,854.17	32,685.10	32,997.97	32,997.97	448,420,000
May 04, 2022	33,171.80	34,117.74	33,021.84	34,061.06	34,061.06	411,060,000
May 03, 2022	33,086.09	33,341.58	32,914.75	33,128.79	33,128.79	364,420,000
May 02, 2022	32,978.49	33,224.95	32,449.87	33,061.50	33,061.50	453,390,000
Apr 29, 2022	33,787.01	33,919.59	32,913.15	32,977.21	32,977.21	501,560,000
Apr 28, 2022	33,425.96	34,054.79	33,248.46	33,916.39	33,916.39	440,380,000
Apr 27, 2022	33,450.92	33,697.18	33,108.89	33,301.93	33,301.93	447,230,000
Apr 26, 2022	33,907.49	33,909.51	33,230.95	33,240.18	33,240.18	400,020,000
Apr 25, 2022	33,731.65	34,106.01	33,323.37	34,049.46	34,049.46	416,900,000
Apr 22, 2022	34,727.38	34,727.38	33,773.39	33,811.40	33,811.40	373,280,000
Apr 21, 2022	35,258.80	35,492.22	34,723.82	34,792.76	34,792.76	366,390,000
Apr 20, 2022	34,962.67	35,315.40	34,962.67	35,160.79	35,160.79	337,620,000
Apr 19, 2022	34,394.62	34,983.11	34,394.62	34,911.20	34,911.20	296,350,000
Apr 18, 2022	34,411.49	34,618.29	34,279.08	34,411.69	34,411.69	264,110,000
Apr 14, 2022	34,628.46	34,889.17	34,437.50	34,451.23	34,451.23	353,680,000
Apr 13, 2022	34,166.64	34,598.36	34,140.64	34,564.59	34,564.59	313,630,000
Apr 12, 2022	34,412.51	34,669.97	34,102.81	34,220.36	34,220.36	333,510,000
Apr 11, 2022	34,630.27	34,701.34	34,272.29	34,308.08	34,308.08	333,290,000
Apr 08, 2022	34,569.24	34,908.46	34,470.19	34,721.12	34,721.12	301,750,000
Apr 07, 2022	34,439.24	34,705.83	34,190.95	34,583.57	34,583.57	319,870,000
Apr 06, 2022	34,520.34	34,609.84	34,277.17	34,496.51	34,496.51	386,250,000
Apr 05, 2022	34,876.33	35,112.21	34,566.04	34,641.18	34,641.18	304,010,000
Apr 04, 2022	34,799.98	34,940.78	34,615.38	34,921.88	34,921.88	316,570,000
Apr 01, 2022	34,740.89	34,847.91	34,538.25	34,818.27	34,818.27	340,400,000
Mar 31, 2022	35,201.52	35,201.52	34,677.99	34,678.35	34,678.35	433,400,000
Mar 30, 2022	35,273.63	35,361.36	35,058.58	35,228.81	35,228.81	317,320,000
Mar 29, 2022	35,114.35	35,372.26	35,030.07	35,294.19	35,294.19	355,050,000
Mar 28, 2022	34,833.03	34,957.93	34,552.23	34,955.89	34,955.89	299,790,000
Mar 25, 2022	34,702.39	34,942.70	34,631.52	34,861.24	34,861.24	285,440,000
Mar 24, 2022	34,406.81	34,713.56	34,355.42	34,707.94	34,707.94	324,870,000
Mar 23, 2022	34,748.84	34,748.84	34,352.96	34,358.50	34,358.50	347,880,000
Mar 22, 2022	34,583.24	34,882.03	34,583.24	34,807.46	34,807.46	368,550,000
Mar 21, 2022	34,669.85	34,808.28	34,341.81	34,552.99	34,552.99	408,490,000
Mar 18, 2022	34,466.72	34,755.20	34,279.83	34,754.93	34,754.93	615,420,000
Mar 17, 2022	33,995.39	34,483.92	33,896.06	34,480.76	34,480.76	339,470,000
Mar 16, 2022	33,653.93	34,075.94	33,391.24	34,063.10	34,063.10	427,400,000
Mar 15, 2022	32,989.27	33,620.84	32,989.27	33,544.34	33,544.34	388,810,000

Mar 14, 2022	33,000.37	33,395.59	32,818.16	32,945.24	32,945.24	394,940,000
Mar 11, 2022	33,279.72	33,515.61	32,911.89	32,944.19	32,944.19	369,480,000
Mar 10, 2022	33,106.77	33,236.59	32,819.76	33,174.07	33,174.07	399,480,000
Mar 09, 2022	32,860.42	33,457.28	32,860.42	33,286.25	33,286.25	418,670,000
Mar 08, 2022	32,885.17	33,402.98	32,578.73	32,632.64	32,632.64	553,750,000
Mar 07, 2022	33,579.75	33,579.75	32,806.15	32,817.38	32,817.38	488,400,000
Mar 04, 2022	33,655.43	33,655.43	33,254.12	33,614.80	33,614.80	408,390,000
Mar 03, 2022	33,972.87	34,179.07	33,641.53	33,794.66	33,794.66	369,740,000
Mar 02, 2022	33,379.51	34,013.67	33,379.51	33,891.35	33,891.35	400,920,000
Mar 01, 2022	33,813.48	33,870.14	33,107.67	33,294.95	33,294.95	438,730,000
Feb 28, 2022	33,870.62	33,963.63	33,469.31	33,892.60	33,892.60	443,460,000
Feb 25, 2022	33,277.22	34,095.74	33,277.22	34,058.75	34,058.75	417,630,000
Feb 24, 2022	32,830.33	33,269.34	32,272.64	33,223.83	33,223.83	590,850,000
Feb 23, 2022	33,680.97	33,832.59	33,084.90	33,131.76	33,131.76	370,600,000
Feb 22, 2022	33,974.09	34,024.74	33,364.29	33,596.61	33,596.61	432,090,000
Feb 18, 2022	34,310.45	34,423.58	33,976.18	34,079.18	34,079.18	414,080,000
Feb 17, 2022	34,858.47	34,858.47	34,246.23	34,312.03	34,312.03	348,720,000
Feb 16, 2022	34,951.83	35,042.09	34,642.64	34,934.27	34,934.27	294,410,000
Feb 15, 2022	34,686.19	35,047.79	34,686.19	34,988.84	34,988.84	305,670,000
Feb 14, 2022	34,694.50	34,744.56	34,304.28	34,566.17	34,566.17	370,320,000
Feb 11, 2022	35,267.89	35,431.15	34,620.52	34,738.06	34,738.06	398,860,000
Feb 10, 2022	35,630.81	35,800.24	35,100.72	35,241.59	35,241.59	411,620,000
Feb 09, 2022	35,614.90	35,824.28	35,614.90	35,768.06	35,768.06	333,610,000
Feb 08, 2022	35,160.68	35,544.89	35,090.42	35,462.78	35,462.78	319,190,000
Feb 07, 2022	35,108.38	35,325.01	34,993.98	35,091.13	35,091.13	328,910,000
Feb 04, 2022	35,095.74	35,333.55	34,799.08	35,089.74	35,089.74	349,880,000
Feb 03, 2022	35,520.08	35,535.94	35,071.06	35,111.16	35,111.16	381,020,000
Feb 02, 2022	35,378.19	35,679.20	35,290.12	35,629.33	35,629.33	359,540,000
Feb 01, 2022	35,151.47	35,441.09	34,977.95	35,405.24	35,405.24	386,080,000
Jan 31, 2022	34,691.17	35,148.14	34,496.10	35,131.86	35,131.86	468,070,000
Jan 28, 2022	34,135.24	34,731.77	33,807.51	34,725.47	34,725.47	568,440,000
Jan 27, 2022	34,261.75	34,773.32	34,007.78	34,160.78	34,160.78	527,780,000
Jan 26, 2022	34,520.82	34,815.67	33,876.48	34,168.09	34,168.09	546,330,000
Jan 25, 2022	34,186.64	34,591.04	33,545.52	34,297.73	34,297.73	506,430,000
Jan 24, 2022	34,070.61	34,420.99	33,150.33	34,364.50	34,364.50	664,010,000
Jan 21, 2022	34,701.69	34,896.67	34,229.55	34,265.37	34,265.37	523,880,000
Jan 20, 2022	35,102.66	35,490.20	34,670.12	34,715.39	34,715.39	369,070,000
Jan 19, 2022	35,412.30	35,547.83	35,015.49	35,028.65	35,028.65	393,080,000
Jan 18, 2022	35,661.76	35,661.76	35,262.02	35,368.47	35,368.47	427,260,000
Jan 14, 2022	35,996.43	35,996.43	35,641.49	35,911.81	35,911.81	389,200,000
Jan 13, 2022	36,312.49	36,513.88	36,044.22	36,113.62	36,113.62	343,640,000

Jan 12, 2022	36,336.16	36,453.49	36,168.15	36,290.32	36,290.32	310,180,000
Jan 11, 2022	36,058.85	36,271.47	35,769.38	36,252.02	36,252.02	356,850,000
Jan 10, 2022	36,175.21	36,175.21	35,639.91	36,068.87	36,068.87	430,680,000
Jan 07, 2022	36,249.59	36,382.84	36,111.53	36,231.66	36,231.66	356,110,000
Jan 06, 2022	36,409.05	36,464.19	36,200.68	36,236.47	36,236.47	385,890,000
Jan 05, 2022	36,722.60	36,952.65	36,400.39	36,407.11	36,407.11	462,040,000
Jan 04, 2022	36,636.00	36,934.84	36,636.00	36,799.65	36,799.65	435,080,000
Jan 03, 2022	36,321.59	36,595.82	36,246.45	36,585.06	36,585.06	347,930,000
Dec 31, 2021	36,385.85	36,484.94	36,303.97	36,338.30	36,338.30	213,200,000
Dec 30, 2021	36,522.48	36,679.44	36,372.13	36,398.08	36,398.08	205,620,000
Dec 29, 2021	36,421.14	36,571.55	36,396.19	36,488.63	36,488.63	213,480,000
Dec 28, 2021	36,302.99	36,527.26	36,302.99	36,398.21	36,398.21	239,090,000
Dec 27, 2021	35,954.48	36,306.61	35,954.48	36,302.38	36,302.38	244,350,000
Dec 23, 2021	35,782.42	36,060.99	35,782.42	35,950.56	35,950.56	247,420,000
Dec 22, 2021	35,491.71	35,764.67	35,433.13	35,753.89	35,753.89	281,930,000
Dec 21, 2021	35,069.50	35,508.96	35,069.50	35,492.70	35,492.70	346,080,000
Dec 20, 2021	35,222.12	35,222.12	34,665.50	34,932.16	34,932.16	386,590,000
Dec 17, 2021	35,800.11	35,800.11	35,284.26	35,365.44	35,365.44	750,480,000
Dec 16, 2021	36,036.28	36,189.83	35,778.21	35,897.64	35,897.64	473,620,000
Dec 15, 2021	35,549.35	35,943.84	35,389.30	35,927.43	35,927.43	424,720,000
Dec 14, 2021	35,605.73	35,779.47	35,441.74	35,544.18	35,544.18	440,900,000
Dec 13, 2021	35,958.93	35,958.93	35,609.53	35,650.95	35,650.95	441,110,000
Dec 10, 2021	35,830.55	35,982.69	35,710.43	35,970.99	35,970.99	361,200,000
Dec 09, 2021	35,722.26	35,864.24	35,577.14	35,754.69	35,754.69	353,020,000
Dec 08, 2021	35,716.85	35,840.52	35,602.65	35,754.75	35,754.75	387,650,000
Dec 07, 2021	35,423.99	35,819.47	35,423.99	35,719.43	35,719.43	474,940,000
Dec 06, 2021	34,633.43	35,356.75	34,633.43	35,227.03	35,227.03	416,720,000
Dec 03, 2021	34,692.78	34,801.31	34,264.57	34,580.08	34,580.08	439,550,000
Dec 02, 2021	34,076.25	34,759.65	34,076.25	34,639.79	34,639.79	466,900,000
Dec 01, 2021	34,678.94	35,004.64	34,006.98	34,022.04	34,022.04	496,000,000
Nov 30, 2021	35,056.99	35,056.99	34,424.44	34,483.72	34,483.72	678,800,000
Nov 29, 2021	35,017.71	35,287.91	34,895.89	35,135.94	35,135.94	416,040,000
Nov 26, 2021	35,366.69	35,366.69	34,749.80	34,899.34	34,899.34	324,580,000
Nov 24, 2021	35,752.31	35,825.47	35,591.03	35,804.38	35,804.38	280,960,000
Nov 23, 2021	35,619.92	35,841.52	35,542.87	35,813.80	35,813.80	392,520,000
Nov 22, 2021	35,631.41	35,929.66	35,615.55	35,619.25	35,619.25	420,810,000
Nov 19, 2021	35,879.09	35,879.09	35,555.37	35,601.98	35,601.98	408,840,000
Nov 18, 2021	35,901.69	35,952.63	35,654.39	35,870.95	35,870.95	435,080,000
Nov 17, 2021	36,159.70	36,159.70	35,909.48	35,931.05	35,931.05	358,310,000
Nov 16, 2021	36,076.18	36,316.61	36,076.18	36,142.22	36,142.22	308,290,000
Nov 15, 2021	36,128.83	36,236.07	36,031.78	36,087.45	36,087.45	280,660,000

Nov 12, 2021	35,963.78	36,147.63	35,934.59	36,100.31	36,100.31	293,180,000
Nov 11, 2021	36,038.78	36,108.17	35,915.27	35,921.23	35,921.23	270,320,000
Nov 10, 2021	36,299.25	36,346.61	36,009.50	36,079.94	36,079.94	278,390,000
Nov 09, 2021	36,404.53	36,416.98	36,173.07	36,319.98	36,319.98	258,020,000
Nov 08, 2021	36,416.46	36,565.73	36,334.42	36,432.22	36,432.22	284,400,000
Nov 05, 2021	36,268.75	36,484.75	36,190.20	36,327.95	36,327.95	344,600,000
Nov 04, 2021	36,107.46	36,168.03	35,986.48	36,124.23	36,124.23	307,640,000
Nov 03, 2021	36,059.09	36,178.51	35,891.73	36,157.58	36,157.58	262,540,000
Nov 02, 2021	35,935.11	36,088.81	35,884.13	36,052.63	36,052.63	304,510,000
Nov 01, 2021	35,833.65	36,009.74	35,797.97	35,913.84	35,913.84	292,880,000
Oct 29, 2021	35,712.28	35,852.53	35,633.20	35,819.56	35,819.56	399,400,000
Oct 28, 2021	35,545.41	35,742.25	35,545.41	35,730.48	35,730.48	337,400,000
Oct 27, 2021	35,835.43	35,835.43	35,490.43	35,490.69	35,490.69	354,180,000
Oct 26, 2021	35,791.05	35,892.92	35,734.73	35,756.88	35,756.88	321,910,000
Oct 25, 2021	35,692.62	35,787.04	35,629.37	35,741.15	35,741.15	267,220,000
Oct 22, 2021	35,607.72	35,765.02	35,533.95	35,677.02	35,677.02	346,290,000
Oct 21, 2021	35,520.32	35,612.36	35,442.53	35,603.08	35,603.08	298,390,000
Oct 20, 2021	35,475.16	35,669.69	35,465.71	35,609.34	35,609.34	265,750,000
Oct 19, 2021	35,325.37	35,462.68	35,290.12	35,457.31	35,457.31	278,510,000
Oct 18, 2021	35,221.02	35,327.57	35,035.94	35,258.61	35,258.61	312,640,000
Oct 15, 2021	35,023.63	35,320.97	35,023.63	35,294.76	35,294.76	327,870,000
Oct 14, 2021	34,443.22	34,923.83	34,443.22	34,912.56	34,912.56	338,540,000
Oct 13, 2021	34,372.71	34,449.83	34,115.10	34,377.81	34,377.81	316,220,000
Oct 12, 2021	34,507.15	34,611.24	34,318.94	34,378.34	34,378.34	319,010,000
Oct 11, 2021	34,723.79	34,951.35	34,486.51	34,496.06	34,496.06	256,820,000
Oct 08, 2021	34,757.57	34,842.62	34,661.95	34,746.25	34,746.25	238,080,000
Oct 07, 2021	34,509.72	34,975.19	34,509.72	34,754.94	34,754.94	273,290,000
Oct 06, 2021	34,198.96	34,432.68	33,854.69	34,416.99	34,416.99	322,170,000
Oct 05, 2021	34,035.25	34,490.95	34,035.25	34,314.67	34,314.67	311,520,000
Oct 04, 2021	34,312.96	34,410.28	33,821.58	34,002.92	34,002.92	396,860,000
Oct 01, 2021	33,930.70	34,490.56	33,785.54	34,326.46	34,326.46	427,580,000
Sep 30, 2021	34,467.98	34,557.37	33,833.32	33,843.92	33,843.92	354,830,000
Sep 29, 2021	34,360.08	34,580.42	34,305.96	34,390.72	34,390.72	290,350,000
Sep 28, 2021	34,747.70	34,798.21	34,254.65	34,299.99	34,299.99	379,020,000
Sep 27, 2021	34,739.27	35,061.12	34,739.27	34,869.37	34,869.37	289,480,000
Sep 24, 2021	34,762.31	34,857.06	34,648.08	34,798.00	34,798.00	254,310,000
Sep 23, 2021	34,296.30	34,879.78	34,296.30	34,764.82	34,764.82	283,890,000
Sep 22, 2021	34,006.87	34,440.42	34,006.87	34,258.32	34,258.32	296,590,000
Sep 21, 2021	34,025.61	34,313.88	33,914.02	33,919.84	33,919.84	306,970,000
Sep 20, 2021	34,459.72	34,459.72	33,613.03	33,970.47	33,970.47	435,780,000
Sep 17, 2021	34,737.86	34,779.86	34,519.63	34,584.88	34,584.88	560,630,000

Sep 16, 2021	34,810.27	34,943.55	34,540.11	34,751.32	34,751.32	278,270,000
Sep 15, 2021	34,580.95	34,880.77	34,522.03	34,814.39	34,814.39	317,490,000
Sep 14, 2021	34,906.90	34,990.36	34,510.30	34,577.57	34,577.57	325,140,000
Sep 13, 2021	34,665.50	34,939.10	34,665.50	34,869.63	34,869.63	334,190,000
Sep 10, 2021	34,949.59	35,104.25	34,599.61	34,607.72	34,607.72	333,230,000
Sep 09, 2021	35,013.08	35,199.89	34,847.30	34,879.38	34,879.38	261,330,000
Sep 08, 2021	35,087.45	35,175.15	34,925.61	35,031.07	35,031.07	261,880,000
Sep 07, 2021	35,373.24	35,373.24	35,060.90	35,100.00	35,100.00	289,830,000
Sep 03, 2021	35,401.73	35,422.71	35,269.40	35,369.09	35,369.09	221,370,000
Sep 02, 2021	35,353.06	35,475.40	35,348.48	35,443.82	35,443.82	263,370,000
Sep 01, 2021	35,387.55	35,407.37	35,287.05	35,312.53	35,312.53	265,520,000
Aug 31, 2021	35,391.38	35,449.79	35,289.86	35,360.73	35,360.73	337,240,000
Aug 30, 2021	35,471.80	35,510.71	35,374.46	35,399.84	35,399.84	245,390,000
Aug 27, 2021	35,231.11	35,479.18	35,231.11	35,455.80	35,455.80	240,990,000
Aug 26, 2021	35,449.68	35,498.45	35,205.10	35,213.12	35,213.12	239,740,000
Aug 25, 2021	35,388.77	35,501.14	35,287.63	35,405.50	35,405.50	237,230,000
Aug 24, 2021	35,382.72	35,445.47	35,356.18	35,366.26	35,366.26	228,710,000
Aug 23, 2021	35,160.97	35,428.91	35,160.97	35,335.71	35,335.71	245,760,000
Aug 20, 2021	34,917.96	35,177.26	34,867.07	35,120.08	35,120.08	269,320,000
Aug 19, 2021	34,874.67	34,997.81	34,690.25	34,894.12	34,894.12	333,550,000
Aug 18, 2021	35,310.20	35,356.80	34,943.41	34,960.69	34,960.69	289,800,000
Aug 17, 2021	35,500.01	35,500.01	35,120.31	35,343.28	35,343.28	309,410,000
Aug 16, 2021	35,490.83	35,631.19	35,231.87	35,625.40	35,625.40	292,590,000
Aug 13, 2021	35,551.56	35,610.57	35,474.78	35,515.38	35,515.38	236,790,000
Aug 12, 2021	35,481.94	35,510.77	35,352.53	35,499.85	35,499.85	231,600,000
Aug 11, 2021	35,300.89	35,501.16	35,300.89	35,484.97	35,484.97	232,800,000
Aug 10, 2021	35,135.17	35,285.16	35,091.91	35,264.67	35,264.67	255,040,000
Aug 09, 2021	35,229.67	35,229.67	35,041.24	35,101.85	35,101.85	207,850,000
Aug 06, 2021	35,077.44	35,246.79	35,077.44	35,208.51	35,208.51	231,570,000
Aug 05, 2021	34,815.61	35,067.54	34,815.61	35,064.25	35,064.25	216,580,000
Aug 04, 2021	35,047.44	35,047.44	34,790.62	34,792.67	34,792.67	245,300,000
Aug 03, 2021	34,864.10	35,120.94	34,714.33	35,116.40	35,116.40	259,820,000
Aug 02, 2021	34,968.56	35,192.11	34,808.20	34,838.16	34,838.16	238,620,000
Jul 30, 2021	35,013.26	35,106.30	34,871.13	34,935.47	34,935.47	276,410,000
Jul 29, 2021	34,985.99	35,171.52	34,985.99	35,084.53	35,084.53	222,680,000
Jul 28, 2021	35,109.95	35,116.37	34,876.84	34,930.93	34,930.93	347,170,000
Jul 27, 2021	35,078.90	35,078.90	34,878.07	35,058.52	35,058.52	326,610,000
Jul 26, 2021	35,055.86	35,150.37	34,950.19	35,144.31	35,144.31	259,790,000
Jul 23, 2021	34,855.11	35,095.33	34,855.11	35,061.55	35,061.55	314,040,000
Jul 22, 2021	34,799.68	34,879.28	34,673.03	34,823.35	34,823.35	291,610,000
Jul 21, 2021	34,556.96	34,820.24	34,556.96	34,798.00	34,798.00	317,090,000

Jul 20, 2021	33,981.79	34,621.79	33,981.79	34,511.99	34,511.99	387,020,000
Jul 19, 2021	34,528.48	34,528.48	33,741.76	33,962.04	33,962.04	440,440,000
Jul 16, 2021	35,007.94	35,090.01	34,647.82	34,687.85	34,687.85	322,980,000
Jul 15, 2021	34,924.81	34,990.41	34,762.87	34,987.02	34,987.02	320,450,000
Jul 14, 2021	34,920.83	35,069.18	34,826.87	34,933.23	34,933.23	344,340,000
Jul 13, 2021	34,959.96	35,018.79	34,876.89	34,888.79	34,888.79	322,430,000
Jul 12, 2021	34,836.75	35,014.90	34,730.15	34,996.18	34,996.18	294,800,000
Jul 09, 2021	34,457.51	34,893.72	34,457.51	34,870.16	34,870.16	301,750,000
Jul 08, 2021	34,569.01	34,569.01	34,145.59	34,421.93	34,421.93	335,690,000
Jul 07, 2021	34,604.17	34,708.78	34,435.59	34,681.79	34,681.79	302,700,000
Jul 06, 2021	34,790.16	34,814.20	34,358.42	34,577.37	34,577.37	348,430,000
Jul 02, 2021	34,642.42	34,821.93	34,613.49	34,786.35	34,786.35	301,860,000
Jul 01, 2021	34,507.32	34,640.28	34,498.85	34,633.53	34,633.53	268,280,000
Jun 30, 2021	34,290.74	34,553.16	34,245.48	34,502.51	34,502.51	295,560,000
Jun 29, 2021	34,338.89	34,469.83	34,266.83	34,292.29	34,292.29	282,360,000
Jun 28, 2021	34,428.10	34,449.65	34,186.13	34,283.27	34,283.27	280,280,000
Jun 25, 2021	34,328.10	34,501.02	34,314.80	34,433.84	34,433.84	432,550,000
Jun 24, 2021	33,933.91	34,233.02	33,933.91	34,196.82	34,196.82	275,870,000
Jun 23, 2021	33,948.54	34,009.40	33,869.83	33,874.24	33,874.24	255,280,000
Jun 22, 2021	33,872.56	34,040.70	33,751.06	33,945.58	33,945.58	284,130,000
Jun 21, 2021	33,312.07	33,908.09	33,312.07	33,876.97	33,876.97	317,260,000
Jun 18, 2021	33,622.70	33,622.70	33,271.93	33,290.08	33,290.08	563,920,000
Jun 17, 2021	34,014.38	34,091.79	33,626.61	33,823.45	33,823.45	354,690,000
Jun 16, 2021	34,308.48	34,333.25	33,917.11	34,033.67	34,033.67	347,150,000
Jun 15, 2021	34,397.10	34,420.38	34,199.16	34,299.33	34,299.33	272,430,000
Jun 14, 2021	34,472.52	34,472.52	34,211.54	34,393.75	34,393.75	295,530,000
Jun 11, 2021	34,499.81	34,618.09	34,328.65	34,479.60	34,479.60	260,840,000
Jun 10, 2021	34,502.51	34,737.79	34,447.25	34,466.24	34,466.24	309,310,000
Jun 09, 2021	34,626.16	34,654.67	34,439.37	34,447.14	34,447.14	245,100,000
Jun 08, 2021	34,645.84	34,665.38	34,452.94	34,599.82	34,599.82	289,440,000
Jun 07, 2021	34,766.20	34,820.91	34,574.51	34,630.24	34,630.24	259,120,000
Jun 04, 2021	34,618.69	34,772.12	34,618.69	34,756.39	34,756.39	270,050,000
Jun 03, 2021	34,550.31	34,667.41	34,334.41	34,577.04	34,577.04	297,380,000
Jun 02, 2021	34,614.62	34,706.65	34,545.96	34,600.38	34,600.38	263,810,000
Jun 01, 2021	34,584.19	34,849.32	34,542.87	34,575.31	34,575.31	287,700,000
May 28, 2021	34,558.50	34,631.11	34,520.09	34,529.45	34,529.45	290,850,000
May 27, 2021	34,432.47	34,608.82	34,389.24	34,464.64	34,464.64	520,630,000
May 26, 2021	34,336.34	34,409.59	34,271.44	34,323.05	34,323.05	258,780,000
May 25, 2021	34,428.66	34,511.35	34,266.03	34,312.46	34,312.46	285,760,000
May 24, 2021	34,253.87	34,472.51	34,253.87	34,393.98	34,393.98	251,010,000
May 21, 2021	34,121.91	34,415.48	34,121.91	34,207.84	34,207.84	306,360,000

May 20, 2021	33,906.30	34,233.40	33,872.41	34,084.15	34,084.15	320,150,000
May 19, 2021	33,945.81	33,945.81	33,473.80	33,896.04	33,896.04	348,480,000
May 18, 2021	34,351.18	34,408.99	34,044.10	34,060.66	34,060.66	289,590,000
May 17, 2021	34,375.29	34,383.84	34,176.65	34,327.79	34,327.79	291,910,000
May 14, 2021	34,050.86	34,454.05	34,050.86	34,382.13	34,382.13	303,230,000
May 13, 2021	33,624.74	34,181.77	33,623.49	34,021.45	34,021.45	350,820,000
May 12, 2021	34,206.40	34,207.87	33,555.22	33,587.66	33,587.66	381,350,000
May 11, 2021	34,572.74	34,572.74	34,075.31	34,269.16	34,269.16	413,180,000
May 10, 2021	34,785.27	35,091.56	34,741.57	34,742.82	34,742.82	361,620,000
May 07, 2021	34,578.27	34,811.39	34,464.31	34,777.76	34,777.76	303,400,000
May 06, 2021	34,245.11	34,561.29	34,185.06	34,548.53	34,548.53	315,430,000
May 05, 2021	34,163.99	34,331.20	34,039.66	34,230.34	34,230.34	253,180,000
May 04, 2021	34,080.20	34,147.78	33,765.68	34,133.03	34,133.03	407,860,000
May 03, 2021	33,904.89	34,221.06	33,904.89	34,113.23	34,113.23	289,770,000
Apr 30, 2021	33,988.75	33,988.75	33,784.96	33,874.85	33,874.85	374,890,000
Apr 29, 2021	33,855.64	34,087.21	33,744.28	34,060.36	34,060.36	408,450,000
Apr 28, 2021	33,860.46	33,946.60	33,805.56	33,820.38	33,820.38	335,820,000
Apr 27, 2021	33,932.13	34,043.98	33,870.46	33,984.93	33,984.93	289,030,000
Apr 26, 2021	34,044.23	34,148.94	33,941.37	33,981.57	33,981.57	279,300,000
Apr 23, 2021	33,804.52	34,157.57	33,713.29	34,043.49	34,043.49	346,380,000
Apr 22, 2021	34,109.88	34,126.57	33,717.19	33,815.90	33,815.90	344,470,000
Apr 21, 2021	33,808.30	34,160.34	33,773.80	34,137.31	34,137.31	290,300,000
Apr 20, 2021	34,034.18	34,034.18	33,687.01	33,821.30	33,821.30	350,960,000
Apr 19, 2021	34,182.38	34,182.38	33,980.56	34,077.63	34,077.63	312,540,000
Apr 16, 2021	34,058.16	34,256.75	34,058.16	34,200.67	34,200.67	343,760,000
Apr 15, 2021	33,788.53	34,068.73	33,788.53	34,035.99	34,035.99	319,680,000
Apr 14, 2021	33,668.95	33,911.25	33,668.95	33,730.89	33,730.89	315,330,000
Apr 13, 2021	33,721.16	33,741.64	33,545.84	33,677.27	33,677.27	324,480,000
Apr 12, 2021	33,775.06	33,786.19	33,665.54	33,745.40	33,745.40	327,370,000
Apr 09, 2021	33,526.19	33,810.87	33,526.19	33,800.60	33,800.60	316,410,000
Apr 08, 2021	33,469.89	33,506.80	33,342.64	33,503.57	33,503.57	304,040,000
Apr 07, 2021	33,444.98	33,521.76	33,347.96	33,446.26	33,446.26	273,440,000
Apr 06, 2021	33,500.02	33,544.79	33,380.43	33,430.24	33,430.24	297,940,000
Apr 05, 2021	33,222.38	33,617.95	33,222.38	33,527.19	33,527.19	344,270,000
Apr 01, 2021	33,054.58	33,167.17	32,985.35	33,153.21	33,153.21	311,880,000
Mar 31, 2021	33,116.04	33,173.77	32,980.57	32,981.55	32,981.55	414,140,000
Mar 30, 2021	33,127.88	33,170.93	32,988.92	33,066.96	33,066.96	307,510,000
Mar 29, 2021	33,087.55	33,259.00	32,905.13	33,171.37	33,171.37	351,610,000
Mar 26, 2021	32,681.07	33,098.83	32,681.07	33,072.88	33,072.88	382,170,000
Mar 25, 2021	32,346.81	32,672.69	32,071.41	32,619.48	32,619.48	411,990,000
Mar 24, 2021	32,470.88	32,787.99	32,418.15	32,420.06	32,420.06	399,390,000

Mar 23, 2021	32,691.50	32,753.77	32,356.28	32,423.15	32,423.15	385,840,000
Mar 22, 2021	32,601.82	32,810.35	32,512.53	32,731.20	32,731.20	383,500,000
Mar 19, 2021	32,858.36	32,858.36	32,505.07	32,627.97	32,627.97	811,890,000
Mar 18, 2021	32,928.16	33,227.78	32,831.25	32,862.30	32,862.30	418,480,000
Mar 17, 2021	32,825.52	33,047.58	32,782.18	33,015.37	33,015.37	391,200,000
Mar 16, 2021	32,966.75	32,966.75	32,778.23	32,825.95	32,825.95	386,370,000
Mar 15, 2021	32,798.84	32,973.40	32,629.75	32,953.46	32,953.46	360,080,000
Mar 12, 2021	32,462.40	32,793.32	32,462.40	32,778.64	32,778.64	346,400,000
Mar 11, 2021	32,354.50	32,661.59	32,345.70	32,485.59	32,485.59	400,020,000
Mar 10, 2021	31,906.96	32,389.50	31,906.96	32,297.02	32,297.02	409,110,000
Mar 09, 2021	31,892.35	32,150.32	31,822.64	31,832.74	31,832.74	458,690,000
Mar 08, 2021	31,512.15	32,148.04	31,512.15	31,802.44	31,802.44	496,980,000
Mar 05, 2021	31,029.18	31,580.33	30,766.81	31,496.30	31,496.30	501,810,000
Mar 04, 2021	31,289.01	31,461.97	30,547.53	30,924.14	30,924.14	537,850,000
Mar 03, 2021	31,352.96	31,563.31	31,260.14	31,270.09	31,270.09	409,660,000
Mar 02, 2021	31,535.37	31,623.27	31,377.16	31,391.52	31,391.52	337,270,000
Mar 01, 2021	31,065.90	31,668.34	31,065.90	31,535.51	31,535.51	385,670,000
Feb 26, 2021	31,401.29	31,450.89	30,911.37	30,932.37	30,932.37	532,250,000
Feb 25, 2021	31,955.94	31,984.77	31,293.32	31,402.01	31,402.01	454,110,000
Feb 24, 2021	31,499.75	32,009.64	31,421.29	31,961.86	31,961.86	410,150,000
Feb 23, 2021	31,501.89	31,653.38	31,158.76	31,537.35	31,537.35	460,410,000
Feb 22, 2021	31,381.12	31,653.48	31,286.57	31,521.69	31,521.69	387,210,000
Feb 19, 2021	31,504.13	31,647.53	31,469.34	31,494.32	31,494.32	353,070,000
Feb 18, 2021	31,558.60	31,558.60	31,285.32	31,493.34	31,493.34	325,990,000
Feb 17, 2021	31,485.61	31,643.70	31,338.76	31,613.02	31,613.02	343,510,000
Feb 16, 2021	31,472.08	31,608.63	31,443.48	31,522.75	31,522.75	329,210,000
Feb 12, 2021	31,420.80	31,475.31	31,347.80	31,458.40	31,458.40	263,090,000
Feb 11, 2021	31,466.49	31,543.82	31,244.36	31,430.70	31,430.70	280,320,000
Feb 10, 2021	31,428.02	31,511.44	31,221.32	31,437.80	31,437.80	326,450,000
Feb 09, 2021	31,359.88	31,439.47	31,245.86	31,375.83	31,375.83	308,210,000
Feb 08, 2021	31,191.20	31,386.10	31,191.20	31,385.76	31,385.76	322,250,000
Feb 05, 2021	31,093.81	31,252.18	31,083.22	31,148.24	31,148.24	293,550,000
Feb 04, 2021	30,737.78	31,065.96	30,737.78	31,055.86	31,055.86	338,330,000
Feb 03, 2021	30,689.65	30,793.41	30,521.31	30,723.60	30,723.60	305,590,000
Feb 02, 2021	30,276.88	30,840.54	30,276.88	30,687.48	30,687.48	333,160,000
Feb 01, 2021	30,054.73	30,335.91	30,014.97	30,211.91	30,211.91	347,060,000
Jan 29, 2021	30,553.91	30,553.91	29,856.30	29,982.62	29,982.62	536,320,000
Jan 28, 2021	30,377.19	30,951.41	30,377.19	30,603.36	30,603.36	465,140,000
Jan 27, 2021	30,893.78	30,893.78	30,206.91	30,303.17	30,303.17	566,470,000
Jan 26, 2021	30,968.55	31,121.42	30,921.71	30,937.04	30,937.04	373,150,000
Jan 25, 2021	30,989.85	30,989.85	30,564.06	30,960.00	30,960.00	472,890,000

Jan 22, 2021	31,141.56	31,141.56	30,908.47	30,996.98	30,996.98	436,040,000
Jan 21, 2021	31,198.01	31,272.22	31,120.92	31,176.01	31,176.01	418,780,000
Jan 20, 2021	31,017.54	31,235.98	30,997.79	31,188.38	31,188.38	384,980,000
Jan 19, 2021	30,887.42	31,086.62	30,865.03	30,930.52	30,930.52	386,400,000
Jan 15, 2021	30,926.77	30,941.98	30,612.67	30,814.26	30,814.26	433,000,000
Jan 14, 2021	31,085.67	31,223.78	30,982.24	30,991.52	30,991.52	427,810,000
Jan 13, 2021	31,084.88	31,153.37	30,992.05	31,060.47	31,060.47	413,250,000
Jan 12, 2021	31,015.01	31,114.56	30,888.76	31,068.69	31,068.69	362,620,000
Jan 11, 2021	31,015.37	31,096.98	30,832.06	31,008.69	31,008.69	356,540,000
Jan 08, 2021	31,069.58	31,140.67	30,793.27	31,097.97	31,097.97	381,150,000
Jan 07, 2021	30,901.18	31,193.40	30,897.86	31,041.13	31,041.13	427,810,000
Jan 06, 2021	30,362.78	31,022.65	30,313.07	30,829.40	30,829.40	500,430,000
Jan 05, 2021	30,204.25	30,504.89	30,141.78	30,391.60	30,391.60	350,910,000
Jan 04, 2021	30,627.47	30,674.28	29,881.82	30,223.89	30,223.89	475,080,000
Dec 31, 2020	30,417.64	30,637.47	30,344.50	30,606.48	30,606.48	284,470,000
Dec 30, 2020	30,415.09	30,525.35	30,393.04	30,409.56	30,409.56	291,890,000
Dec 29, 2020	30,492.07	30,588.79	30,274.24	30,335.67	30,335.67	357,610,000
Dec 28, 2020	30,283.23	30,525.56	30,283.23	30,403.97	30,403.97	302,490,000
Dec 24, 2020	30,155.92	30,209.67	30,099.30	30,199.87	30,199.87	145,570,000
Dec 23, 2020	30,046.73	30,292.53	30,046.73	30,129.83	30,129.83	274,050,000
Dec 22, 2020	30,200.20	30,200.20	29,992.89	30,015.51	30,015.51	388,810,000
Dec 21, 2020	30,159.00	30,304.14	29,755.53	30,216.45	30,216.45	437,480,000
Dec 18, 2020	30,314.26	30,343.59	30,029.44	30,179.05	30,179.05	782,070,000
Dec 17, 2020	30,216.00	30,323.78	30,216.00	30,303.37	30,303.37	345,530,000
Dec 16, 2020	30,191.38	30,236.03	30,080.11	30,154.54	30,154.54	344,590,000
Dec 15, 2020	29,919.09	30,243.26	29,894.95	30,199.31	30,199.31	417,770,000
Dec 14, 2020	30,123.91	30,325.79	29,849.15	29,861.55	29,861.55	371,980,000
Dec 11, 2020	29,988.21	30,071.13	29,820.84	30,046.37	30,046.37	393,870,000
Dec 10, 2020	30,032.55	30,063.87	29,876.82	29,999.26	29,999.26	325,550,000
Dec 09, 2020	30,229.81	30,319.70	29,951.85	30,068.81	30,068.81	380,520,000
Dec 08, 2020	29,997.95	30,246.22	29,972.07	30,173.88	30,173.88	311,190,000
Dec 07, 2020	30,233.03	30,233.03	29,967.22	30,069.79	30,069.79	365,810,000
Dec 04, 2020	29,989.56	30,218.26	29,989.56	30,218.26	30,218.26	356,590,000
Dec 03, 2020	29,920.83	30,110.88	29,877.27	29,969.52	29,969.52	405,680,000
Dec 02, 2020	29,695.09	29,902.51	29,599.29	29,883.79	29,883.79	385,280,000
Dec 01, 2020	29,797.50	30,083.31	29,797.50	29,823.92	29,823.92	429,510,000
Nov 30, 2020	29,854.51	29,854.51	29,463.64	29,638.64	29,638.64	551,350,000
Nov 27, 2020	29,911.33	30,015.13	29,819.98	29,910.37	29,910.37	177,040,000
Nov 25, 2020	30,058.87	30,058.87	29,806.25	29,872.47	29,872.47	323,030,000
Nov 24, 2020	29,746.11	30,116.51	29,746.11	30,046.24	30,046.24	432,760,000
Nov 23, 2020	29,332.82	29,667.75	29,332.82	29,591.27	29,591.27	387,200,000

Nov 20, 2020	29,437.57	29,469.89	29,231.20	29,263.48	29,263.48	305,350,000
Nov 19, 2020	29,370.92	29,524.90	29,228.38	29,483.23	29,483.23	331,330,000
Nov 18, 2020	29,875.40	29,930.85	29,432.84	29,438.42	29,438.42	382,850,000
Nov 17, 2020	29,800.15	29,872.42	29,520.29	29,783.35	29,783.35	372,380,000
Nov 16, 2020	29,672.36	29,964.29	29,672.36	29,950.44	29,950.44	418,290,000
Nov 13, 2020	29,203.90	29,559.25	29,203.90	29,479.81	29,479.81	348,810,000
Nov 12, 2020	29,231.91	29,311.83	28,902.13	29,080.17	29,080.17	369,320,000
Nov 11, 2020	29,524.35	29,593.11	29,281.22	29,397.63	29,397.63	376,700,000
Nov 10, 2020	29,254.17	29,478.86	29,126.99	29,420.92	29,420.92	526,980,000
Nov 09, 2020	29,467.90	29,933.83	29,130.66	29,157.97	29,157.97	673,310,000
Nov 06, 2020	28,399.13	28,431.96	28,189.50	28,323.40	28,323.40	326,710,000
Nov 05, 2020	28,083.37	28,495.05	28,083.37	28,390.18	28,390.18	390,140,000
Nov 04, 2020	27,512.83	28,301.50	27,512.83	27,847.66	27,847.66	447,860,000
Nov 03, 2020	27,138.69	27,640.21	27,138.69	27,480.03	27,480.03	350,980,000
Nov 02, 2020	26,691.28	27,043.48	26,691.28	26,925.05	26,925.05	379,990,000
Oct 30, 2020	26,572.27	26,639.18	26,143.77	26,501.60	26,501.60	504,820,000
Oct 29, 2020	26,480.98	26,891.89	26,290.78	26,659.11	26,659.11	419,770,000
Oct 28, 2020	27,102.14	27,102.14	26,497.39	26,519.95	26,519.95	496,150,000
Oct 27, 2020	27,651.18	27,707.69	27,457.96	27,463.19	27,463.19	366,400,000
Oct 26, 2020	28,185.82	28,185.82	27,370.16	27,685.38	27,685.38	404,150,000
Oct 23, 2020	28,409.65	28,436.52	28,149.82	28,335.57	28,335.57	347,300,000
Oct 22, 2020	28,197.42	28,421.31	28,040.18	28,363.66	28,363.66	330,320,000
Oct 21, 2020	28,270.66	28,450.64	28,196.39	28,210.82	28,210.82	299,420,000
Oct 20, 2020	28,245.85	28,575.03	28,243.04	28,308.79	28,308.79	350,220,000
Oct 19, 2020	28,633.55	28,711.93	28,139.40	28,195.42	28,195.42	358,700,000
Oct 16, 2020	28,570.72	28,842.90	28,570.72	28,606.31	28,606.31	356,140,000
Oct 15, 2020	28,323.40	28,535.85	28,181.54	28,494.20	28,494.20	328,940,000
Oct 14, 2020	28,731.30	28,792.56	28,461.73	28,514.00	28,514.00	370,800,000
Oct 13, 2020	28,764.95	28,808.84	28,604.35	28,679.81	28,679.81	526,110,000
Oct 12, 2020	28,671.12	28,957.90	28,659.67	28,837.52	28,837.52	493,680,000
Oct 09, 2020	28,533.61	28,676.29	28,440.63	28,586.90	28,586.90	324,050,000
Oct 08, 2020	28,348.86	28,459.13	28,265.56	28,425.51	28,425.51	314,750,000
Oct 07, 2020	27,971.36	28,369.66	27,971.36	28,303.46	28,303.46	328,750,000
Oct 06, 2020	28,214.24	28,354.48	27,728.03	27,772.76	27,772.76	435,030,000
Oct 05, 2020	27,825.42	28,162.64	27,825.42	28,148.64	28,148.64	318,210,000
Oct 02, 2020	27,536.39	27,861.43	27,382.94	27,682.81	27,682.81	392,770,000
Oct 01, 2020	27,940.63	28,041.46	27,669.26	27,816.90	27,816.90	373,450,000
Sep 30, 2020	27,514.64	28,026.33	27,511.06	27,781.70	27,781.70	452,550,000
Sep 29, 2020	27,560.24	27,605.60	27,338.09	27,452.66	27,452.66	303,820,000
Sep 28, 2020	27,362.14	27,722.60	27,362.14	27,584.06	27,584.06	401,860,000
Sep 25, 2020	26,694.51	27,239.29	26,635.38	27,173.96	27,173.96	388,900,000

Sep 24, 2020	26,716.09	27,094.85	26,537.01	26,815.44	26,815.44	447,810,000
Sep 23, 2020	27,413.60	27,464.46	26,716.12	26,763.13	26,763.13	459,470,000
Sep 22, 2020	27,170.34	27,333.09	26,989.93	27,288.18	27,288.18	420,720,000
Sep 21, 2020	27,484.71	27,484.71	26,715.15	27,147.70	27,147.70	523,660,000
Sep 18, 2020	27,864.87	27,946.69	27,487.97	27,657.42	27,657.42	698,710,000
Sep 17, 2020	27,834.18	28,056.79	27,647.93	27,901.98	27,901.98	437,230,000
Sep 16, 2020	28,031.69	28,364.77	28,021.67	28,032.38	28,032.38	416,650,000
Sep 15, 2020	28,139.76	28,231.06	27,931.45	27,995.60	27,995.60	414,520,000
Sep 14, 2020	27,718.74	28,086.06	27,718.74	27,993.33	27,993.33	380,750,000
Sep 11, 2020	27,613.86	27,828.82	27,448.15	27,665.64	27,665.64	435,330,000
Sep 10, 2020	28,022.51	28,174.78	27,447.08	27,534.58	27,534.58	442,850,000
Sep 09, 2020	27,711.71	28,206.21	27,704.54	27,940.47	27,940.47	460,590,000
Sep 08, 2020	27,925.23	27,925.23	27,464.90	27,500.89	27,500.89	580,860,000
Sep 04, 2020	28,341.05	28,539.75	27,664.68	28,133.31	28,133.31	694,640,000
Sep 03, 2020	29,090.70	29,199.35	28,074.76	28,292.73	28,292.73	650,080,000
Sep 02, 2020	28,736.79	29,162.88	28,713.53	29,100.50	29,100.50	539,510,000
Sep 01, 2020	28,439.61	28,659.26	28,290.72	28,645.66	28,645.66	423,410,000
Aug 31, 2020	28,643.66	28,643.66	28,363.55	28,430.05	28,430.05	517,320,000
Aug 28, 2020	28,601.29	28,733.35	28,487.98	28,653.87	28,653.87	369,740,000
Aug 27, 2020	28,384.07	28,634.22	28,363.93	28,492.27	28,492.27	404,340,000
Aug 26, 2020	28,257.88	28,353.80	28,153.91	28,331.92	28,331.92	327,070,000
Aug 25, 2020	28,347.42	28,400.74	28,094.57	28,248.44	28,248.44	338,420,000
Aug 24, 2020	28,077.58	28,314.94	28,041.75	28,308.46	28,308.46	383,880,000
Aug 21, 2020	27,758.13	27,959.48	27,686.78	27,930.33	27,930.33	372,720,000
Aug 20, 2020	27,622.68	27,781.46	27,526.25	27,739.73	27,739.73	304,550,000
Aug 19, 2020	27,811.26	27,920.42	27,647.67	27,692.88	27,692.88	311,460,000
Aug 18, 2020	27,853.48	27,891.12	27,668.79	27,778.07	27,778.07	276,170,000
Aug 17, 2020	27,970.05	27,999.81	27,816.40	27,844.91	27,844.91	286,860,000
Aug 14, 2020	27,828.93	27,977.81	27,759.39	27,931.02	27,931.02	275,540,000
Aug 13, 2020	27,922.51	27,986.10	27,789.78	27,896.72	27,896.72	360,160,000
Aug 12, 2020	27,860.24	28,043.89	27,843.32	27,976.84	27,976.84	340,210,000
Aug 11, 2020	27,961.64	28,154.88	27,624.51	27,686.91	27,686.91	428,220,000
Aug 10, 2020	27,488.21	27,803.86	27,488.21	27,791.44	27,791.44	360,340,000
Aug 07, 2020	27,321.68	27,456.24	27,223.55	27,433.48	27,433.48	321,170,000
Aug 06, 2020	27,170.82	27,394.10	27,145.25	27,386.98	27,386.98	322,460,000
Aug 05, 2020	26,924.78	27,221.67	26,924.78	27,201.52	27,201.52	366,000,000
Aug 04, 2020	26,664.61	26,832.72	26,597.82	26,828.47	26,828.47	346,350,000
Aug 03, 2020	26,542.32	26,707.26	26,534.38	26,664.40	26,664.40	439,380,000
Jul 31, 2020	26,409.33	26,440.02	26,013.59	26,428.32	26,428.32	488,050,000
Jul 30, 2020	26,367.42	26,374.93	25,992.28	26,313.65	26,313.65	351,540,000
Jul 29, 2020	26,388.44	26,602.45	26,375.39	26,539.57	26,539.57	347,580,000

Jul 28, 2020	26,529.45	26,556.84	26,361.71	26,379.28	26,379.28	361,090,000
Jul 27, 2020	26,447.67	26,625.46	26,426.92	26,584.77	26,584.77	400,930,000
Jul 24, 2020	26,533.41	26,625.70	26,402.86	26,469.89	26,469.89	494,170,000
Jul 23, 2020	26,955.97	26,973.85	26,560.04	26,652.33	26,652.33	419,800,000
Jul 22, 2020	26,824.56	27,035.24	26,794.19	27,005.84	27,005.84	404,340,000
Jul 21, 2020	26,833.14	27,025.38	26,766.22	26,840.40	26,840.40	364,930,000
Jul 20, 2020	26,660.29	26,765.02	26,504.20	26,680.87	26,680.87	309,390,000
Jul 17, 2020	26,774.62	26,808.43	26,619.88	26,671.95	26,671.95	296,460,000
Jul 16, 2020	26,746.57	26,879.16	26,590.01	26,734.71	26,734.71	287,330,000
Jul 15, 2020	27,009.81	27,071.33	26,692.48	26,870.10	26,870.10	384,900,000
Jul 14, 2020	26,044.17	26,690.52	25,994.98	26,642.59	26,642.59	401,640,000
Jul 13, 2020	26,225.07	26,639.09	26,044.23	26,085.80	26,085.80	425,320,000
Jul 10, 2020	25,690.35	26,101.32	25,637.50	26,075.30	26,075.30	338,170,000
Jul 09, 2020	26,094.92	26,103.28	25,523.51	25,706.09	25,706.09	387,610,000
Jul 08, 2020	25,950.06	26,109.49	25,816.25	26,067.28	26,067.28	337,300,000
Jul 07, 2020	26,172.01	26,174.93	25,866.58	25,890.18	25,890.18	346,820,000
Jul 06, 2020	25,996.08	26,297.53	25,996.08	26,287.03	26,287.03	340,690,000
Jul 02, 2020	25,936.45	26,204.41	25,778.12	25,827.36	25,827.36	349,130,000
Jul 01, 2020	25,879.38	26,019.31	25,713.61	25,734.97	25,734.97	373,180,000
Jun 30, 2020	25,512.43	25,905.38	25,475.14	25,812.88	25,812.88	410,780,000
Jun 29, 2020	25,152.45	25,601.15	25,096.16	25,595.80	25,595.80	378,070,000
Jun 26, 2020	25,641.69	25,641.69	24,971.03	25,015.55	25,015.55	640,860,000
Jun 25, 2020	25,365.22	25,769.61	25,209.79	25,745.60	25,745.60	384,560,000
Jun 24, 2020	25,992.96	25,992.96	25,296.73	25,445.94	25,445.94	450,120,000
Jun 23, 2020	26,159.39	26,314.97	26,105.97	26,156.10	26,156.10	389,980,000
Jun 22, 2020	25,865.08	26,059.81	25,667.68	26,024.96	26,024.96	351,780,000
Jun 19, 2020	26,213.10	26,451.44	25,759.66	25,871.46	25,871.46	669,390,000
Jun 18, 2020	26,016.45	26,154.20	25,848.53	26,080.10	26,080.10	328,390,000
Jun 17, 2020	26,330.52	26,400.07	26,068.41	26,119.61	26,119.61	328,830,000
Jun 16, 2020	26,326.68	26,611.03	25,811.70	26,289.98	26,289.98	489,500,000
Jun 15, 2020	25,270.39	25,891.58	24,843.18	25,763.16	25,763.16	461,230,000
Jun 12, 2020	25,659.42	25,965.55	25,078.41	25,605.54	25,605.54	514,440,000
Jun 11, 2020	26,282.51	26,294.08	25,082.72	25,128.17	25,128.17	647,780,000
Jun 10, 2020	27,251.89	27,355.22	26,938.05	26,989.99	26,989.99	448,930,000
Jun 09, 2020	27,447.37	27,447.37	27,151.06	27,272.30	27,272.30	381,430,000
Jun 08, 2020	27,232.93	27,580.21	27,232.48	27,572.44	27,572.44	430,200,000
Jun 05, 2020	26,836.80	27,338.30	26,836.80	27,110.98	27,110.98	542,310,000
Jun 04, 2020	26,226.49	26,384.10	26,082.31	26,281.82	26,281.82	372,190,000
Jun 03, 2020	25,906.88	26,337.75	25,906.88	26,269.89	26,269.89	398,790,000
Jun 02, 2020	25,582.52	25,743.13	25,523.74	25,742.65	25,742.65	352,610,000
Jun 01, 2020	25,342.99	25,508.83	25,220.66	25,475.02	25,475.02	338,310,000

May 29, 2020	25,324.15	25,482.80	25,031.67	25,383.11	25,383.11	527,620,000
May 28, 2020	25,697.36	25,758.79	25,358.73	25,400.64	25,400.64	377,800,000
May 27, 2020	25,298.63	25,551.56	25,009.87	25,548.27	25,548.27	411,230,000
May 26, 2020	24,781.84	25,176.42	24,781.84	24,995.11	24,995.11	424,380,000
May 22, 2020	24,461.98	24,481.64	24,294.07	24,465.16	24,465.16	255,660,000
May 21, 2020	24,564.27	24,718.46	24,370.88	24,474.12	24,474.12	346,770,000
May 20, 2020	24,455.94	24,649.48	24,455.94	24,575.90	24,575.90	366,510,000
May 19, 2020	24,577.48	24,599.50	24,202.96	24,206.86	24,206.86	374,100,000
May 18, 2020	24,059.98	24,708.54	24,059.98	24,597.37	24,597.37	484,190,000
May 15, 2020	23,454.83	23,730.08	23,354.15	23,685.42	23,685.42	491,510,000
May 14, 2020	23,049.06	23,630.86	22,789.62	23,625.34	23,625.34	472,700,000
May 13, 2020	23,702.16	23,708.90	23,067.64	23,247.97	23,247.97	469,950,000
May 12, 2020	24,292.84	24,382.09	23,761.58	23,764.78	23,764.78	359,480,000
May 11, 2020	24,256.45	24,366.21	24,070.22	24,221.99	24,221.99	352,400,000
May 08, 2020	24,107.82	24,349.90	24,107.05	24,331.32	24,331.32	335,760,000
May 07, 2020	23,837.21	24,094.62	23,834.39	23,875.89	23,875.89	369,890,000
May 06, 2020	23,978.88	24,054.59	23,661.14	23,664.64	23,664.64	377,450,000
May 05, 2020	23,958.88	24,169.72	23,868.91	23,883.09	23,883.09	369,710,000
May 04, 2020	23,581.55	23,769.56	23,361.16	23,749.76	23,749.76	355,360,000
May 01, 2020	24,120.78	24,120.78	23,645.30	23,723.69	23,723.69	418,160,000
Apr 30, 2020	24,585.57	24,585.57	24,186.90	24,345.72	24,345.72	487,840,000
Apr 29, 2020	24,490.37	24,764.77	24,453.99	24,633.86	24,633.86	455,290,000
Apr 28, 2020	24,357.17	24,512.24	24,031.20	24,101.55	24,101.55	400,250,000
Apr 27, 2020	23,866.15	24,207.65	23,840.61	24,133.78	24,133.78	389,390,000
Apr 24, 2020	23,628.24	23,826.00	23,417.68	23,775.27	23,775.27	376,020,000
Apr 23, 2020	23,543.09	23,885.36	23,483.35	23,515.26	23,515.26	389,290,000
Apr 22, 2020	23,437.34	23,613.10	23,339.60	23,475.82	23,475.82	352,880,000
Apr 21, 2020	23,365.25	23,365.25	22,941.88	23,018.88	23,018.88	485,140,000
Apr 20, 2020	24,095.10	24,108.69	23,627.19	23,650.44	23,650.44	423,410,000
Apr 17, 2020	23,817.15	24,264.21	23,817.15	24,242.49	24,242.49	525,950,000
Apr 16, 2020	23,543.66	23,598.08	23,211.38	23,537.68	23,537.68	471,050,000
Apr 15, 2020	23,600.72	23,649.72	23,233.32	23,504.35	23,504.35	437,630,000
Apr 14, 2020	23,690.57	24,040.58	23,683.44	23,949.76	23,949.76	485,910,000
Apr 13, 2020	23,698.93	23,698.93	23,095.35	23,390.77	23,390.77	394,320,000
Apr 09, 2020	23,690.66	24,008.99	23,504.09	23,719.37	23,719.37	566,400,000
Apr 08, 2020	22,893.47	23,513.40	22,682.99	23,433.57	23,433.57	472,740,000
Apr 07, 2020	23,537.44	23,617.24	22,634.45	22,653.86	22,653.86	594,660,000
Apr 06, 2020	21,693.63	22,783.45	21,693.63	22,679.99	22,679.99	610,760,000
Apr 03, 2020	21,285.93	21,447.81	20,863.09	21,052.53	21,052.53	450,010,000
Apr 02, 2020	20,819.46	21,477.77	20,735.02	21,413.44	21,413.44	529,540,000
Apr 01, 2020	21,227.38	21,487.24	20,784.43	20,943.51	20,943.51	506,680,000

Mar 31, 2020	22,208.42	22,480.37	21,852.08	21,917.16	21,917.16	571,210,000
Mar 30, 2020	21,678.22	22,378.09	21,522.08	22,327.48	22,327.48	545,540,000
Mar 27, 2020	21,898.47	22,327.57	21,469.27	21,636.78	21,636.78	588,830,000
Mar 26, 2020	21,468.38	22,595.06	21,427.10	22,552.17	22,552.17	705,180,000
Mar 25, 2020	21,050.34	22,019.93	20,538.34	21,200.55	21,200.55	796,320,000
Mar 24, 2020	19,722.19	20,737.70	19,649.25	20,704.91	20,704.91	799,340,000
Mar 23, 2020	19,028.36	19,121.01	18,213.65	18,591.93	18,591.93	787,970,000
Mar 20, 2020	20,253.15	20,531.26	19,094.27	19,173.98	19,173.98	872,290,000
Mar 19, 2020	19,830.01	20,442.63	19,177.13	20,087.19	20,087.19	780,300,000
Mar 18, 2020	20,188.69	20,489.33	18,917.46	19,898.92	19,898.92	871,360,000
Mar 17, 2020	20,487.05	21,379.35	19,882.26	21,237.38	21,237.38	793,060,000
Mar 16, 2020	20,917.53	21,768.28	20,116.46	20,188.52	20,188.52	770,130,000
Mar 13, 2020	21,973.82	23,189.76	21,285.37	23,185.62	23,185.62	843,080,000
Mar 12, 2020	22,184.71	22,837.95	21,154.46	21,200.62	21,200.62	908,260,000
Mar 11, 2020	24,604.63	24,604.63	23,328.32	23,553.22	23,553.22	663,960,000
Mar 10, 2020	24,453.00	25,020.99	23,690.34	25,018.16	25,018.16	654,860,000
Mar 09, 2020	24,992.36	24,992.36	23,706.07	23,851.02	23,851.02	750,430,000
Mar 06, 2020	25,457.21	25,994.38	25,226.62	25,864.78	25,864.78	599,780,000
Mar 05, 2020	26,671.92	26,671.92	25,943.33	26,121.28	26,121.28	477,370,000
Mar 04, 2020	26,383.68	27,102.34	26,286.31	27,090.86	27,090.86	457,590,000
Mar 03, 2020	26,762.47	27,084.59	25,706.28	25,917.41	25,917.41	647,080,000
Mar 02, 2020	25,590.51	26,706.17	25,391.96	26,703.32	26,703.32	637,200,000
Feb 28, 2020	25,270.83	25,494.24	24,681.01	25,409.36	25,409.36	915,990,000
Feb 27, 2020	26,526.00	26,775.31	25,752.82	25,766.64	25,766.64	664,980,000
Feb 26, 2020	27,159.46	27,542.78	26,890.97	26,957.59	26,957.59	472,450,000
Feb 25, 2020	28,037.65	28,149.20	26,997.62	27,081.36	27,081.36	513,270,000
Feb 24, 2020	28,402.93	28,402.93	27,912.44	27,960.80	27,960.80	452,580,000
Feb 21, 2020	29,146.53	29,146.53	28,892.70	28,992.41	28,992.41	311,210,000
Feb 20, 2020	29,296.25	29,368.45	28,959.65	29,219.98	29,219.98	287,780,000
Feb 19, 2020	29,312.70	29,409.09	29,274.38	29,348.03	29,348.03	240,640,000
Feb 18, 2020	29,282.78	29,330.16	29,116.81	29,232.19	29,232.19	256,600,000
Feb 14, 2020	29,440.47	29,463.04	29,283.18	29,398.08	29,398.08	231,000,000
Feb 13, 2020	29,436.03	29,535.40	29,345.93	29,423.31	29,423.31	291,150,000
Feb 12, 2020	29,406.75	29,568.57	29,406.75	29,551.42	29,551.42	309,530,000
Feb 11, 2020	29,390.71	29,415.39	29,210.47	29,276.34	29,276.34	279,540,000
Feb 10, 2020	28,995.66	29,278.07	28,995.66	29,276.82	29,276.82	250,510,000
Feb 07, 2020	29,286.92	29,286.92	29,056.98	29,102.51	29,102.51	252,860,000
Feb 06, 2020	29,388.58	29,408.05	29,246.93	29,379.77	29,379.77	263,700,000
Feb 05, 2020	29,048.73	29,308.89	29,000.85	29,290.85	29,290.85	357,540,000
Feb 04, 2020	28,696.74	28,904.88	28,696.74	28,807.63	28,807.63	332,750,000
Feb 03, 2020	28,319.65	28,630.39	28,319.65	28,399.81	28,399.81	307,910,000

Jan 31, 2020	28,813.04	28,813.04	28,169.53	28,256.03	28,256.03	403,890,000
Jan 30, 2020	28,640.16	28,879.71	28,489.76	28,859.44	28,859.44	326,850,000
Jan 29, 2020	28,820.53	28,944.24	28,728.19	28,734.45	28,734.45	302,290,000
Jan 28, 2020	28,594.28	28,823.23	28,575.75	28,722.85	28,722.85	330,140,000
Jan 27, 2020	28,542.49	28,671.79	28,440.47	28,535.80	28,535.80	337,270,000
Jan 24, 2020	29,230.39	29,288.79	28,843.31	28,989.73	28,989.73	380,010,000
Jan 23, 2020	29,111.02	29,190.47	28,966.98	29,160.09	29,160.09	307,060,000
Jan 22, 2020	29,263.63	29,320.20	29,172.26	29,186.27	29,186.27	283,440,000
Jan 21, 2020	29,269.05	29,341.21	29,146.47	29,196.04	29,196.04	320,640,000
Jan 17, 2020	29,313.31	29,373.62	29,289.91	29,348.10	29,348.10	321,820,000
Jan 16, 2020	29,131.95	29,300.32	29,131.95	29,297.64	29,297.64	252,110,000
Jan 15, 2020	28,901.80	29,127.59	28,897.35	29,030.22	29,030.22	260,270,000
Jan 14, 2020	28,895.50	29,054.16	28,872.27	28,939.67	28,939.67	287,440,000
Jan 13, 2020	28,869.01	28,909.91	28,819.43	28,907.05	28,907.05	249,830,000
Jan 10, 2020	28,977.52	29,009.07	28,789.10	28,823.77	28,823.77	237,830,000
Jan 09, 2020	28,851.97	28,988.01	28,844.31	28,956.90	28,956.90	275,060,000
Jan 08, 2020	28,556.14	28,866.18	28,522.51	28,745.09	28,745.09	291,750,000
Jan 07, 2020	28,639.18	28,685.50	28,565.28	28,583.68	28,583.68	258,900,000
Jan 06, 2020	28,465.50	28,708.02	28,418.63	28,703.38	28,703.38	252,760,000
Jan 03, 2020	28,553.33	28,716.31	28,500.36	28,634.88	28,634.88	239,590,000
Jan 02, 2020	28,638.97	28,872.80	28,627.77	28,868.80	28,868.80	251,820,000
Dec 31, 2019	28,414.64	28,547.35	28,376.49	28,538.44	28,538.44	193,340,000
Dec 30, 2019	28,654.76	28,664.69	28,428.98	28,462.14	28,462.14	181,600,000
Dec 27, 2019	28,675.34	28,701.66	28,608.98	28,645.26	28,645.26	182,280,000
Dec 26, 2019	28,539.46	28,624.10	28,535.15	28,621.39	28,621.39	155,970,000
Dec 24, 2019	28,572.57	28,576.80	28,503.21	28,515.45	28,515.45	86,150,000
Dec 23, 2019	28,491.78	28,582.49	28,491.78	28,551.53	28,551.53	223,530,000
Dec 20, 2019	28,608.64	28,608.64	28,445.60	28,455.09	28,455.09	603,780,000
Dec 19, 2019	28,278.31	28,381.48	28,278.24	28,376.96	28,376.96	262,570,000
Dec 18, 2019	28,291.44	28,323.25	28,239.28	28,239.28	28,239.28	289,890,000
Dec 17, 2019	28,221.75	28,328.63	28,220.56	28,267.16	28,267.16	286,770,000
Dec 16, 2019	28,191.67	28,337.49	28,191.67	28,235.89	28,235.89	286,770,000
Dec 13, 2019	28,123.64	28,290.73	28,028.32	28,135.38	28,135.38	250,660,000
Dec 12, 2019	27,898.34	28,224.95	27,859.87	28,132.05	28,132.05	277,740,000
Dec 11, 2019	27,867.31	27,925.50	27,801.80	27,911.30	27,911.30	213,510,000
Dec 10, 2019	27,900.65	27,949.02	27,804.00	27,881.72	27,881.72	213,250,000
Dec 09, 2019	27,987.05	28,010.42	27,906.14	27,909.60	27,909.60	217,000,000
Dec 06, 2019	27,839.68	28,035.85	27,839.68	28,015.06	28,015.06	223,310,000
Dec 05, 2019	27,736.05	27,745.20	27,562.80	27,677.79	27,677.79	208,600,000
Dec 04, 2019	27,634.63	27,727.45	27,612.08	27,649.78	27,649.78	218,200,000
Dec 03, 2019	27,501.98	27,524.74	27,325.13	27,502.81	27,502.81	275,350,000

Dec 02, 2019	28,109.74	28,109.84	27,782.35	27,783.04	27,783.04	236,610,000
Nov 29, 2019	28,103.16	28,119.51	28,042.53	28,051.41	28,051.41	120,630,000
Nov 27, 2019	28,156.47	28,174.97	28,075.23	28,164.00	28,164.00	198,210,000
Nov 26, 2019	28,080.75	28,146.02	28,042.21	28,121.68	28,121.68	324,050,000
Nov 25, 2019	27,917.77	28,068.69	27,917.77	28,066.47	28,066.47	248,420,000
Nov 22, 2019	27,831.23	27,898.46	27,773.98	27,875.62	27,875.62	214,780,000
Nov 21, 2019	27,820.28	27,828.33	27,708.34	27,766.29	27,766.29	232,020,000
Nov 20, 2019	27,879.55	27,897.28	27,675.28	27,821.09	27,821.09	258,140,000
Nov 19, 2019	28,079.76	28,090.21	27,894.52	27,934.02	27,934.02	245,890,000
Nov 18, 2019	27,993.22	28,040.97	27,969.24	28,036.22	28,036.22	252,320,000
Nov 15, 2019	27,843.54	28,004.89	27,843.54	28,004.89	28,004.89	283,720,000
Nov 14, 2019	27,757.20	27,800.71	27,676.97	27,781.96	27,781.96	303,970,000
Nov 13, 2019	27,622.04	27,806.40	27,587.20	27,783.59	27,783.59	278,390,000
Nov 12, 2019	27,701.59	27,770.86	27,635.32	27,691.49	27,691.49	213,670,000
Nov 11, 2019	27,580.66	27,714.39	27,517.67	27,691.49	27,691.49	202,350,000
Nov 08, 2019	27,686.20	27,694.95	27,578.97	27,681.24	27,681.24	221,440,000
Nov 07, 2019	27,590.16	27,774.67	27,590.16	27,674.80	27,674.80	259,020,000
Nov 06, 2019	27,502.74	27,526.05	27,407.81	27,492.56	27,492.56	237,910,000
Nov 05, 2019	27,500.23	27,560.36	27,453.55	27,492.63	27,492.63	286,350,000
Nov 04, 2019	27,402.06	27,517.58	27,402.06	27,462.11	27,462.11	273,030,000
Nov 01, 2019	27,142.95	27,347.43	27,142.95	27,347.36	27,347.36	270,870,000
Oct 31, 2019	27,188.37	27,188.37	26,918.29	27,046.23	27,046.23	273,830,000
Oct 30, 2019	27,110.71	27,204.36	26,999.64	27,186.69	27,186.69	231,750,000
Oct 29, 2019	27,061.07	27,165.94	27,039.76	27,071.46	27,071.46	269,610,000
Oct 28, 2019	27,040.33	27,167.88	27,028.71	27,090.72	27,090.72	290,770,000
Oct 25, 2019	26,789.61	27,015.37	26,765.68	26,958.06	26,958.06	274,610,000
Oct 24, 2019	26,893.93	26,931.78	26,714.34	26,805.53	26,805.53	253,590,000
Oct 23, 2019	26,835.24	26,896.89	26,745.00	26,833.95	26,833.95	247,680,000
Oct 22, 2019	26,850.43	26,946.64	26,782.61	26,788.10	26,788.10	265,510,000
Oct 21, 2019	26,852.67	26,852.67	26,747.62	26,827.64	26,827.64	241,030,000
Oct 18, 2019	27,004.49	27,018.25	26,770.13	26,770.20	26,770.20	288,970,000
Oct 17, 2019	27,032.38	27,112.16	26,970.29	27,025.88	27,025.88	222,540,000
Oct 16, 2019	26,972.31	27,058.34	26,943.29	27,001.98	27,001.98	214,660,000
Oct 15, 2019	26,811.20	27,120.11	26,811.20	27,024.80	27,024.80	245,510,000
Oct 14, 2019	26,766.43	26,874.33	26,749.18	26,787.36	26,787.36	178,620,000
Oct 11, 2019	26,694.20	27,013.97	26,694.20	26,816.59	26,816.59	282,080,000
Oct 10, 2019	26,317.35	26,603.31	26,314.51	26,496.67	26,496.67	217,680,000
Oct 09, 2019	26,308.23	26,424.31	26,249.75	26,346.01	26,346.01	190,060,000
Oct 08, 2019	26,276.59	26,421.81	26,139.80	26,164.04	26,164.04	244,590,000
Oct 07, 2019	26,502.33	26,655.84	26,424.54	26,478.02	26,478.02	195,200,000
Oct 04, 2019	26,271.70	26,590.74	26,271.70	26,573.72	26,573.72	221,310,000

Oct 03, 2019	26,039.02	26,205.20	25,743.46	26,201.04	26,201.04	241,610,000
Oct 02, 2019	26,425.86	26,438.04	25,974.12	26,078.62	26,078.62	309,640,000
Oct 01, 2019	26,962.54	27,046.21	26,562.22	26,573.04	26,573.04	260,110,000
Sep 30, 2019	26,852.33	26,998.86	26,852.33	26,916.83	26,916.83	222,680,000
Sep 27, 2019	26,987.26	27,012.54	26,715.82	26,820.25	26,820.25	217,780,000
Sep 26, 2019	27,004.11	27,015.07	26,803.84	26,891.12	26,891.12	229,180,000
Sep 25, 2019	26,866.71	27,016.56	26,755.86	26,970.71	26,970.71	237,220,000
Sep 24, 2019	27,034.07	27,079.68	26,704.96	26,807.77	26,807.77	301,750,000
Sep 23, 2019	26,851.45	27,011.07	26,831.34	26,949.99	26,949.99	204,240,000
Sep 20, 2019	27,102.18	27,194.75	26,926.68	26,935.07	26,935.07	497,640,000
Sep 19, 2019	27,186.05	27,272.17	27,064.21	27,094.79	27,094.79	212,360,000
Sep 18, 2019	27,075.39	27,161.93	26,899.15	27,147.08	27,147.08	212,860,000
Sep 17, 2019	27,010.12	27,110.80	26,984.14	27,110.80	27,110.80	223,580,000
Sep 16, 2019	27,146.06	27,172.87	27,032.56	27,076.82	27,076.82	220,620,000
Sep 13, 2019	27,216.67	27,277.55	27,193.95	27,219.52	27,219.52	255,260,000
Sep 12, 2019	27,197.32	27,306.73	27,105.01	27,182.45	27,182.45	250,010,000
Sep 11, 2019	26,928.05	27,137.04	26,885.48	27,137.04	27,137.04	272,590,000
Sep 10, 2019	26,805.83	26,909.43	26,717.05	26,909.43	26,909.43	322,340,000
Sep 09, 2019	26,866.23	26,900.83	26,762.18	26,835.51	26,835.51	273,120,000
Sep 06, 2019	26,790.25	26,860.87	26,708.39	26,797.46	26,797.46	209,700,000
Sep 05, 2019	26,603.15	26,836.30	26,603.15	26,728.15	26,728.15	256,670,000
Sep 04, 2019	26,301.99	26,362.35	26,244.44	26,355.47	26,355.47	202,710,000
Sep 03, 2019	26,198.26	26,198.26	25,978.22	26,118.02	26,118.02	223,210,000
Aug 30, 2019	26,476.39	26,514.62	26,295.59	26,403.28	26,403.28	216,160,000
Aug 29, 2019	26,249.09	26,408.84	26,185.71	26,362.25	26,362.25	208,650,000
Aug 28, 2019	25,712.99	26,041.57	25,637.43	26,036.10	26,036.10	207,070,000
Aug 27, 2019	26,014.46	26,054.02	25,721.85	25,777.90	25,777.90	263,530,000
Aug 26, 2019	25,826.05	25,941.25	25,716.39	25,898.83	25,898.83	222,780,000
Aug 23, 2019	26,134.21	26,320.29	25,507.18	25,628.90	25,628.90	364,260,000
Aug 22, 2019	26,271.64	26,388.78	26,099.01	26,252.24	26,252.24	222,810,000
Aug 21, 2019	26,145.36	26,268.32	26,141.77	26,202.73	26,202.73	208,790,000
Aug 20, 2019	26,086.86	26,160.12	25,952.00	25,962.44	25,962.44	238,100,000
Aug 19, 2019	26,020.06	26,222.32	26,020.06	26,135.79	26,135.79	252,640,000
Aug 16, 2019	25,678.17	25,929.65	25,678.17	25,886.01	25,886.01	276,210,000
Aug 15, 2019	25,514.25	25,639.69	25,339.60	25,579.39	25,579.39	332,600,000
Aug 14, 2019	26,035.08	26,035.08	25,471.59	25,479.42	25,479.42	352,250,000
Aug 13, 2019	25,888.88	26,426.97	25,833.25	26,279.91	26,279.91	308,900,000
Aug 12, 2019	26,169.91	26,178.95	25,824.94	25,896.44	25,896.44	199,390,000
Aug 09, 2019	26,337.09	26,413.36	26,097.64	26,287.44	26,287.44	240,260,000
Aug 08, 2019	26,086.52	26,383.61	26,038.10	26,378.19	26,378.19	282,530,000
Aug 07, 2019	25,814.22	26,073.21	25,440.39	26,007.07	26,007.07	344,790,000

Aug 06, 2019	25,810.62	26,038.68	25,710.87	26,029.52	26,029.52	318,160,000
Aug 05, 2019	26,259.23	26,259.23	25,523.38	25,717.74	25,717.74	422,240,000
Aug 02, 2019	26,528.66	26,570.02	26,249.22	26,485.01	26,485.01	327,640,000
Aug 01, 2019	26,879.86	27,175.59	26,548.71	26,583.42	26,583.42	386,320,000
Jul 31, 2019	27,244.67	27,281.65	26,719.60	26,864.27	26,864.27	385,150,000
Jul 30, 2019	27,145.39	27,224.36	27,069.86	27,198.02	27,198.02	293,250,000
Jul 29, 2019	27,192.24	27,275.85	27,178.06	27,221.35	27,221.35	250,850,000
Jul 26, 2019	27,166.00	27,213.70	27,123.25	27,192.45	27,192.45	233,650,000
Jul 25, 2019	27,247.39	27,298.43	27,062.48	27,140.98	27,140.98	241,420,000
Jul 24, 2019	27,262.24	27,291.04	27,191.12	27,269.97	27,269.97	244,320,000
Jul 23, 2019	27,231.86	27,368.81	27,204.58	27,349.19	27,349.19	248,850,000
Jul 22, 2019	27,174.18	27,227.77	27,088.90	27,171.90	27,171.90	216,310,000
Jul 19, 2019	27,246.38	27,342.96	27,145.78	27,154.20	27,154.20	287,690,000
Jul 18, 2019	27,191.98	27,266.81	27,068.79	27,222.97	27,222.97	216,070,000
Jul 17, 2019	27,320.91	27,343.06	27,218.38	27,219.85	27,219.85	193,930,000
Jul 16, 2019	27,349.32	27,398.68	27,290.24	27,335.63	27,335.63	226,000,000
Jul 15, 2019	27,364.69	27,364.69	27,294.17	27,359.16	27,359.16	199,730,000
Jul 12, 2019	27,139.49	27,333.79	27,135.45	27,332.03	27,332.03	248,390,000
Jul 11, 2019	26,950.16	27,088.45	26,916.32	27,088.08	27,088.08	252,300,000
Jul 10, 2019	26,851.96	26,983.45	26,813.11	26,860.20	26,860.20	207,990,000
Jul 09, 2019	26,725.12	26,807.70	26,665.57	26,783.49	26,783.49	218,330,000
Jul 08, 2019	26,835.64	26,839.14	26,744.87	26,806.14	26,806.14	214,290,000
Jul 05, 2019	26,867.75	26,950.81	26,733.33	26,922.12	26,922.12	192,170,000
Jul 03, 2019	26,832.32	26,966.00	26,831.44	26,966.00	26,966.00	149,840,000
Jul 02, 2019	26,719.53	26,787.56	26,632.65	26,786.68	26,786.68	219,270,000
Jul 01, 2019	26,805.86	26,890.64	26,616.21	26,717.43	26,717.43	267,670,000
Jun 28, 2019	26,605.93	26,638.35	26,522.27	26,599.96	26,599.96	499,350,000
Jun 27, 2019	26,523.72	26,607.24	26,465.32	26,526.58	26,526.58	246,930,000
Jun 26, 2019	26,599.42	26,660.04	26,536.33	26,536.82	26,536.82	274,340,000
Jun 25, 2019	26,731.61	26,752.31	26,527.66	26,548.22	26,548.22	282,160,000
Jun 24, 2019	26,727.61	26,806.52	26,723.37	26,727.54	26,727.54	232,390,000
Jun 21, 2019	26,749.12	26,907.37	26,705.87	26,719.13	26,719.13	485,560,000
Jun 20, 2019	26,665.38	26,798.63	26,539.69	26,753.17	26,753.17	278,170,000
Jun 19, 2019	26,490.16	26,569.75	26,415.05	26,504.00	26,504.00	240,950,000
Jun 18, 2019	26,228.90	26,527.19	26,227.80	26,465.54	26,465.54	273,490,000
Jun 17, 2019	26,108.53	26,165.78	26,049.80	26,112.53	26,112.53	199,500,000
Jun 14, 2019	26,076.36	26,162.28	25,988.09	26,089.61	26,089.61	208,020,000
Jun 13, 2019	26,036.94	26,146.91	25,995.71	26,106.77	26,106.77	213,400,000
Jun 12, 2019	26,040.30	26,082.10	25,958.66	26,004.83	26,004.83	223,690,000
Jun 11, 2019	26,180.59	26,248.67	25,998.87	26,048.51	26,048.51	253,490,000
Jun 10, 2019	26,090.22	26,210.61	26,054.31	26,062.68	26,062.68	249,280,000

Jun 07, 2019	25,768.72	26,072.75	25,768.72	25,983.94	25,983.94	263,010,000
Jun 06, 2019	25,567.45	25,800.30	25,518.05	25,720.66	25,720.66	248,070,000
Jun 05, 2019	25,451.00	25,544.66	25,373.58	25,539.57	25,539.57	262,660,000
Jun 04, 2019	24,962.82	25,343.77	24,962.82	25,332.18	25,332.18	281,690,000
Jun 03, 2019	24,830.16	24,935.21	24,680.57	24,819.78	24,819.78	328,240,000
May 31, 2019	25,046.31	25,046.31	24,809.51	24,815.04	24,815.04	292,230,000
May 30, 2019	25,139.94	25,218.54	25,066.75	25,169.88	25,169.88	215,240,000
May 29, 2019	25,231.46	25,231.46	24,938.24	25,126.41	25,126.41	298,930,000
May 28, 2019	25,616.55	25,717.63	25,342.28	25,347.77	25,347.77	358,980,000
May 24, 2019	25,551.07	25,670.81	25,496.20	25,585.69	25,585.69	201,370,000
May 23, 2019	25,657.99	25,657.99	25,328.09	25,490.47	25,490.47	316,940,000
May 22, 2019	25,818.46	25,878.21	25,755.11	25,776.61	25,776.61	241,760,000
May 21, 2019	25,782.34	25,898.27	25,779.61	25,877.33	25,877.33	260,870,000
May 20, 2019	25,655.31	25,751.71	25,560.55	25,679.90	25,679.90	279,560,000
May 17, 2019	25,719.95	25,948.74	25,657.78	25,764.00	25,764.00	283,760,000
May 16, 2019	25,692.14	25,957.63	25,692.14	25,862.68	25,862.68	320,960,000
May 15, 2019	25,400.13	25,724.89	25,341.94	25,648.02	25,648.02	270,720,000
May 14, 2019	25,384.03	25,688.96	25,384.03	25,532.05	25,532.05	291,570,000
May 13, 2019	25,568.06	25,568.06	25,222.51	25,324.99	25,324.99	361,240,000
May 10, 2019	25,763.72	26,019.32	25,469.86	25,942.37	25,942.37	294,580,000
May 09, 2019	25,878.85	25,884.89	25,517.39	25,828.36	25,828.36	334,760,000
May 08, 2019	25,933.79	26,118.10	25,889.41	25,967.33	25,967.33	276,420,000
May 07, 2019	26,276.90	26,276.90	25,789.71	25,965.09	25,965.09	334,960,000
May 06, 2019	26,160.62	26,476.27	26,033.95	26,438.48	26,438.48	283,020,000
May 03, 2019	26,379.14	26,534.96	26,370.09	26,504.95	26,504.95	248,900,000
May 02, 2019	26,407.15	26,454.69	26,180.36	26,307.79	26,307.79	287,610,000
May 01, 2019	26,639.06	26,689.39	26,426.38	26,430.14	26,430.14	303,850,000
Apr 30, 2019	26,594.56	26,614.04	26,419.47	26,592.91	26,592.91	340,320,000
Apr 29, 2019	26,559.87	26,602.54	26,520.75	26,554.39	26,554.39	279,450,000
Apr 26, 2019	26,454.62	26,543.56	26,392.55	26,543.33	26,543.33	317,290,000
Apr 25, 2019	26,426.37	26,536.48	26,310.28	26,462.08	26,462.08	296,730,000
Apr 24, 2019	26,652.56	26,680.58	26,582.86	26,597.05	26,597.05	283,370,000
Apr 23, 2019	26,513.83	26,695.96	26,503.56	26,656.39	26,656.39	311,690,000
Apr 22, 2019	26,510.77	26,553.05	26,458.61	26,511.05	26,511.05	232,570,000
Apr 18, 2019	26,463.37	26,602.42	26,444.53	26,559.54	26,559.54	332,850,000
Apr 17, 2019	26,468.53	26,501.02	26,391.86	26,449.54	26,449.54	367,780,000
Apr 16, 2019	26,482.19	26,530.71	26,397.19	26,452.66	26,452.66	302,340,000
Apr 15, 2019	26,407.76	26,424.85	26,316.42	26,384.77	26,384.77	250,620,000
Apr 12, 2019	26,357.79	26,436.68	26,309.72	26,412.30	26,412.30	369,910,000
Apr 11, 2019	26,188.21	26,229.88	26,062.59	26,143.05	26,143.05	211,680,000
Apr 10, 2019	26,173.71	26,209.49	26,101.24	26,157.16	26,157.16	203,130,000

Apr 09, 2019	26,243.54	26,246.34	26,103.14	26,150.58	26,150.58	244,410,000
Apr 08, 2019	26,312.67	26,344.65	26,246.03	26,341.02	26,341.02	229,380,000
Apr 05, 2019	26,427.56	26,487.57	26,370.82	26,424.99	26,424.99	229,550,000
Apr 04, 2019	26,213.42	26,398.90	26,212.78	26,384.63	26,384.63	233,040,000
Apr 03, 2019	26,238.03	26,282.17	26,138.47	26,218.13	26,218.13	271,980,000
Apr 02, 2019	26,213.55	26,221.24	26,122.31	26,179.13	26,179.13	250,940,000
Apr 01, 2019	26,075.10	26,280.90	26,071.69	26,258.42	26,258.42	295,570,000
Mar 29, 2019	25,827.31	25,949.32	25,771.67	25,928.68	25,928.68	300,620,000
Mar 28, 2019	25,693.32	25,743.41	25,576.69	25,717.46	25,717.46	237,170,000
Mar 27, 2019	25,676.34	25,758.17	25,425.27	25,625.59	25,625.59	270,340,000
Mar 26, 2019	25,649.56	25,796.29	25,544.78	25,657.73	25,657.73	305,110,000
Mar 25, 2019	25,490.72	25,603.27	25,372.26	25,516.83	25,516.83	303,120,000
Mar 22, 2019	25,844.65	25,877.01	25,501.45	25,502.32	25,502.32	386,510,000
Mar 21, 2019	25,688.44	26,009.90	25,657.78	25,962.51	25,962.51	335,250,000
Mar 20, 2019	25,867.79	25,929.52	25,670.63	25,745.67	25,745.67	321,980,000
Mar 19, 2019	25,987.87	26,109.68	25,814.92	25,887.38	25,887.38	342,250,000
Mar 18, 2019	25,801.88	25,924.77	25,785.66	25,914.10	25,914.10	303,170,000
Mar 15, 2019	25,720.96	25,927.91	25,649.70	25,848.87	25,848.87	616,910,000
Mar 14, 2019	25,692.31	25,752.84	25,621.31	25,709.94	25,709.94	302,790,000
Mar 13, 2019	25,637.23	25,776.49	25,571.31	25,702.89	25,702.89	331,410,000
Mar 12, 2019	25,600.30	25,675.44	25,522.17	25,554.66	25,554.66	317,730,000
Mar 11, 2019	25,208.00	25,661.63	25,208.00	25,650.88	25,650.88	309,790,000
Mar 08, 2019	25,347.38	25,466.14	25,252.46	25,450.24	25,450.24	270,500,000
Mar 07, 2019	25,645.45	25,645.45	25,352.55	25,473.23	25,473.23	289,140,000
Mar 06, 2019	25,818.76	25,837.61	25,633.71	25,673.46	25,673.46	256,250,000
Mar 05, 2019	25,829.07	25,877.15	25,725.63	25,806.63	25,806.63	252,260,000
Mar 04, 2019	26,122.19	26,155.98	25,611.55	25,819.65	25,819.65	295,510,000
Mar 01, 2019	26,019.67	26,143.92	25,914.37	26,026.32	26,026.32	289,890,000
Feb 28, 2019	25,984.28	26,029.21	25,896.56	25,916.00	25,916.00	338,520,000
Feb 27, 2019	25,995.60	26,039.68	25,877.24	25,985.16	25,985.16	255,380,000
Feb 26, 2019	26,051.61	26,155.29	25,966.01	26,057.98	26,057.98	257,850,000
Feb 25, 2019	26,126.15	26,241.42	26,080.66	26,091.95	26,091.95	281,600,000
Feb 22, 2019	25,906.27	26,052.90	25,906.27	26,031.81	26,031.81	279,760,000
Feb 21, 2019	25,922.41	25,938.88	25,762.21	25,850.63	25,850.63	255,710,000
Feb 20, 2019	25,872.26	25,986.20	25,846.48	25,954.44	25,954.44	287,880,000
Feb 19, 2019	25,849.85	25,961.44	25,820.01	25,891.32	25,891.32	280,620,000
Feb 15, 2019	25,564.63	25,883.72	25,564.63	25,883.25	25,883.25	328,530,000
Feb 14, 2019	25,460.65	25,558.90	25,308.09	25,439.39	25,439.39	313,010,000
Feb 13, 2019	25,480.86	25,625.95	25,480.86	25,543.27	25,543.27	263,330,000
Feb 12, 2019	25,152.03	25,458.98	25,152.03	25,425.76	25,425.76	278,110,000
Feb 11, 2019	25,142.81	25,196.75	25,009.10	25,053.11	25,053.11	241,700,000

Feb 08, 2019	25,042.36	25,106.39	24,883.04	25,106.33	25,106.33	248,210,000
Feb 07, 2019	25,265.81	25,314.26	25,000.52	25,169.53	25,169.53	295,460,000
Feb 06, 2019	25,371.57	25,439.04	25,312.06	25,390.30	25,390.30	247,590,000
Feb 05, 2019	25,287.93	25,427.32	25,287.65	25,411.52	25,411.52	310,650,000
Feb 04, 2019	25,062.12	25,239.91	24,977.67	25,239.37	25,239.37	298,540,000
Feb 01, 2019	25,025.31	25,193.15	24,982.49	25,063.89	25,063.89	372,090,000
Jan 31, 2019	24,954.48	25,049.62	24,842.09	24,999.67	24,999.67	507,350,000
Jan 30, 2019	24,826.52	25,109.62	24,790.90	25,014.86	25,014.86	397,330,000
Jan 29, 2019	24,519.62	24,674.87	24,504.04	24,579.96	24,579.96	330,870,000
Jan 28, 2019	24,596.98	24,596.98	24,323.94	24,528.22	24,528.22	347,170,000
Jan 25, 2019	24,687.21	24,860.15	24,676.75	24,737.20	24,737.20	376,890,000
Jan 24, 2019	24,579.96	24,626.30	24,422.73	24,553.24	24,553.24	320,170,000
Jan 23, 2019	24,577.25	24,700.98	24,307.17	24,575.62	24,575.62	318,600,000
Jan 22, 2019	24,607.76	24,607.76	24,244.31	24,404.48	24,404.48	338,480,000
Jan 18, 2019	24,534.19	24,750.22	24,459.03	24,706.35	24,706.35	372,970,000
Jan 17, 2019	24,147.09	24,474.46	24,088.90	24,370.10	24,370.10	288,590,000
Jan 16, 2019	24,139.91	24,288.61	24,119.72	24,207.16	24,207.16	302,830,000
Jan 15, 2019	23,914.11	24,099.14	23,887.93	24,065.59	24,065.59	291,570,000
Jan 14, 2019	23,880.53	23,964.90	23,765.24	23,909.84	23,909.84	277,560,000
Jan 11, 2019	23,940.01	23,996.32	23,798.16	23,995.95	23,995.95	262,650,000
Jan 10, 2019	23,811.11	24,014.78	23,703.25	24,001.92	24,001.92	338,150,000
Jan 09, 2019	23,844.27	24,084.71	23,703.16	23,879.12	23,879.12	324,570,000
Jan 08, 2019	23,680.32	23,864.65	23,581.45	23,787.45	23,787.45	317,420,000
Jan 07, 2019	23,474.26	23,687.74	23,301.59	23,531.35	23,531.35	334,200,000
Jan 04, 2019	22,894.92	23,518.64	22,894.92	23,433.16	23,433.16	396,020,000
Jan 03, 2019	23,176.39	23,176.39	22,638.41	22,686.22	22,686.22	424,240,000
Jan 02, 2019	23,058.61	23,413.47	22,928.59	23,346.24	23,346.24	321,570,000
Dec 31, 2018	23,153.94	23,333.18	23,118.30	23,327.46	23,327.46	288,830,000
Dec 28, 2018	23,213.61	23,381.88	22,981.33	23,062.40	23,062.40	336,510,000
Dec 27, 2018	22,629.06	23,138.89	22,267.42	23,138.82	23,138.82	407,940,000
Dec 26, 2018	21,857.73	22,878.92	21,712.53	22,878.45	22,878.45	433,080,000
Dec 24, 2018	22,317.28	22,339.87	21,792.20	21,792.20	21,792.20	308,420,000
Dec 21, 2018	22,871.74	23,254.59	22,396.34	22,445.37	22,445.37	900,510,000
Dec 20, 2018	23,224.12	23,282.20	22,644.31	22,859.60	22,859.60	576,700,000
Dec 19, 2018	23,693.33	24,057.34	23,162.64	23,323.66	23,323.66	503,550,000
Dec 18, 2018	23,769.13	23,927.77	23,516.02	23,675.64	23,675.64	408,070,000
Dec 17, 2018	23,986.83	24,088.08	23,456.88	23,592.98	23,592.98	466,740,000
Dec 14, 2018	24,408.04	24,431.14	24,033.79	24,100.51	24,100.51	416,940,000
Dec 13, 2018	24,575.86	24,740.96	24,473.29	24,597.38	24,597.38	299,970,000
Dec 12, 2018	24,509.09	24,828.29	24,509.09	24,527.27	24,527.27	344,330,000
Dec 11, 2018	24,719.91	24,791.26	24,221.06	24,370.24	24,370.24	342,960,000

Dec 10, 2018	24,360.95	24,500.81	23,881.37	24,423.26	24,423.26	389,690,000
Dec 07, 2018	24,918.82	25,095.62	24,284.78	24,388.95	24,388.95	398,230,000
Dec 06, 2018	24,737.42	24,951.01	24,242.22	24,947.67	24,947.67	471,690,000
Dec 04, 2018	25,752.56	25,773.12	25,008.11	25,027.07	25,027.07	418,900,000
Dec 03, 2018	25,779.57	25,980.21	25,670.51	25,826.43	25,826.43	388,480,000
Nov 30, 2018	25,307.14	25,549.71	25,250.97	25,538.46	25,538.46	482,250,000
Nov 29, 2018	25,343.65	25,479.04	25,202.79	25,338.84	25,338.84	305,110,000
Nov 28, 2018	24,832.84	25,368.93	24,832.84	25,366.43	25,366.43	362,160,000
Nov 27, 2018	24,557.02	24,750.73	24,416.03	24,748.73	24,748.73	308,250,000
Nov 26, 2018	24,364.13	24,673.29	24,364.13	24,640.24	24,640.24	306,280,000
Nov 23, 2018	24,336.40	24,408.80	24,268.74	24,285.95	24,285.95	155,940,000
Nov 21, 2018	24,541.65	24,669.79	24,463.28	24,464.69	24,464.69	289,370,000
Nov 20, 2018	24,618.68	24,707.26	24,368.98	24,465.64	24,465.64	445,920,000
Nov 19, 2018	25,392.61	25,392.61	24,900.98	25,017.44	25,017.44	358,090,000
Nov 16, 2018	25,242.35	25,510.23	25,147.80	25,413.22	25,413.22	354,460,000
Nov 15, 2018	25,061.48	25,354.56	24,787.79	25,289.27	25,289.27	392,750,000
Nov 14, 2018	25,388.08	25,501.29	24,935.82	25,080.50	25,080.50	384,240,000
Nov 13, 2018	25,321.21	25,511.03	25,193.78	25,286.49	25,286.49	339,690,000
Nov 12, 2018	25,959.33	25,966.71	25,340.51	25,387.18	25,387.18	346,910,000
Nov 09, 2018	26,149.11	26,161.49	25,882.91	25,989.30	25,989.30	324,450,000
Nov 08, 2018	26,139.59	26,277.82	26,081.90	26,191.22	26,191.22	295,650,000
Nov 07, 2018	25,788.46	26,200.14	25,765.88	26,180.30	26,180.30	361,780,000
Nov 06, 2018	25,452.83	25,651.86	25,444.90	25,635.01	25,635.01	265,450,000
Nov 05, 2018	25,261.47	25,507.35	25,261.47	25,461.70	25,461.70	334,740,000
Nov 02, 2018	25,443.60	25,578.98	25,078.72	25,270.83	25,270.83	429,480,000
Nov 01, 2018	25,142.08	25,396.47	25,108.11	25,380.74	25,380.74	383,970,000
Oct 31, 2018	25,008.82	25,336.55	25,008.82	25,115.76	25,115.76	448,930,000
Oct 30, 2018	24,482.04	24,906.68	24,415.69	24,874.64	24,874.64	470,050,000
Oct 29, 2018	24,818.98	25,040.58	24,122.23	24,442.92	24,442.92	443,850,000
Oct 26, 2018	24,770.25	24,916.16	24,445.19	24,688.31	24,688.31	505,310,000
Oct 25, 2018	24,736.54	25,104.29	24,645.56	24,984.55	24,984.55	439,670,000
Oct 24, 2018	25,172.88	25,306.23	24,533.19	24,583.42	24,583.42	496,400,000
Oct 23, 2018	25,038.46	25,307.70	24,768.79	25,191.43	25,191.43	436,430,000
Oct 22, 2018	25,492.14	25,561.34	25,236.05	25,317.41	25,317.41	314,940,000
Oct 19, 2018	25,421.09	25,608.71	25,350.02	25,444.34	25,444.34	350,810,000
Oct 18, 2018	25,645.56	25,691.46	25,236.01	25,379.45	25,379.45	328,020,000
Oct 17, 2018	25,705.87	25,810.09	25,479.16	25,706.68	25,706.68	298,210,000
Oct 16, 2018	25,351.53	25,817.68	25,351.53	25,798.42	25,798.42	305,800,000
Oct 15, 2018	25,332.46	25,482.42	25,243.88	25,250.55	25,250.55	287,320,000
Oct 12, 2018	25,407.63	25,467.55	25,000.83	25,339.99	25,339.99	397,510,000
Oct 11, 2018	25,518.39	25,683.64	24,899.77	25,052.83	25,052.83	522,880,000

Oct 10, 2018	26,441.73	26,487.87	25,567.10	25,598.74	25,598.74	438,200,000
Oct 09, 2018	26,469.19	26,539.94	26,324.16	26,430.57	26,430.57	282,560,000
Oct 08, 2018	26,399.45	26,529.35	26,223.09	26,486.78	26,486.78	273,020,000
Oct 05, 2018	26,632.77	26,676.16	26,301.81	26,447.05	26,447.05	267,100,000
Oct 04, 2018	26,784.08	26,793.82	26,471.61	26,627.48	26,627.48	289,600,000
Oct 03, 2018	26,833.47	26,951.81	26,789.08	26,828.39	26,828.39	280,130,000
Oct 02, 2018	26,648.91	26,824.78	26,627.66	26,773.94	26,773.94	258,290,000
Oct 01, 2018	26,598.36	26,737.98	26,596.05	26,651.21	26,651.21	238,320,000
Sep 28, 2018	26,407.66	26,515.76	26,383.57	26,458.31	26,458.31	264,030,000
Sep 27, 2018	26,418.27	26,557.01	26,379.95	26,439.93	26,439.93	226,350,000
Sep 26, 2018	26,536.86	26,606.09	26,349.34	26,385.28	26,385.28	262,620,000
Sep 25, 2018	26,601.58	26,634.85	26,475.58	26,492.21	26,492.21	234,110,000
Sep 24, 2018	26,705.25	26,709.94	26,548.68	26,562.05	26,562.05	254,850,000
Sep 21, 2018	26,726.25	26,769.16	26,680.37	26,743.50	26,743.50	657,620,000
Sep 20, 2018	26,519.39	26,697.49	26,519.39	26,656.98	26,656.98	260,480,000
Sep 19, 2018	26,287.84	26,464.41	26,280.76	26,405.76	26,405.76	251,330,000
Sep 18, 2018	26,076.21	26,317.34	26,076.21	26,246.96	26,246.96	243,000,000
Sep 17, 2018	26,151.66	26,184.47	26,030.35	26,062.12	26,062.12	230,230,000
Sep 14, 2018	26,169.56	26,211.11	26,068.29	26,154.67	26,154.67	228,640,000
Sep 13, 2018	26,083.94	26,191.64	26,067.08	26,145.99	26,145.99	260,800,000
Sep 12, 2018	25,989.07	26,145.72	25,929.43	25,998.92	25,998.92	282,770,000
Sep 11, 2018	25,841.14	26,019.92	25,754.32	25,971.06	25,971.06	265,960,000
Sep 10, 2018	25,991.91	26,039.96	25,854.13	25,857.07	25,857.07	252,170,000
Sep 07, 2018	25,951.02	26,002.99	25,818.39	25,916.54	25,916.54	281,350,000
Sep 06, 2018	25,973.02	26,073.68	25,880.84	25,995.87	25,995.87	262,380,000
Sep 05, 2018	25,919.84	26,011.22	25,871.04	25,974.99	25,974.99	289,210,000
Sep 04, 2018	25,916.07	25,971.77	25,805.95	25,952.48	25,952.48	254,600,000
Aug 31, 2018	25,964.85	26,028.83	25,879.77	25,964.82	25,964.82	273,440,000
Aug 30, 2018	26,099.01	26,104.37	25,934.80	25,986.92	25,986.92	240,440,000
Aug 29, 2018	26,082.53	26,167.94	26,035.30	26,124.57	26,124.57	214,960,000
Aug 28, 2018	26,092.70	26,122.24	26,037.85	26,064.02	26,064.02	211,470,000
Aug 27, 2018	25,882.71	26,067.57	25,882.71	26,049.64	26,049.64	220,510,000
Aug 24, 2018	25,688.58	25,826.05	25,688.58	25,790.35	25,790.35	200,330,000
Aug 23, 2018	25,714.86	25,762.56	25,608.02	25,656.98	25,656.98	213,270,000
Aug 22, 2018	25,825.06	25,836.16	25,722.43	25,733.60	25,733.60	200,480,000
Aug 21, 2018	25,786.99	25,888.82	25,784.90	25,822.29	25,822.29	241,620,000
Aug 20, 2018	25,727.70	25,790.17	25,716.41	25,758.69	25,758.69	248,570,000
Aug 17, 2018	25,550.80	25,728.16	25,521.66	25,669.32	25,669.32	284,160,000
Aug 16, 2018	25,294.97	25,607.34	25,294.97	25,558.73	25,558.73	342,430,000
Aug 15, 2018	25,235.37	25,235.37	24,965.77	25,162.41	25,162.41	295,810,000
Aug 14, 2018	25,215.69	25,339.51	25,201.87	25,299.92	25,299.92	219,210,000

Aug 13, 2018	25,327.19	25,381.39	25,153.93	25,187.70	25,187.70	219,990,000
Aug 10, 2018	25,401.19	25,401.19	25,222.88	25,313.14	25,313.14	234,480,000
Aug 09, 2018	25,589.79	25,613.31	25,492.69	25,509.23	25,509.23	214,970,000
Aug 08, 2018	25,615.72	25,634.11	25,557.48	25,583.75	25,583.75	217,770,000
Aug 07, 2018	25,551.65	25,692.72	25,551.65	25,628.91	25,628.91	239,910,000
Aug 06, 2018	25,437.43	25,540.02	25,381.38	25,502.18	25,502.18	238,990,000
Aug 03, 2018	25,360.37	25,467.90	25,325.17	25,462.58	25,462.58	239,020,000
Aug 02, 2018	25,256.45	25,360.48	25,120.07	25,326.16	25,326.16	307,700,000
Aug 01, 2018	25,461.63	25,488.80	25,278.32	25,333.82	25,333.82	315,880,000
Jul 31, 2018	25,345.21	25,490.98	25,345.21	25,415.19	25,415.19	348,700,000
Jul 30, 2018	25,439.32	25,500.16	25,287.38	25,306.83	25,306.83	296,880,000
Jul 27, 2018	25,520.52	25,580.22	25,370.07	25,451.06	25,451.06	329,680,000
Jul 26, 2018	25,468.55	25,587.24	25,463.16	25,527.07	25,527.07	297,530,000
Jul 25, 2018	25,183.70	25,432.87	25,113.55	25,414.10	25,414.10	273,670,000
Jul 24, 2018	25,092.43	25,286.62	25,092.43	25,241.94	25,241.94	274,940,000
Jul 23, 2018	25,036.90	25,081.45	24,983.33	25,044.29	25,044.29	228,820,000
Jul 20, 2018	25,041.14	25,124.10	24,986.35	25,058.12	25,058.12	274,080,000
Jul 19, 2018	25,139.15	25,154.17	25,052.62	25,064.50	25,064.50	285,760,000
Jul 18, 2018	25,133.79	25,215.32	25,101.12	25,199.29	25,199.29	252,870,000
Jul 17, 2018	25,033.92	25,155.39	24,989.61	25,119.89	25,119.89	250,450,000
Jul 16, 2018	25,025.58	25,072.41	24,979.64	25,064.36	25,064.36	231,270,000
Jul 13, 2018	24,926.07	25,043.21	24,890.06	25,019.41	25,019.41	255,520,000
Jul 12, 2018	24,802.90	24,939.97	24,802.90	24,924.89	24,924.89	233,150,000
Jul 11, 2018	24,789.48	24,815.16	24,663.82	24,700.45	24,700.45	237,370,000
Jul 10, 2018	24,806.97	24,945.38	24,806.97	24,919.66	24,919.66	235,040,000
Jul 09, 2018	24,519.20	24,796.52	24,518.43	24,776.59	24,776.59	240,580,000
Jul 06, 2018	24,352.47	24,520.29	24,281.47	24,456.48	24,456.48	219,450,000
Jul 05, 2018	24,285.82	24,372.80	24,177.44	24,356.74	24,356.74	237,670,000
Jul 03, 2018	24,359.39	24,444.88	24,150.85	24,174.82	24,174.82	160,960,000
Jul 02, 2018	24,161.53	24,319.42	24,077.56	24,307.18	24,307.18	244,110,000
Jun 29, 2018	24,323.93	24,509.52	24,269.71	24,271.41	24,271.41	329,070,000
Jun 28, 2018	24,064.19	24,308.16	23,997.21	24,216.05	24,216.05	305,020,000
Jun 27, 2018	24,303.11	24,569.02	24,115.82	24,117.59	24,117.59	313,790,000
Jun 26, 2018	24,281.89	24,384.21	24,241.22	24,283.11	24,283.11	301,130,000
Jun 25, 2018	24,463.73	24,463.73	24,084.39	24,252.80	24,252.80	439,130,000
Jun 22, 2018	24,526.97	24,663.18	24,526.97	24,580.89	24,580.89	473,970,000
Jun 21, 2018	24,639.21	24,639.21	24,406.63	24,461.70	24,461.70	349,570,000
Jun 20, 2018	24,771.17	24,804.76	24,628.39	24,657.80	24,657.80	359,520,000
Jun 19, 2018	24,763.59	24,763.59	24,567.75	24,700.21	24,700.21	376,380,000
Jun 18, 2018	24,944.28	25,003.10	24,825.77	24,987.47	24,987.47	337,070,000
Jun 15, 2018	25,116.71	25,130.82	24,894.38	25,090.48	25,090.48	659,370,000

Jun 14, 2018	25,254.65	25,332.50	25,138.60	25,175.31	25,175.31	341,170,000
Jun 13, 2018	25,328.65	25,362.07	25,191.42	25,201.20	25,201.20	329,700,000
Jun 12, 2018	25,346.82	25,364.37	25,247.04	25,320.73	25,320.73	265,750,000
Jun 11, 2018	25,336.67	25,402.83	25,290.20	25,322.31	25,322.31	276,740,000
Jun 08, 2018	25,209.29	25,325.63	25,165.70	25,316.53	25,316.53	317,510,000
Jun 07, 2018	25,192.14	25,326.09	25,164.48	25,241.41	25,241.41	357,680,000
Jun 06, 2018	24,854.14	25,146.46	24,854.14	25,146.39	25,146.39	316,090,000
Jun 05, 2018	24,820.12	24,838.78	24,710.82	24,799.98	24,799.98	306,340,000
Jun 04, 2018	24,727.55	24,859.37	24,722.14	24,813.69	24,813.69	340,400,000
Jun 01, 2018	24,542.09	24,673.60	24,542.09	24,635.21	24,635.21	315,710,000
May 31, 2018	24,620.79	24,620.79	24,352.15	24,415.84	24,415.84	429,740,000
May 30, 2018	24,467.83	24,714.48	24,459.09	24,667.78	24,667.78	324,870,000
May 29, 2018	24,606.59	24,635.18	24,247.84	24,361.45	24,361.45	395,810,000
May 25, 2018	24,781.29	24,824.22	24,687.81	24,753.09	24,753.09	257,210,000
May 24, 2018	24,877.36	24,877.36	24,605.90	24,811.76	24,811.76	347,050,000
May 23, 2018	24,757.71	24,889.46	24,667.12	24,886.81	24,886.81	399,610,000
May 22, 2018	25,047.55	25,064.99	24,812.06	24,834.41	24,834.41	288,200,000
May 21, 2018	24,883.06	25,086.49	24,883.06	25,013.29	25,013.29	308,920,000
May 18, 2018	24,707.72	24,774.97	24,664.87	24,715.09	24,715.09	269,700,000
May 17, 2018	24,752.40	24,839.49	24,639.40	24,713.98	24,713.98	314,650,000
May 16, 2018	24,722.32	24,801.19	24,672.79	24,768.93	24,768.93	280,810,000
May 15, 2018	24,809.55	24,809.55	24,629.39	24,706.41	24,706.41	301,900,000
May 14, 2018	24,879.37	24,994.19	24,862.52	24,899.41	24,899.41	282,860,000
May 11, 2018	24,758.64	24,868.65	24,717.50	24,831.17	24,831.17	274,150,000
May 10, 2018	24,591.66	24,794.99	24,575.91	24,739.53	24,739.53	304,210,000
May 09, 2018	24,399.18	24,586.48	24,323.87	24,542.54	24,542.54	361,580,000
May 08, 2018	24,341.35	24,412.34	24,198.34	24,360.21	24,360.21	344,940,000
May 07, 2018	24,317.66	24,479.45	24,263.42	24,357.32	24,357.32	307,670,000
May 04, 2018	23,865.22	24,333.35	23,778.87	24,262.51	24,262.51	329,480,000
May 03, 2018	23,836.23	23,996.15	23,531.31	23,930.15	23,930.15	389,240,000
May 02, 2018	24,097.63	24,185.52	23,886.30	23,924.98	23,924.98	385,350,000
May 01, 2018	24,117.29	24,117.29	23,808.19	24,099.05	24,099.05	380,070,000

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